



US012403235B2

(12) **United States Patent**
Gonzalez et al.

(10) **Patent No.:** **US 12,403,235 B2**

(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **WOUND THERAPY SYSTEM WITH
INTERNAL ALTERNATING ORIFICE**

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(71) Applicant: **Solventum Intellectual Properties
Company**, Maplewood, MN (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 301 days.

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This patent is subject to a terminal dis-
claimer.

Japanese Office Action for corresponding application 2021-545678,
dated Jan. 31, 2023.

(Continued)

(21) Appl. No.: **17/957,739**

(22) Filed: **Sep. 30, 2022**

Primary Examiner — Kai H Weng

(65) **Prior Publication Data**

US 2023/0016336 A1 Jan. 19, 2023

Related U.S. Application Data

(62) Division of application No. 16/363,928, filed on Mar.
25, 2019, now Pat. No. 11,491,054.
(Continued)

(51) **Int. Cl.**
A61M 1/00 (2006.01)
A61B 5/00 (2006.01)
(Continued)

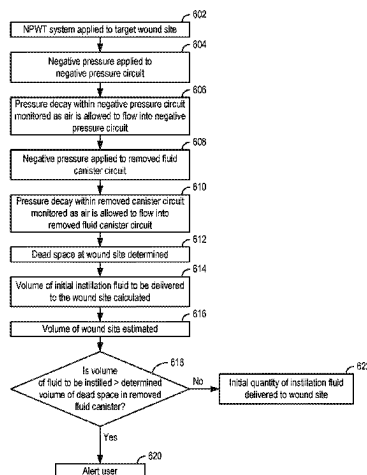
(52) **U.S. Cl.**
CPC **A61M 1/742** (2021.05); **A61B 5/445**
(2013.01); **A61F 13/05** (2024.01); **A61M 1/71**
(2021.05);
(Continued)

(58) **Field of Classification Search**
CPC A61M 1/0023; A61M 1/0058; A61M 1/73;
A61M 1/742; A61M 1/85; A61M 1/962;
(Continued)

(57) **ABSTRACT**

A wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, tubing fluidly communicable with the wound space, and a canister fluidly communicable with the tubing. The canister, the tubing, and the dressing define a sealed space that includes the wound space. The wound therapy system also includes a therapy unit coupled to the canister. The therapy unit includes a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and control-able between an open position and a closed position, and a control circuit. The control circuit is configured to control the valve to alternate between the open position and the closed position to allow airflow through the valve, receive measurements from the sensor, and determine a volume of the wound space based on the measurements.

12 Claims, 25 Drawing Sheets



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(51)	Int. Cl. <i>A61F 13/05</i> (2024.01) <i>A61M 3/02</i> (2006.01) <i>A61F 13/00</i> (2006.01) <i>A61F 13/84</i> (2006.01) <i>A61M 35/00</i> (2006.01)		4,655,754	A	4/1987 Richmond et al.
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(52)	U.S. Cl. CPC <i>A61M 1/77</i> (2021.05); <i>A61M 1/777</i> (2021.05); <i>A61M 1/90</i> (2021.05); <i>A61M 1/92</i> (2021.05); <i>A61M 1/96</i> (2021.05); <i>A61M 1/962</i> (2021.05); <i>A61M 3/0258</i> (2013.01); <i>A61F 2013/00174</i> (2013.01); <i>A61F 2013/8494</i> (2013.01); <i>A61M 1/73</i> (2021.05); <i>A61M 1/74</i> (2021.05); <i>A61M 1/964</i> (2021.05); <i>A61M 1/966</i> (2021.05); <i>A61M 3/0202</i> (2021.05); <i>A61M 35/006</i> (2013.01); <i>A61M 2205/3337</i> (2013.01); <i>A61M 2205/3344</i> (2013.01)		4,826,494	A	5/1989 Richmond et al.
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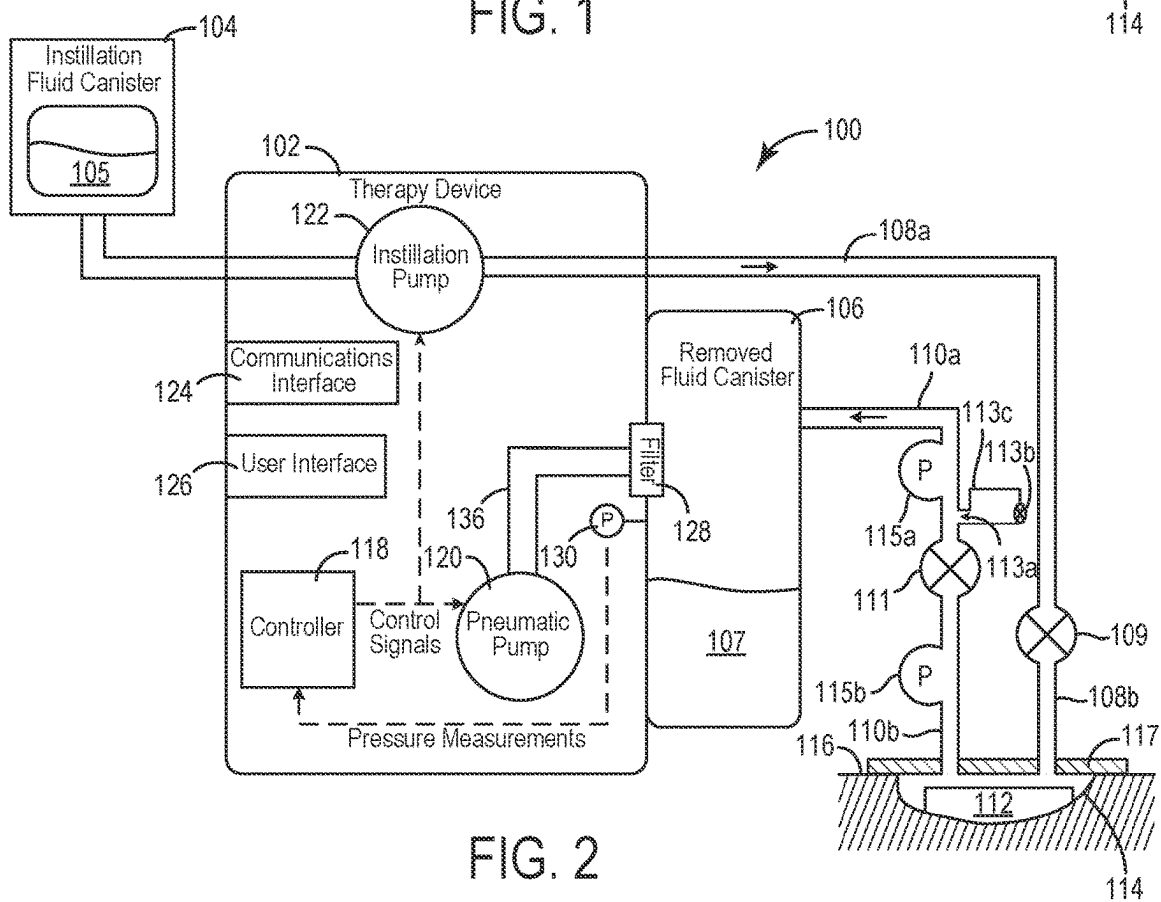
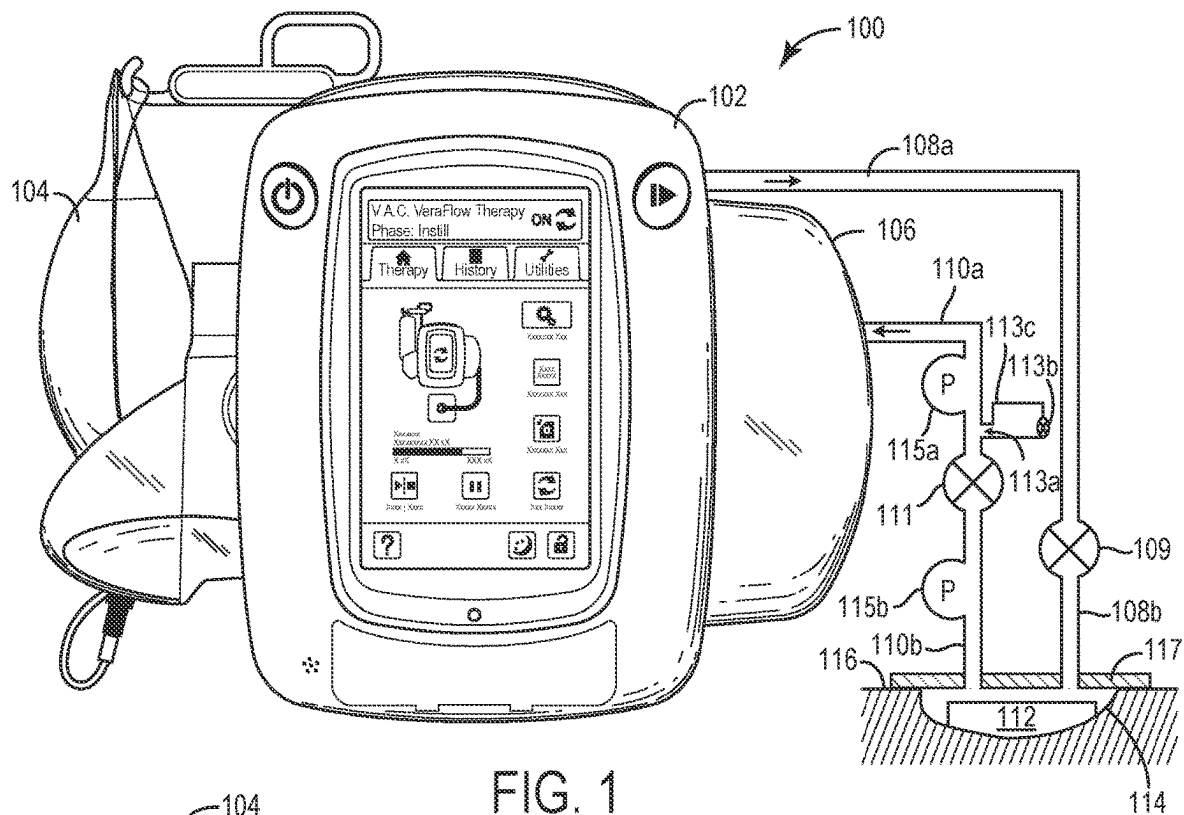
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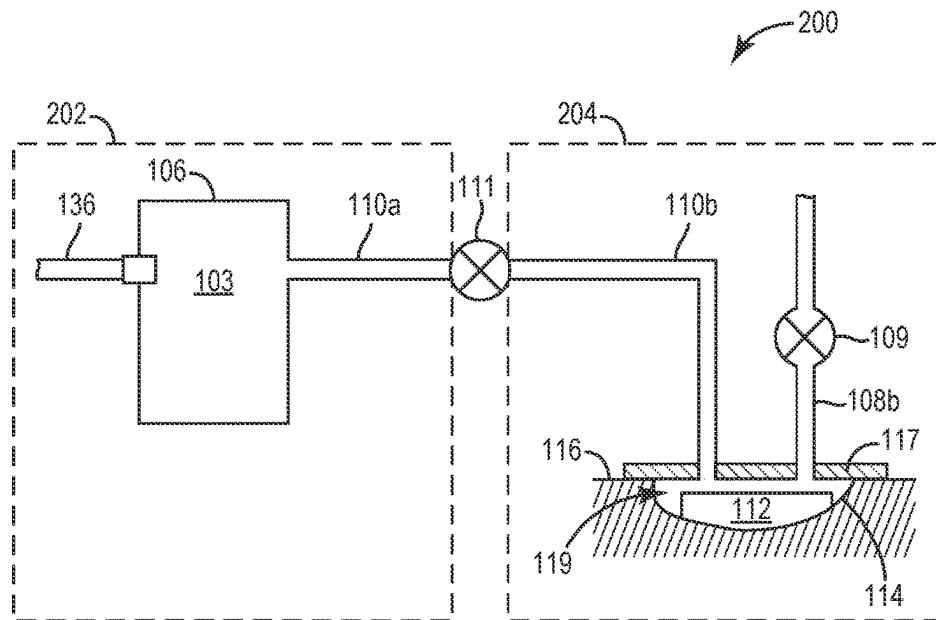


FIG. 3

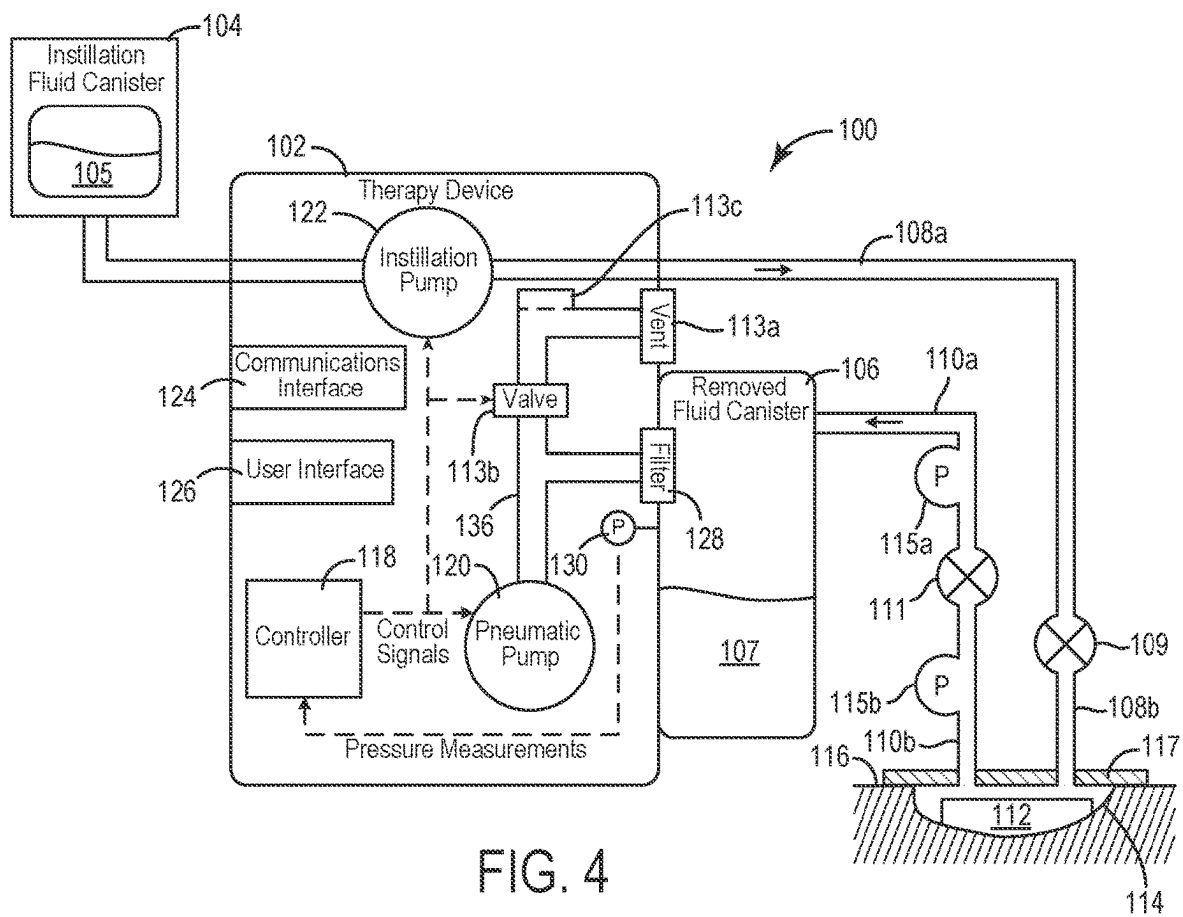


FIG. 4

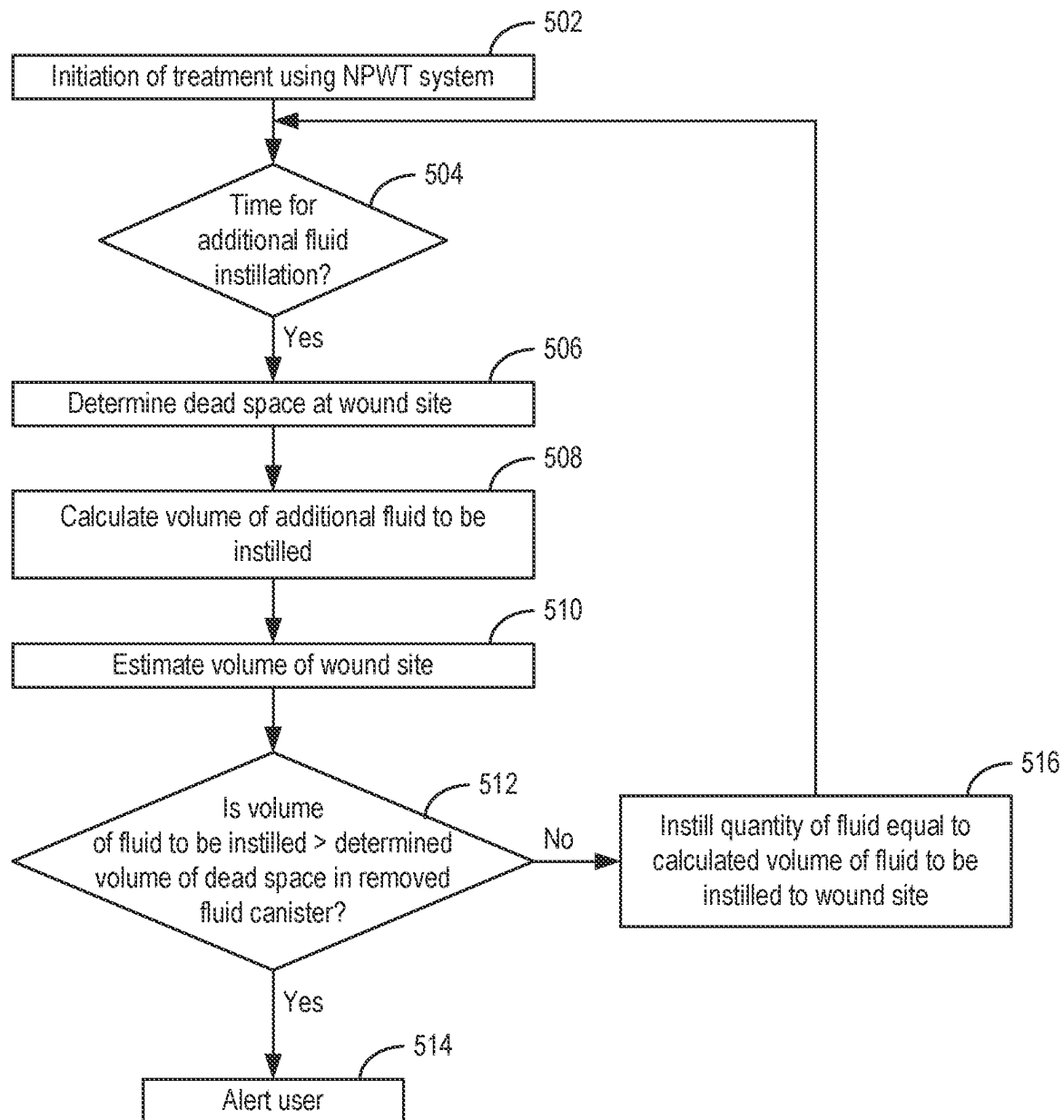


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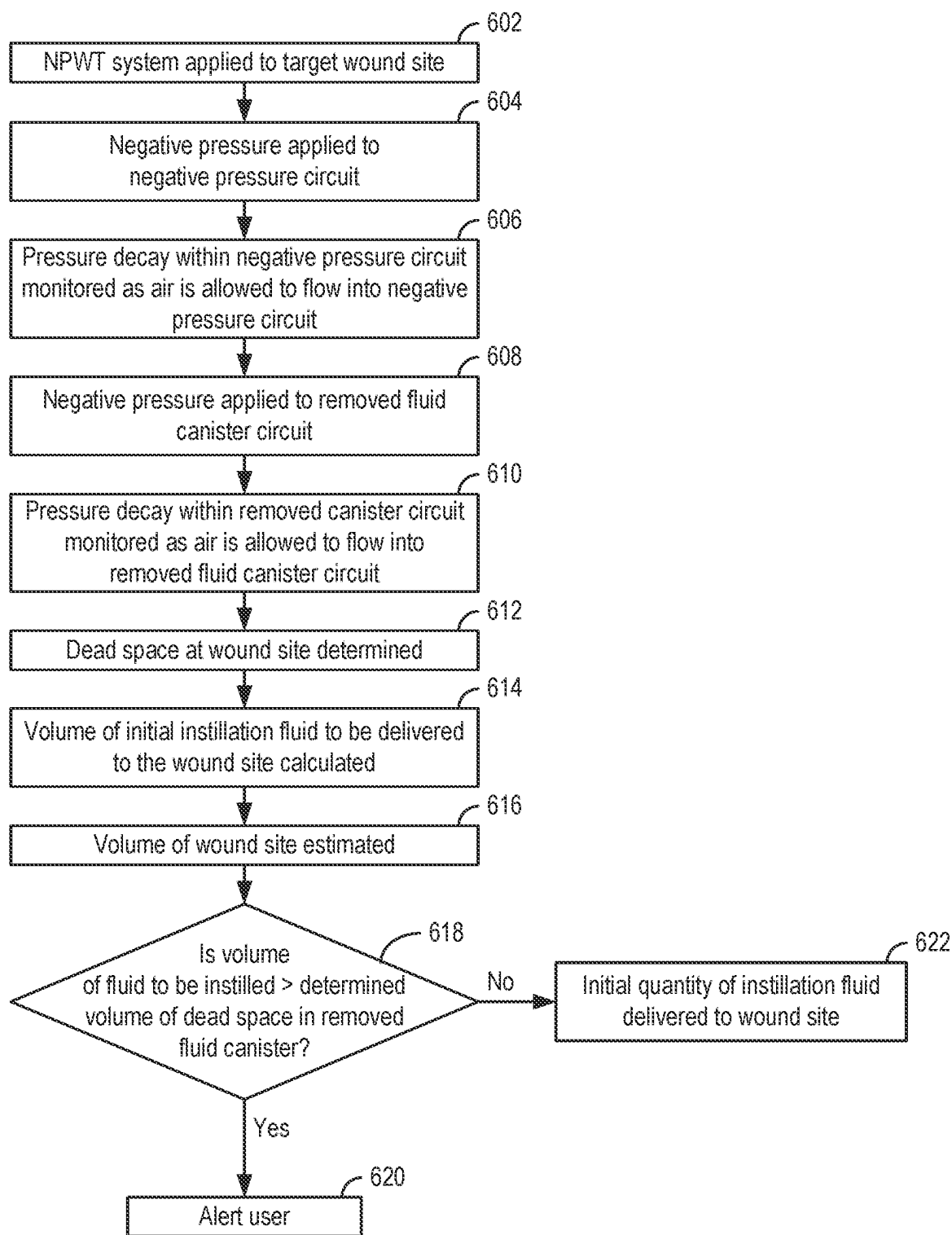


FIG. 6A

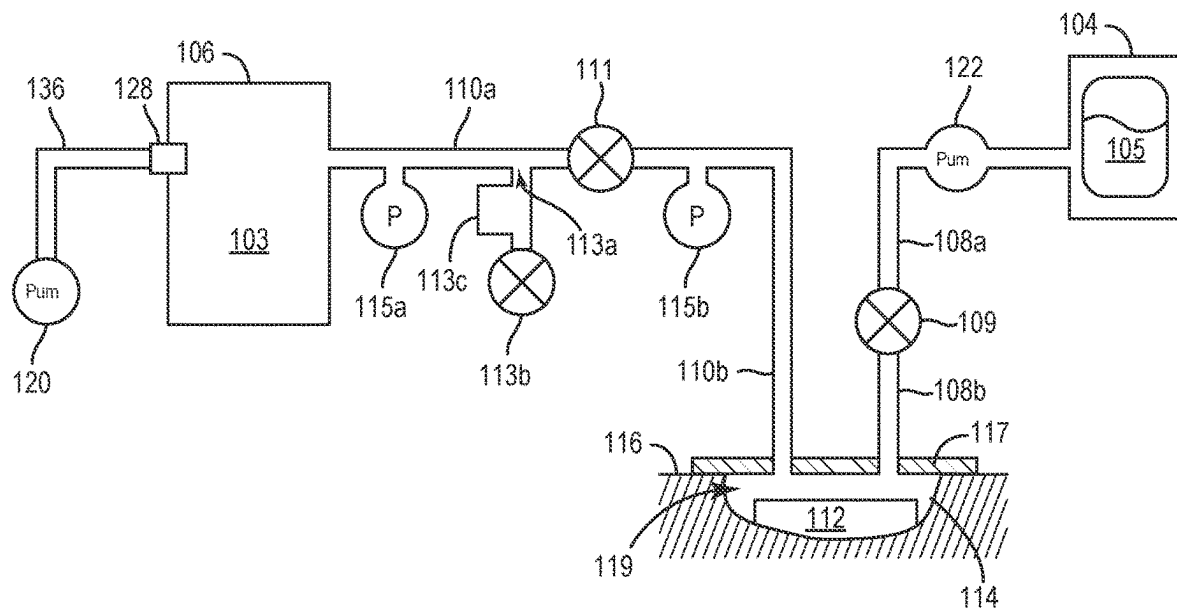


FIG. 6B

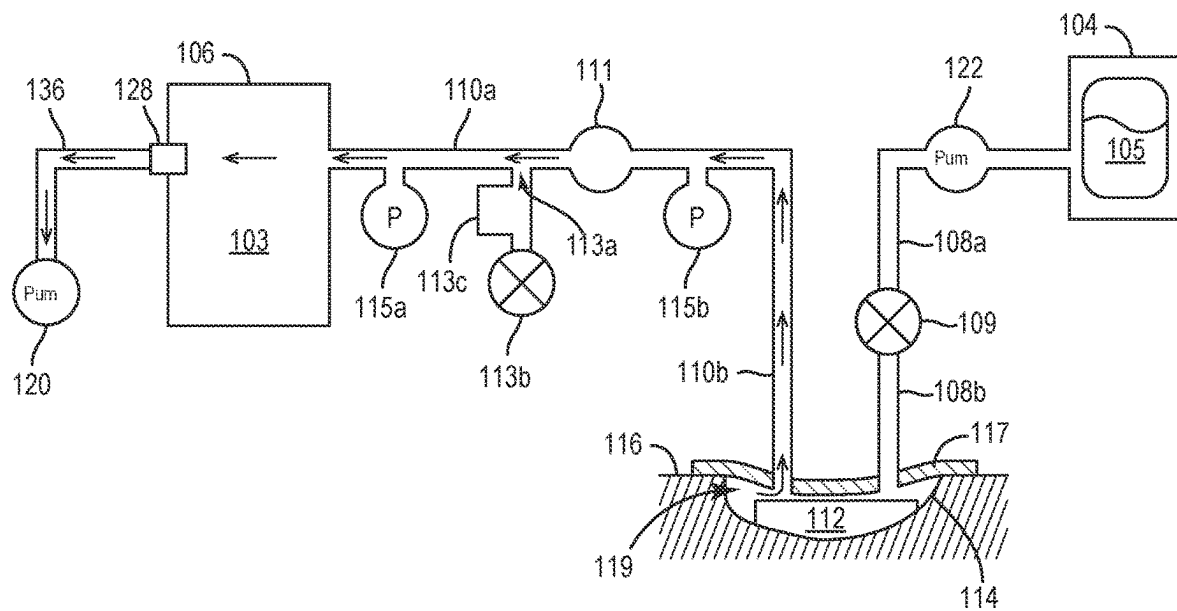


FIG. 6C

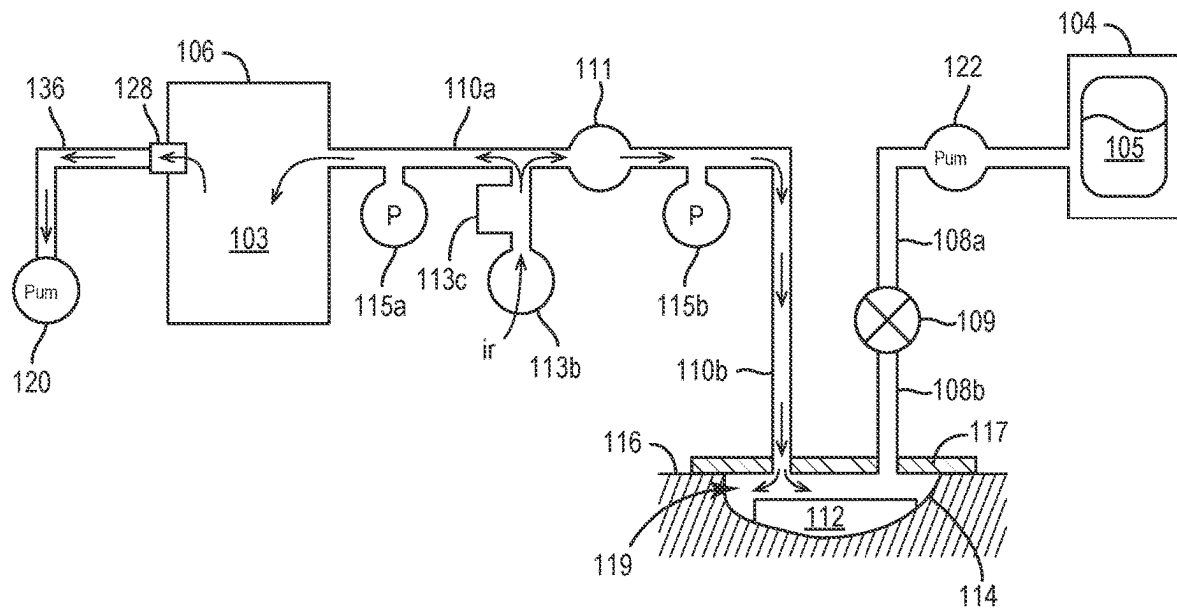


FIG. 6D

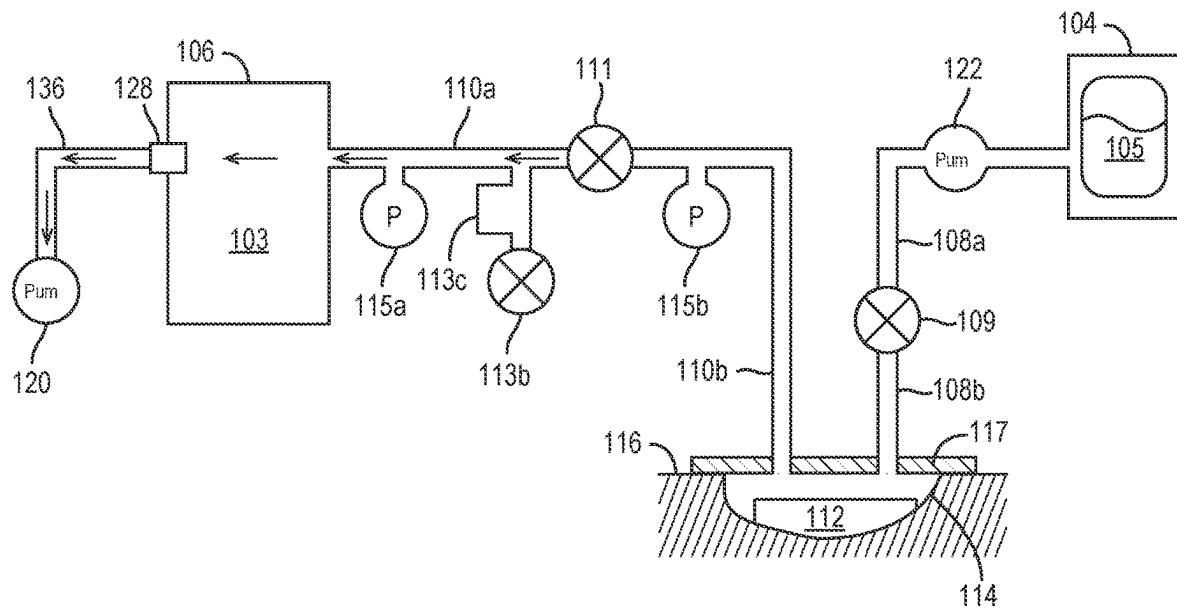


FIG. 6E

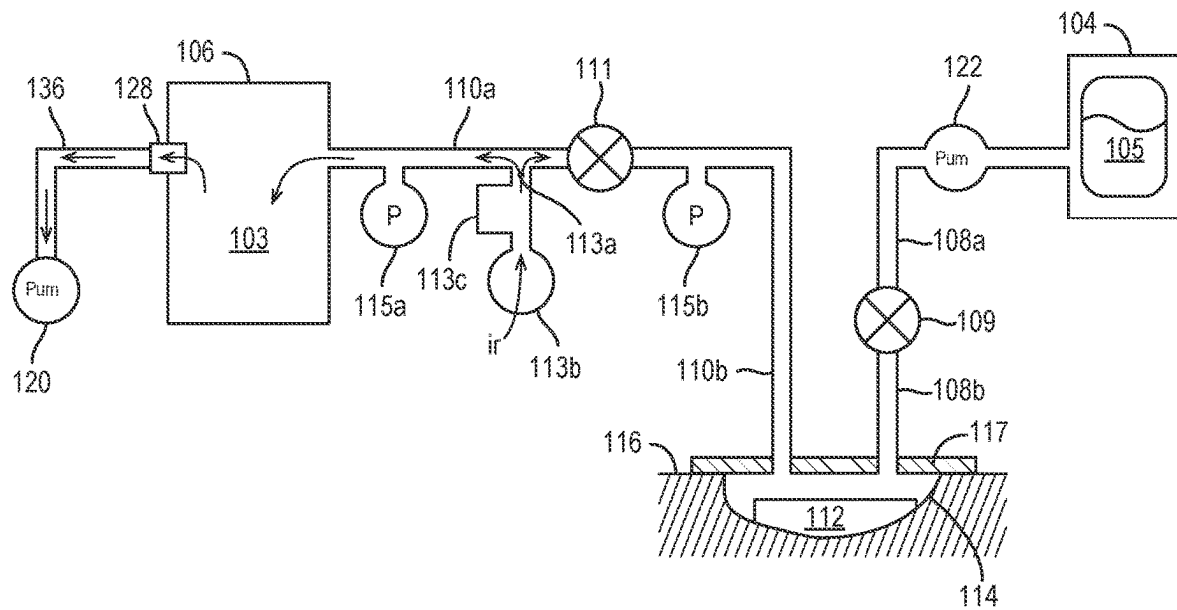


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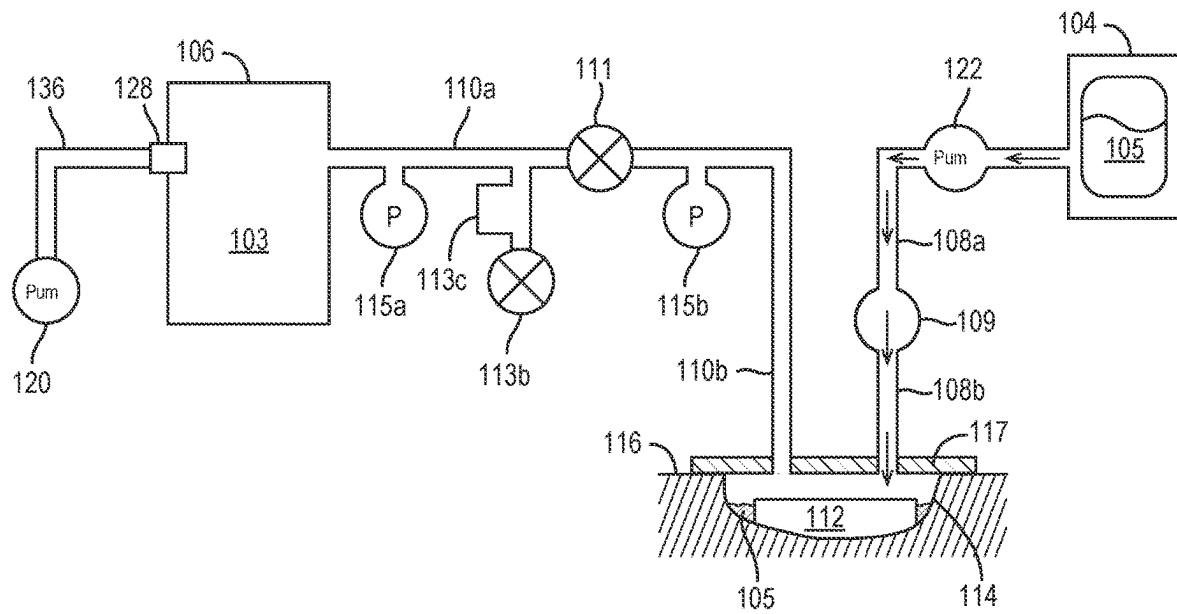


FIG. 6G

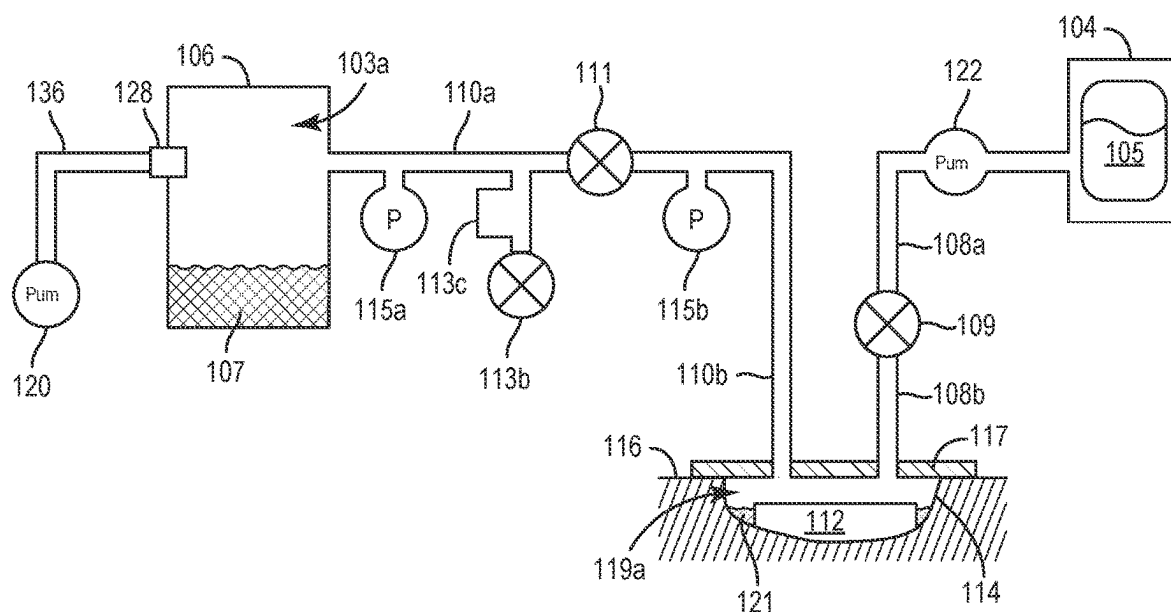


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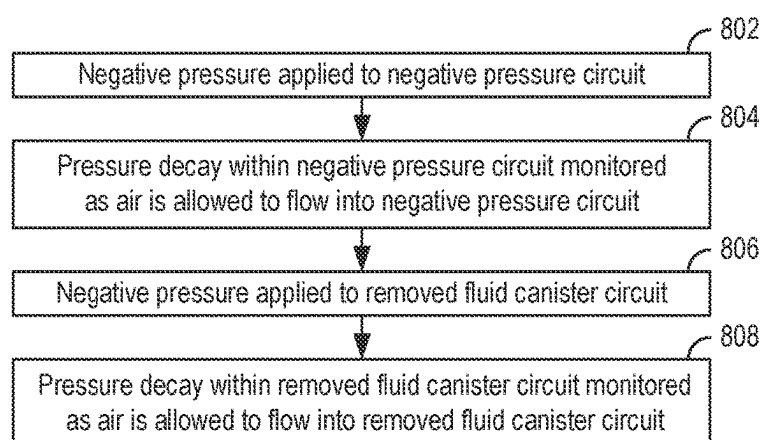


FIG. 8A

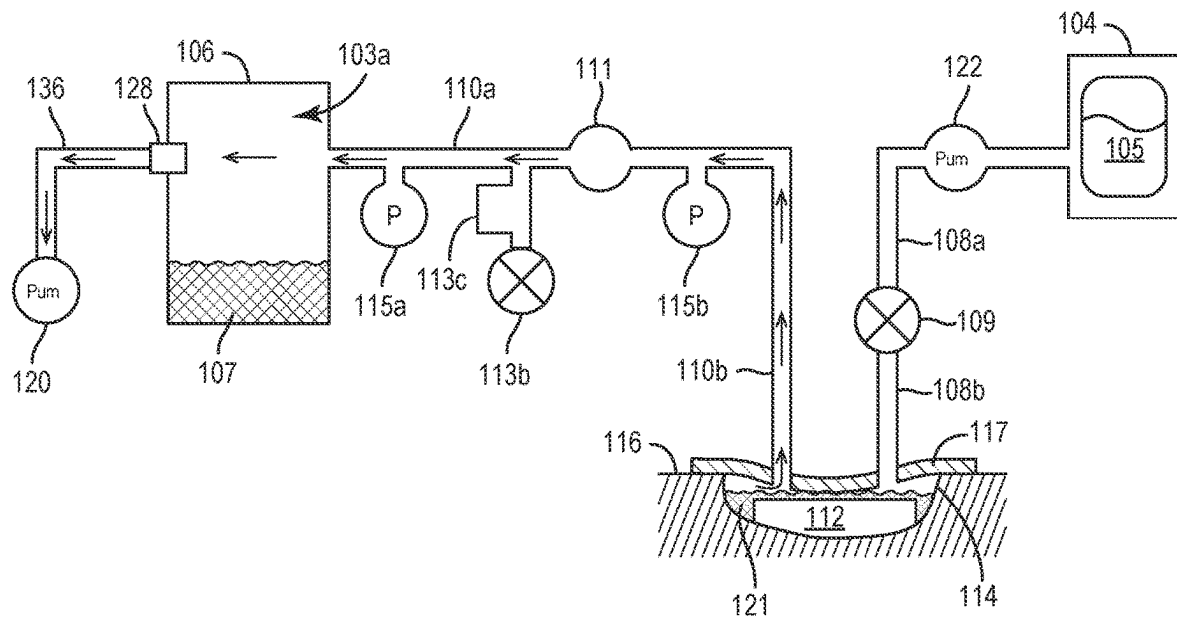


FIG. 8B

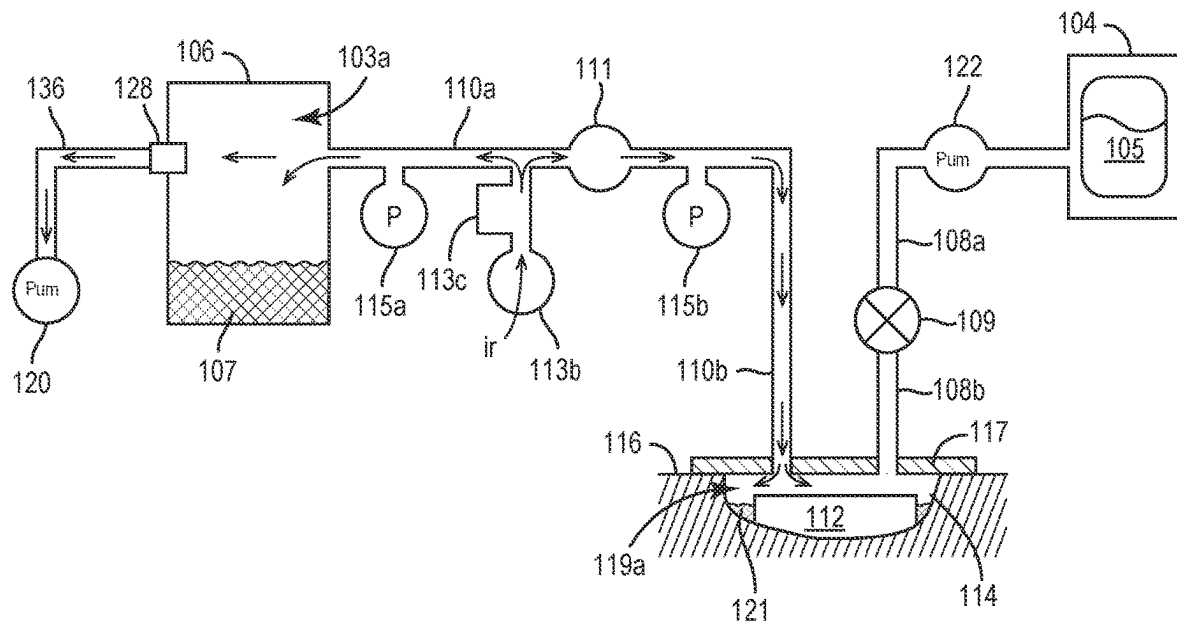


FIG. 8C

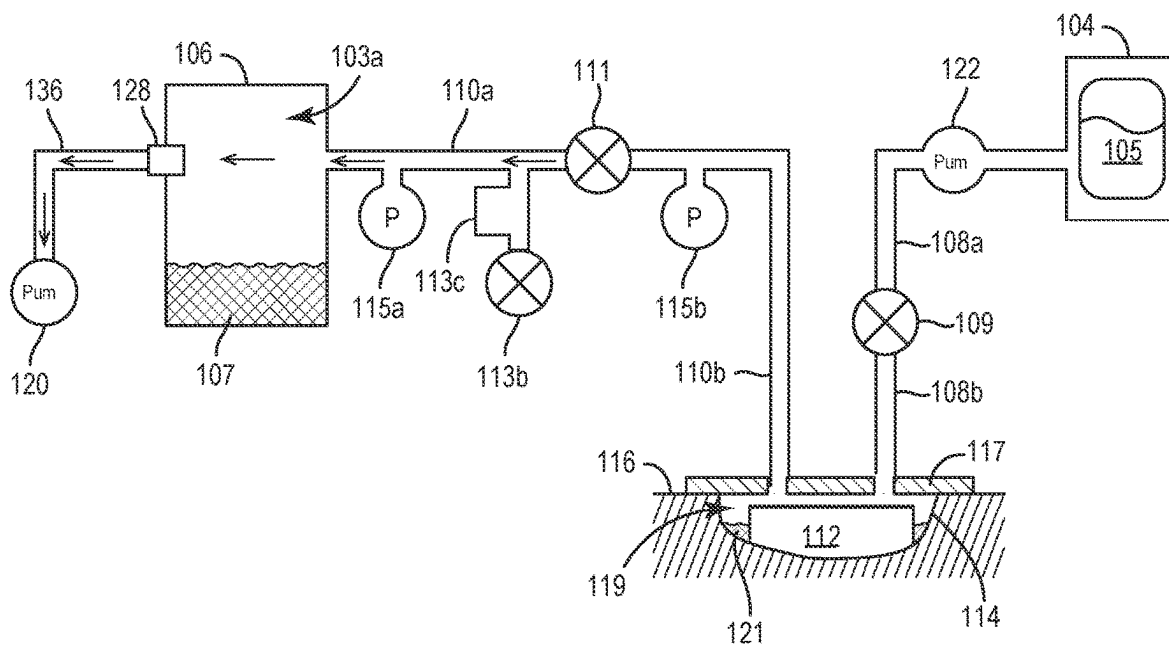


FIG. 8D

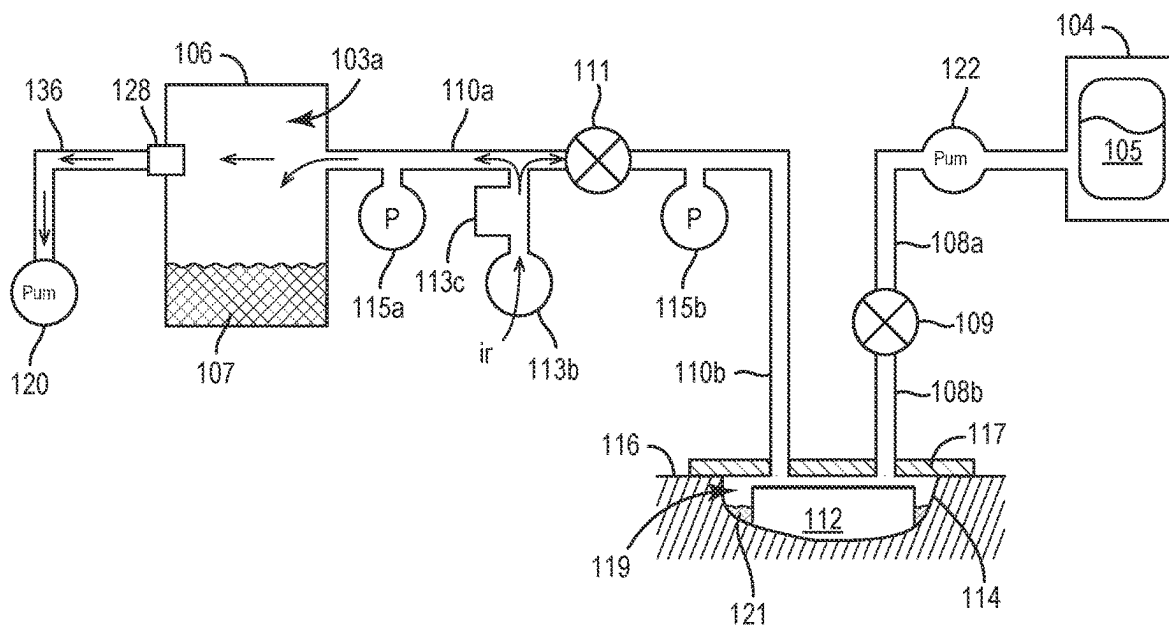


FIG. 8E

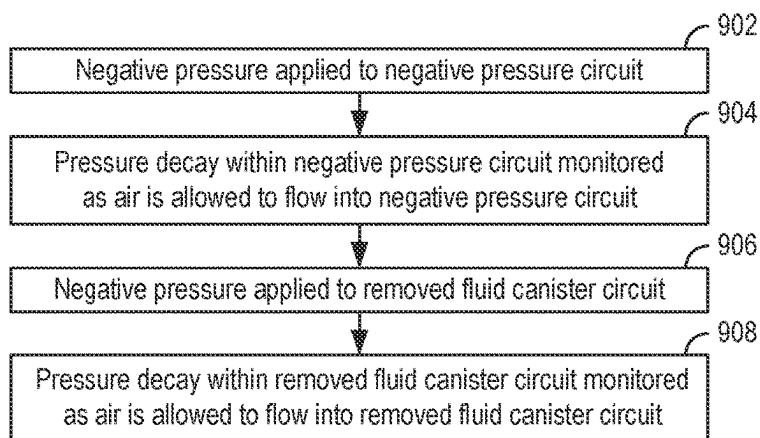


FIG. 9A

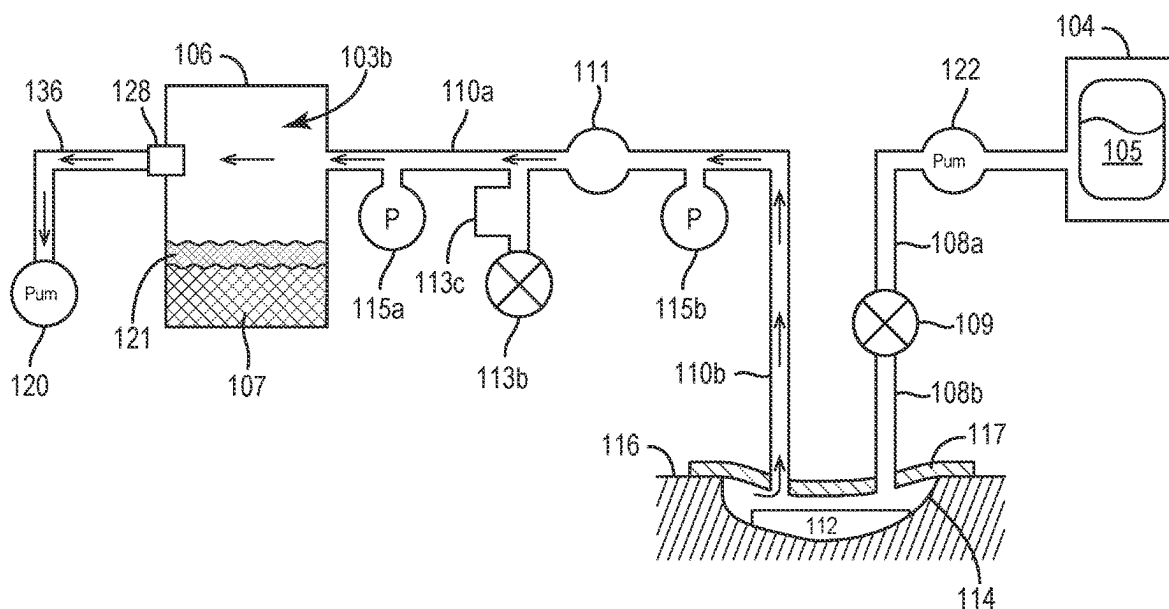


FIG. 9B

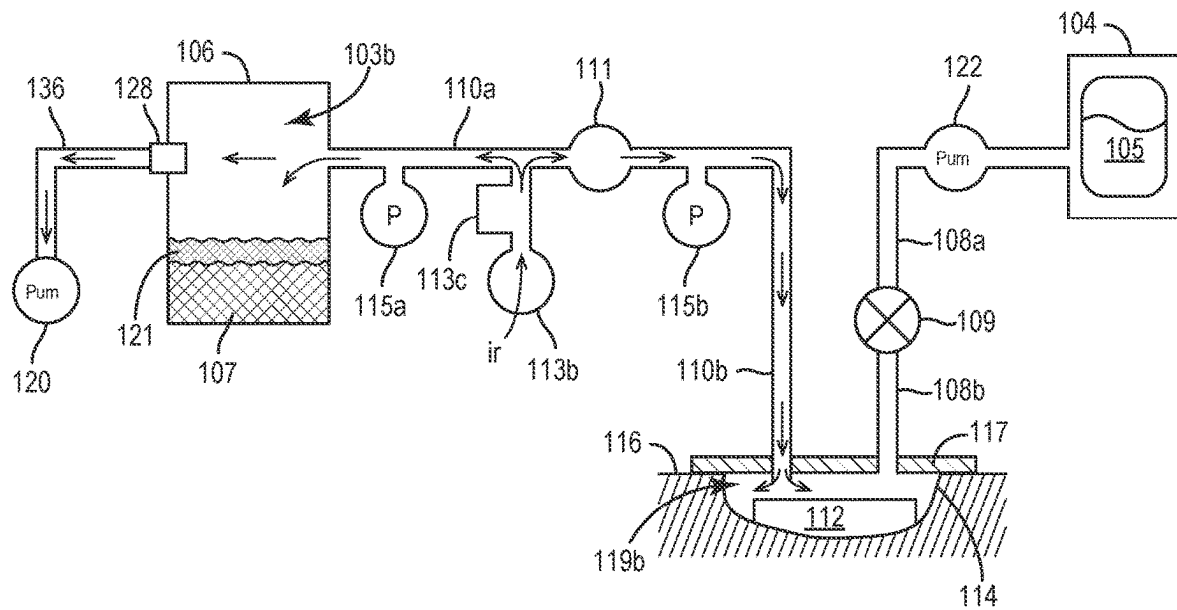


FIG. 9C

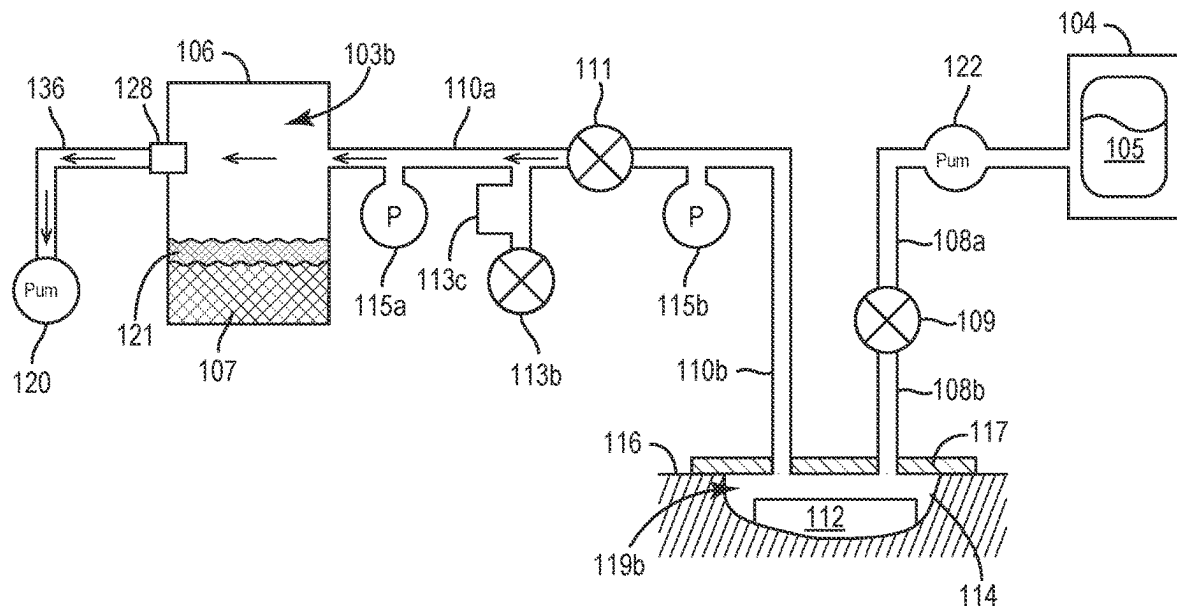


FIG. 9D

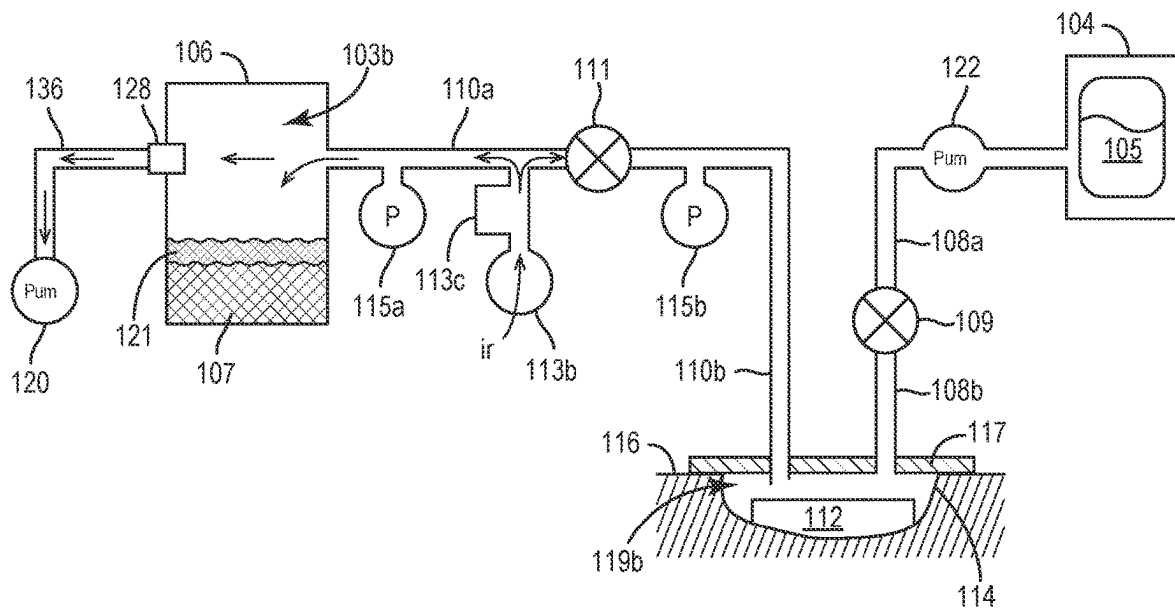


FIG. 9E

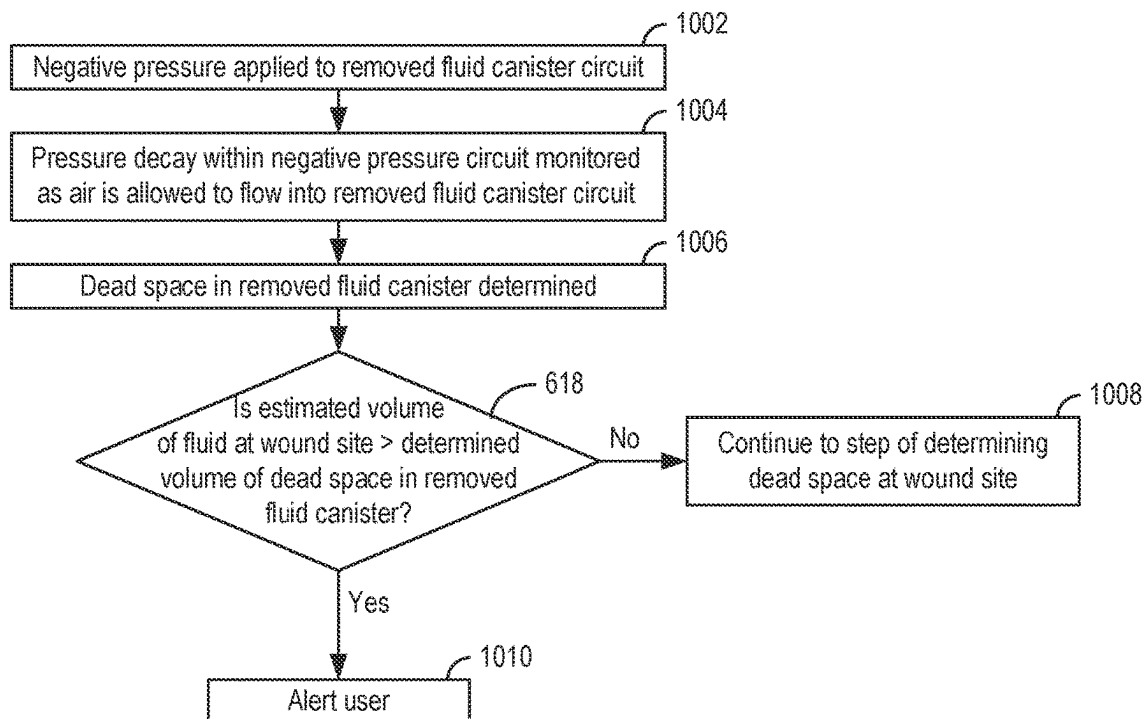


FIG. 10A

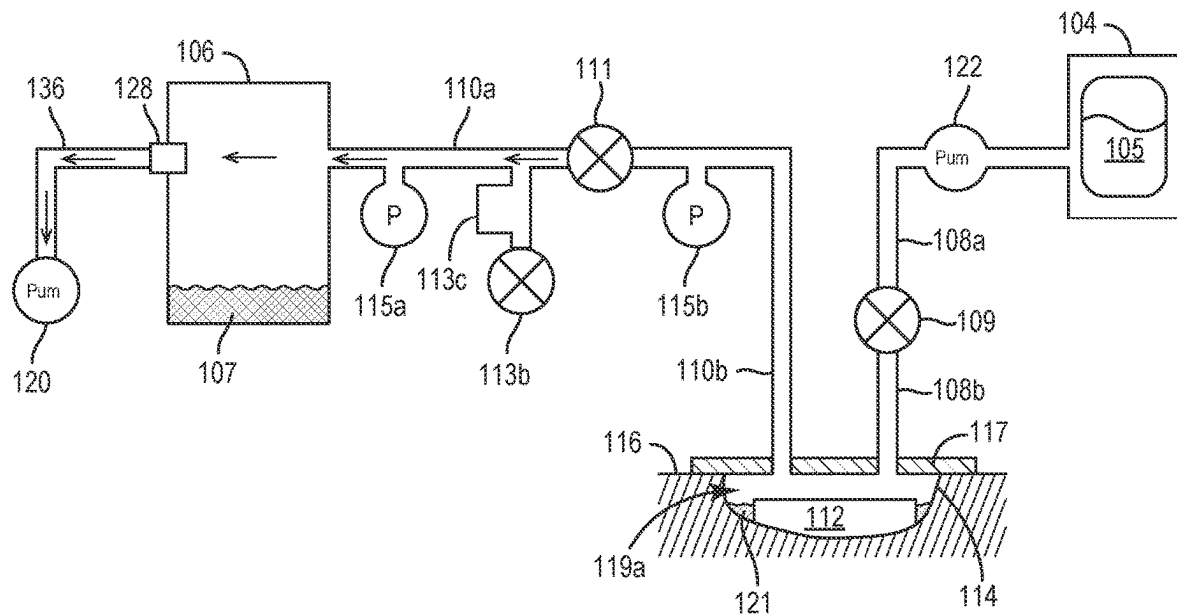


FIG. 10B

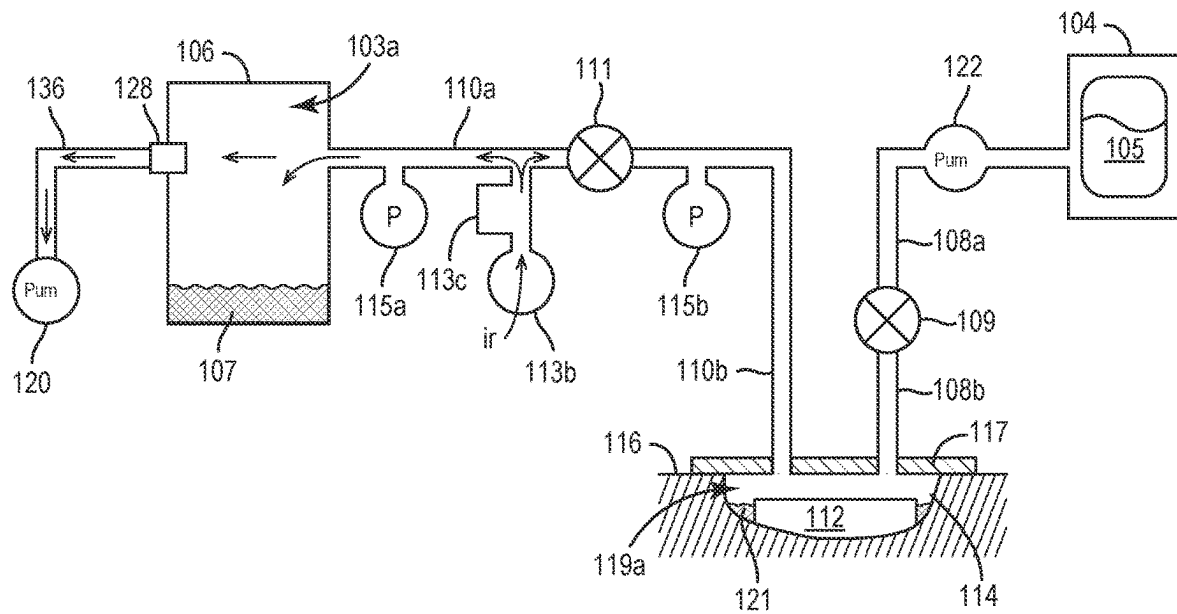


FIG. 10C

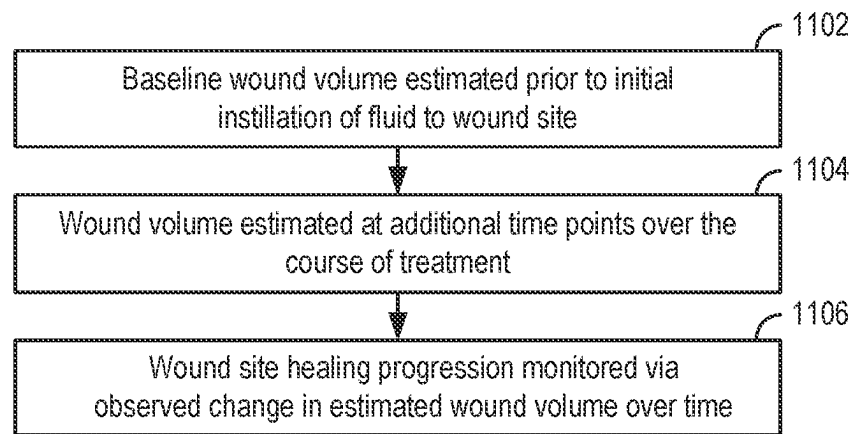


FIG. 11

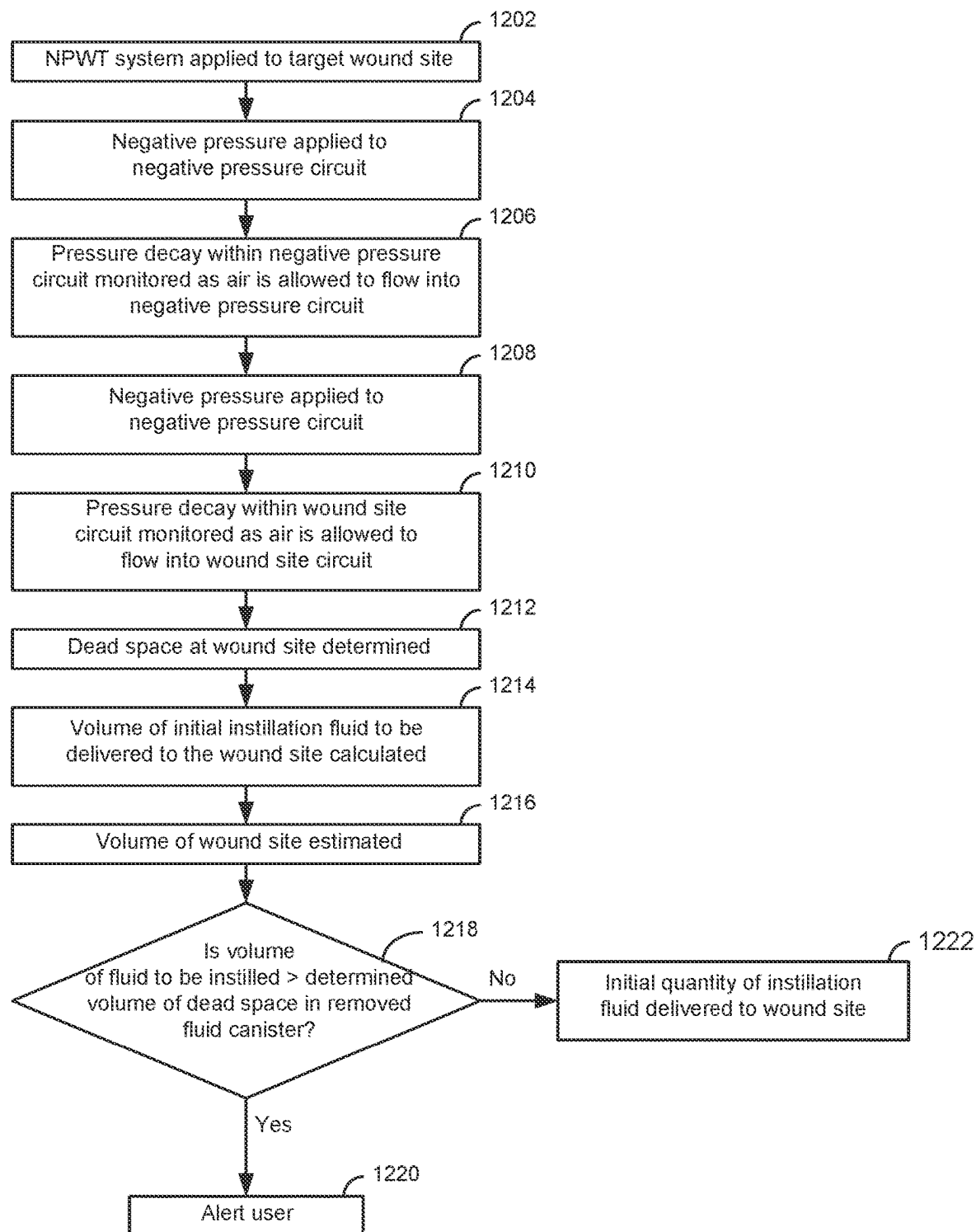
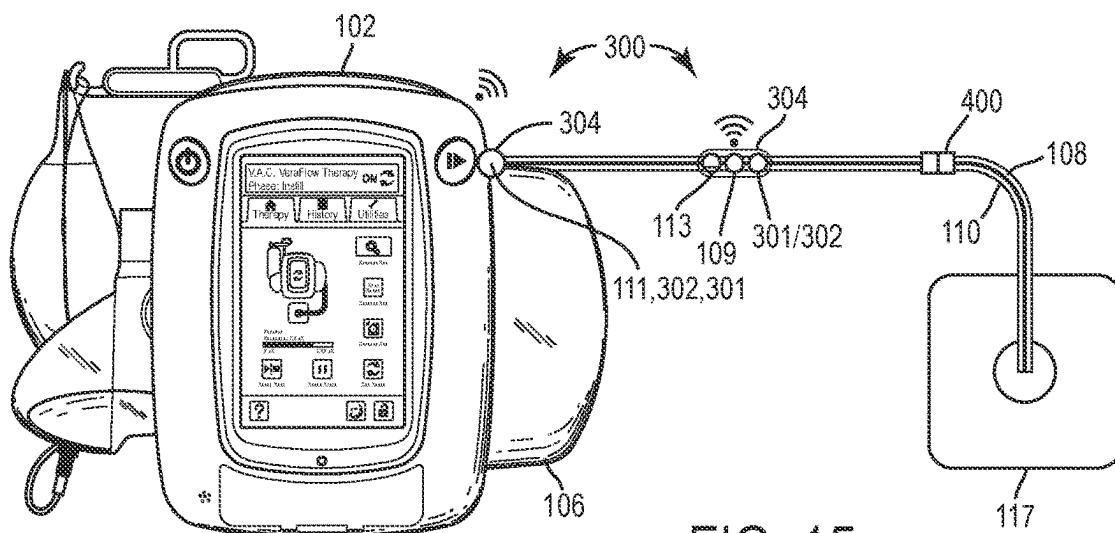
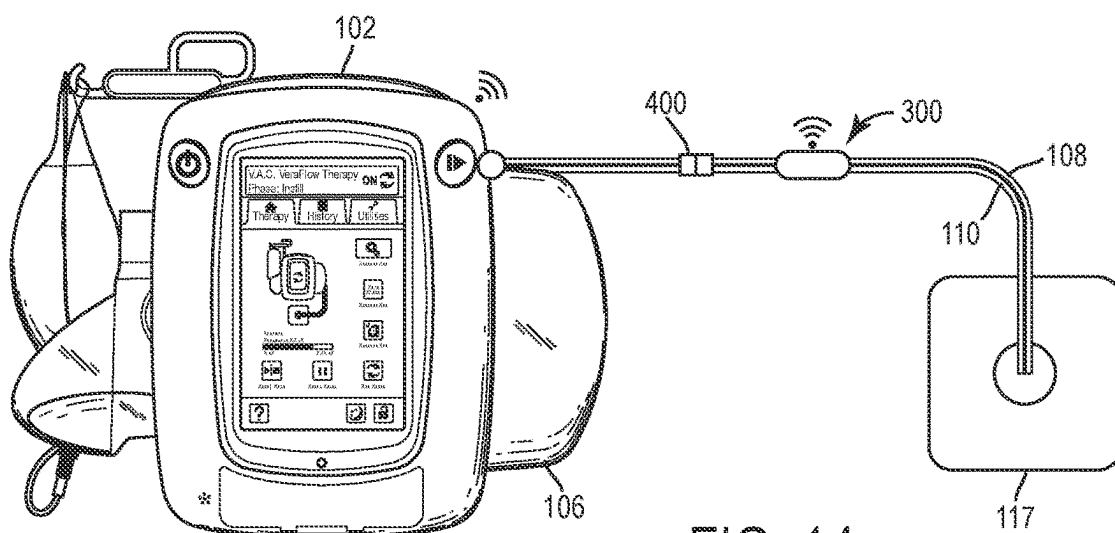
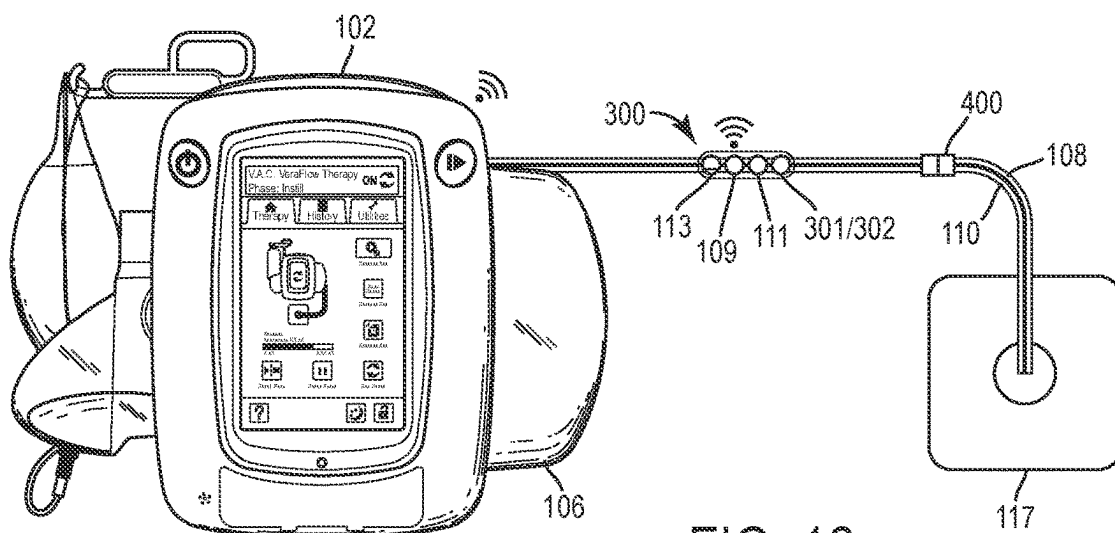


FIG. 12



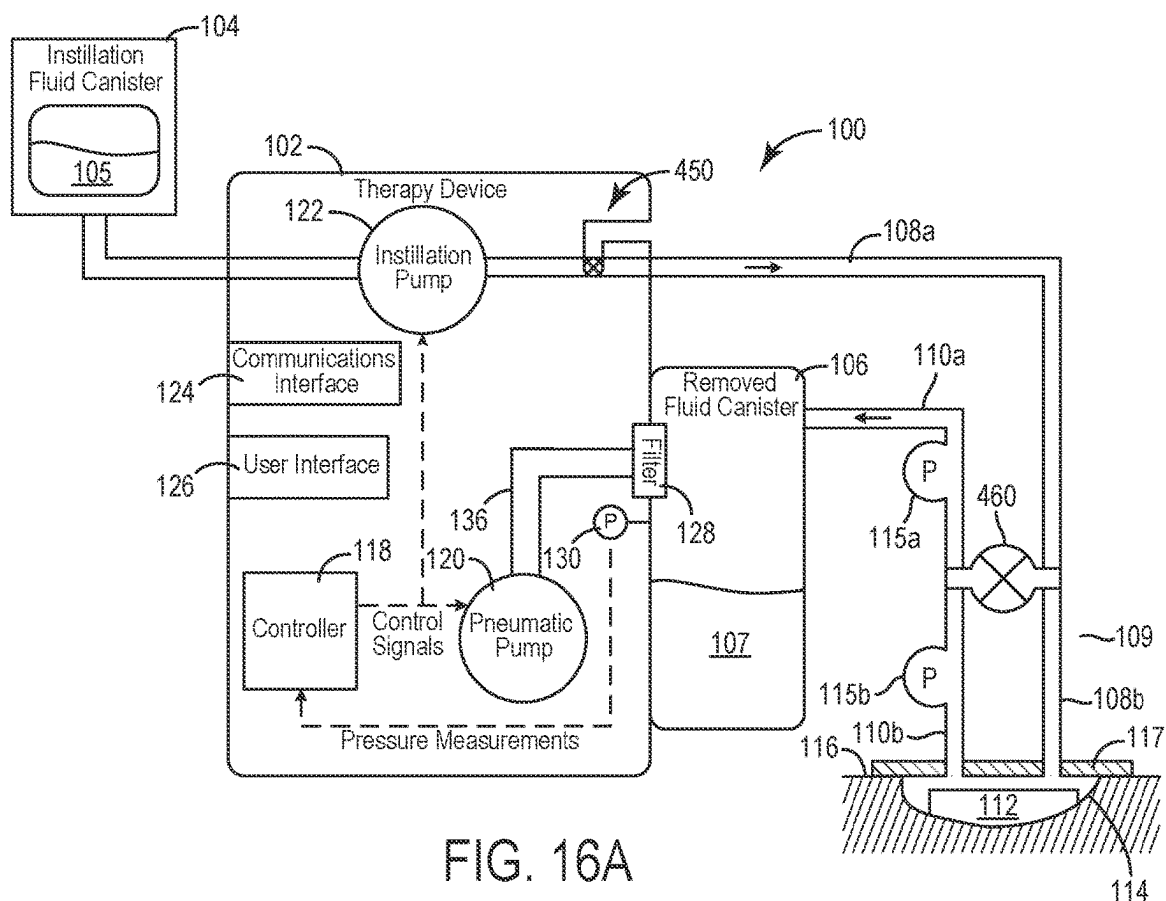


FIG. 16A

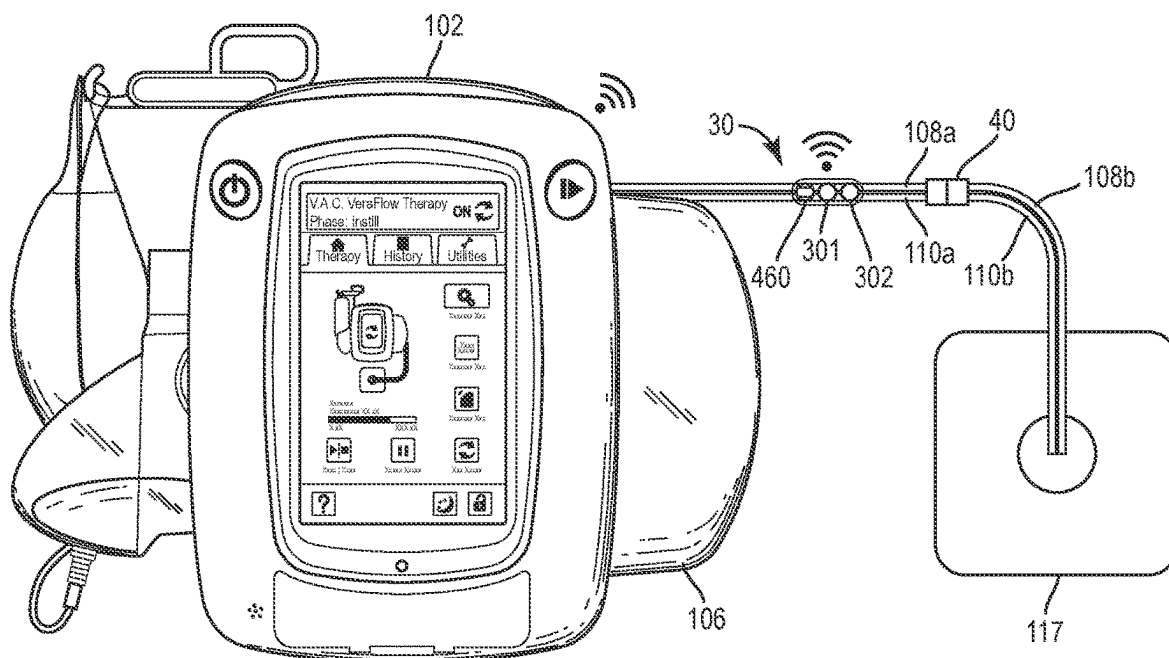


FIG. 16B

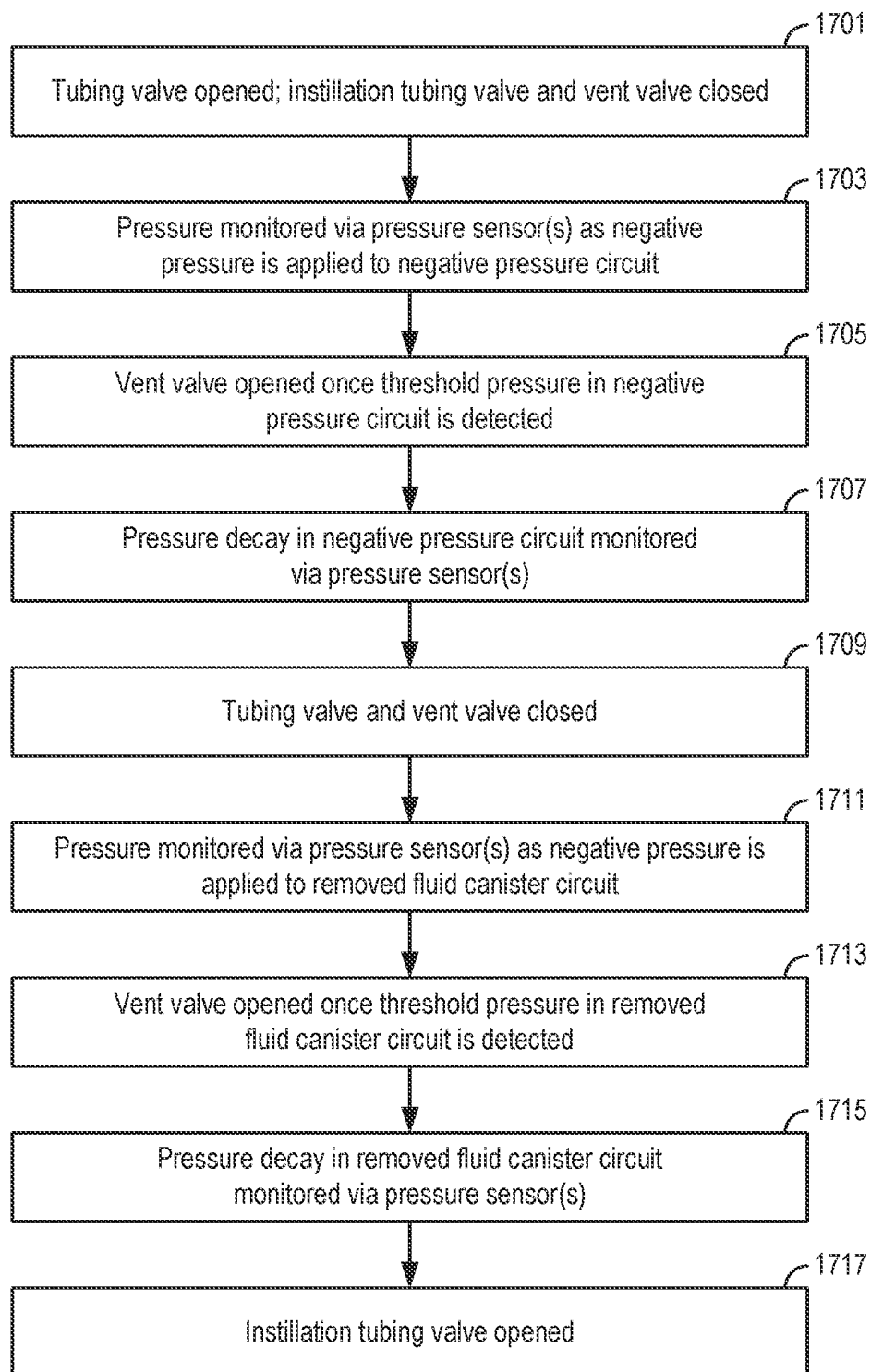
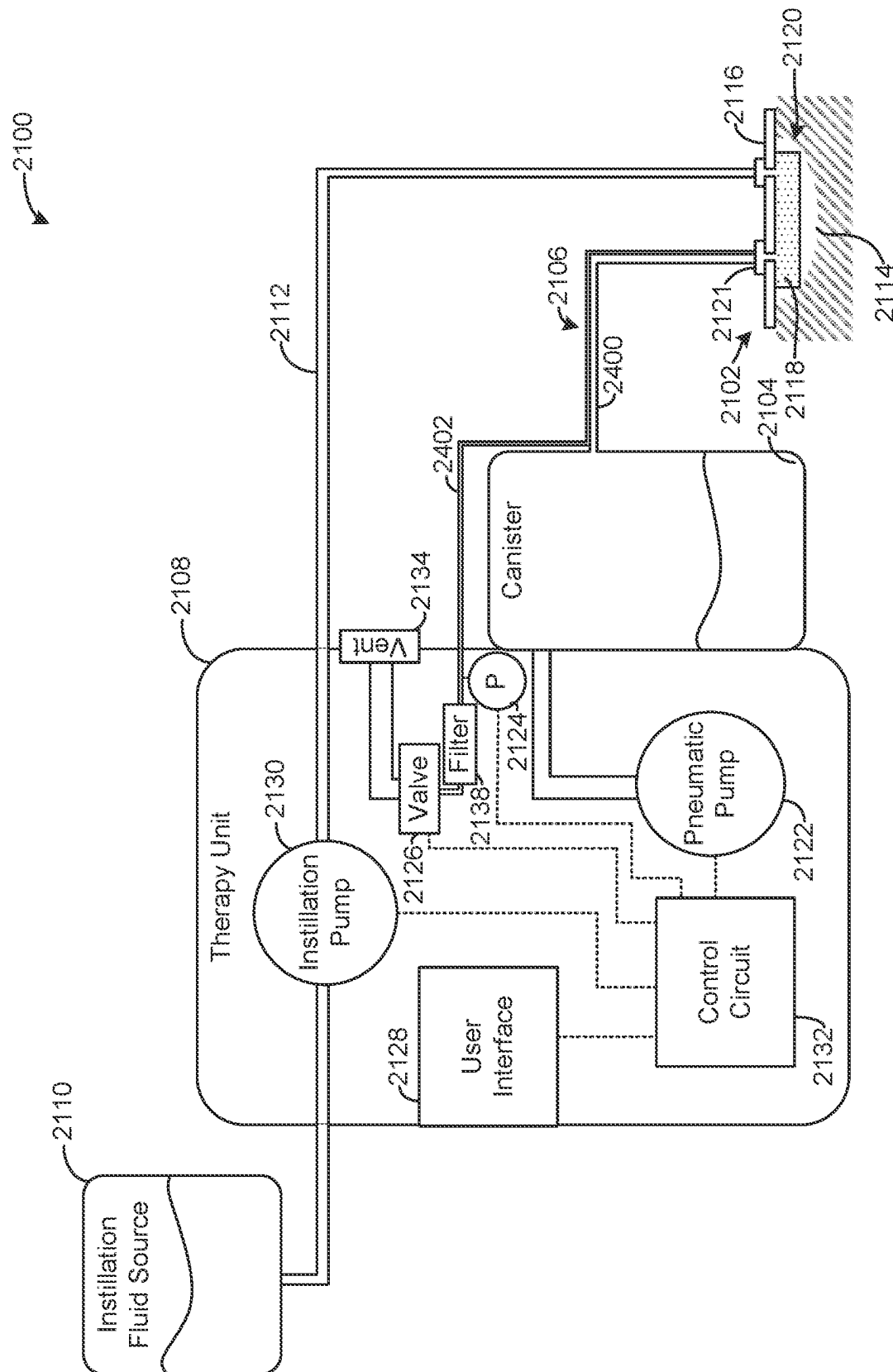


FIG. 17



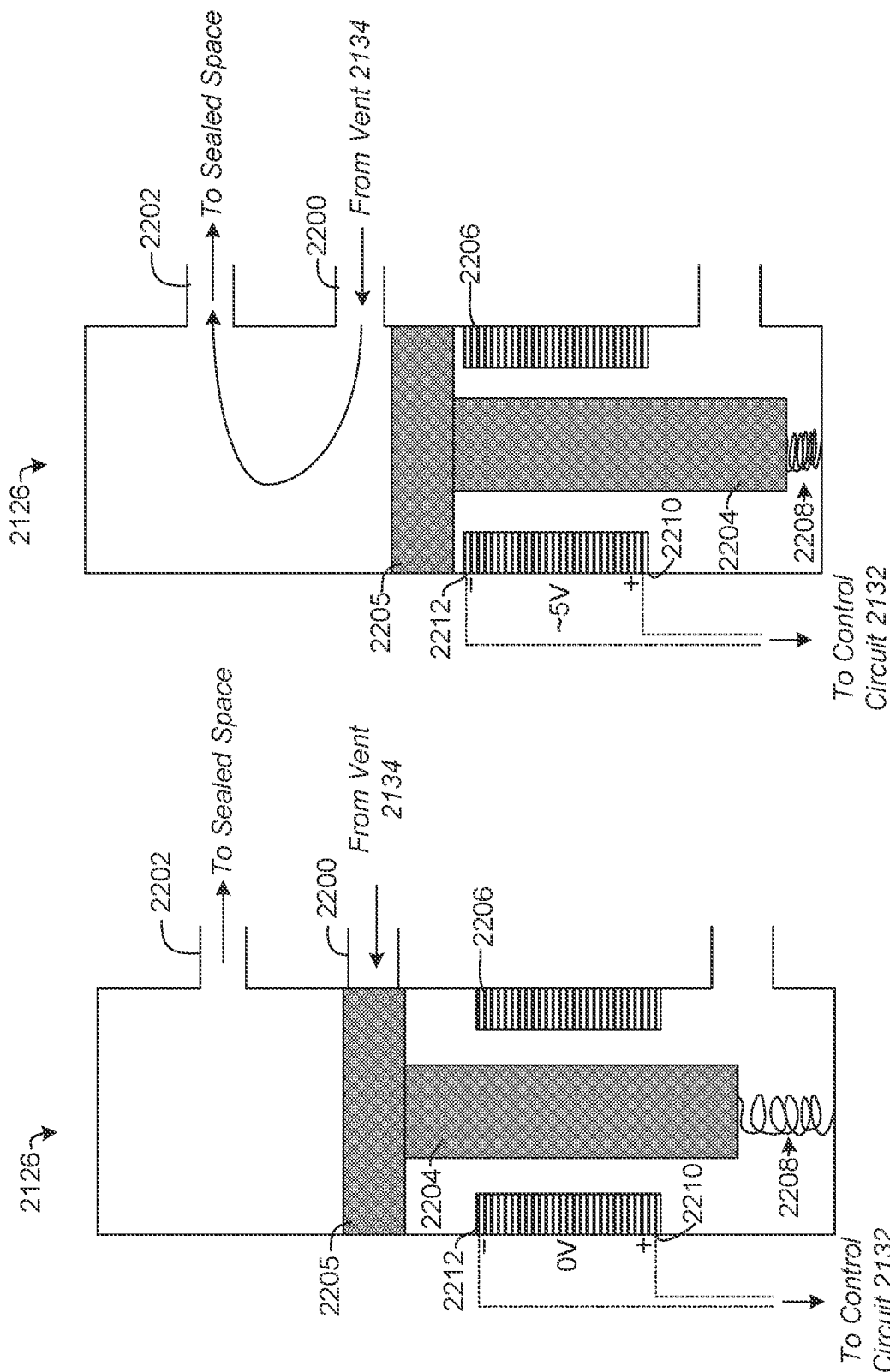


FIG. 20

FIG. 19

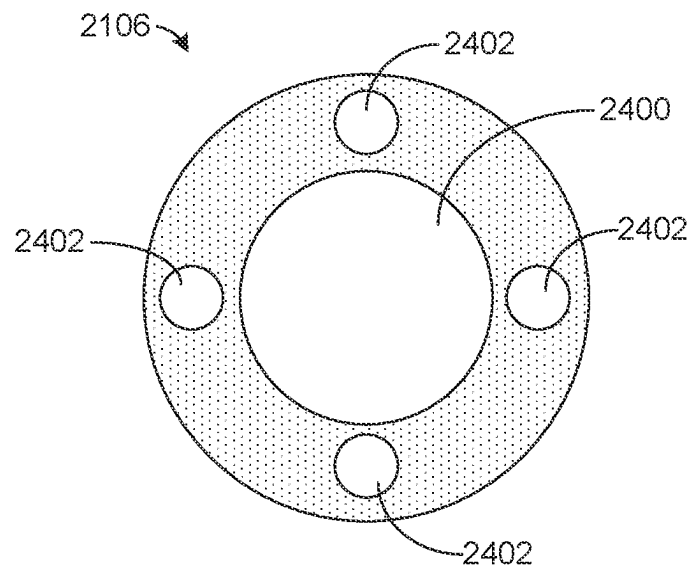


FIG. 21

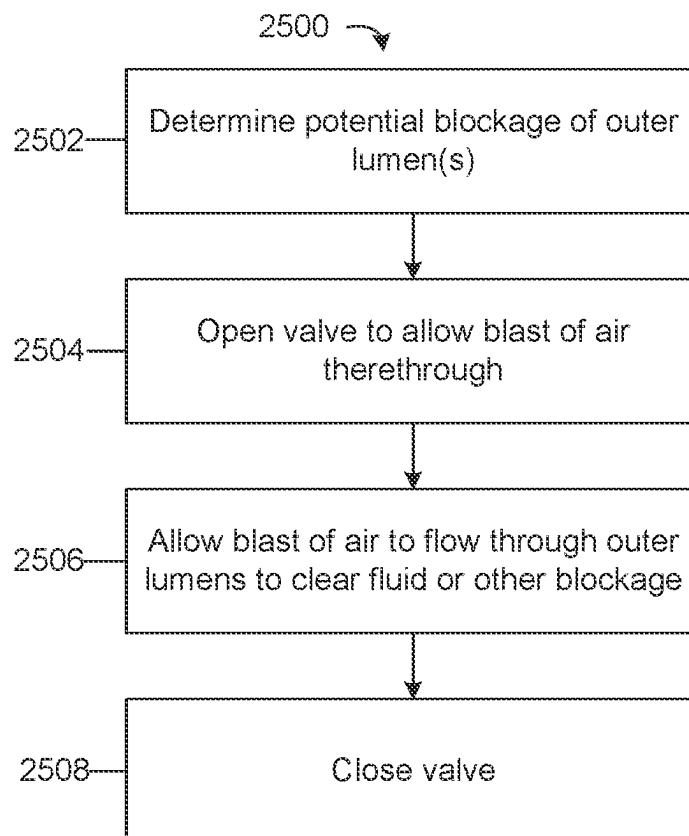


FIG. 22

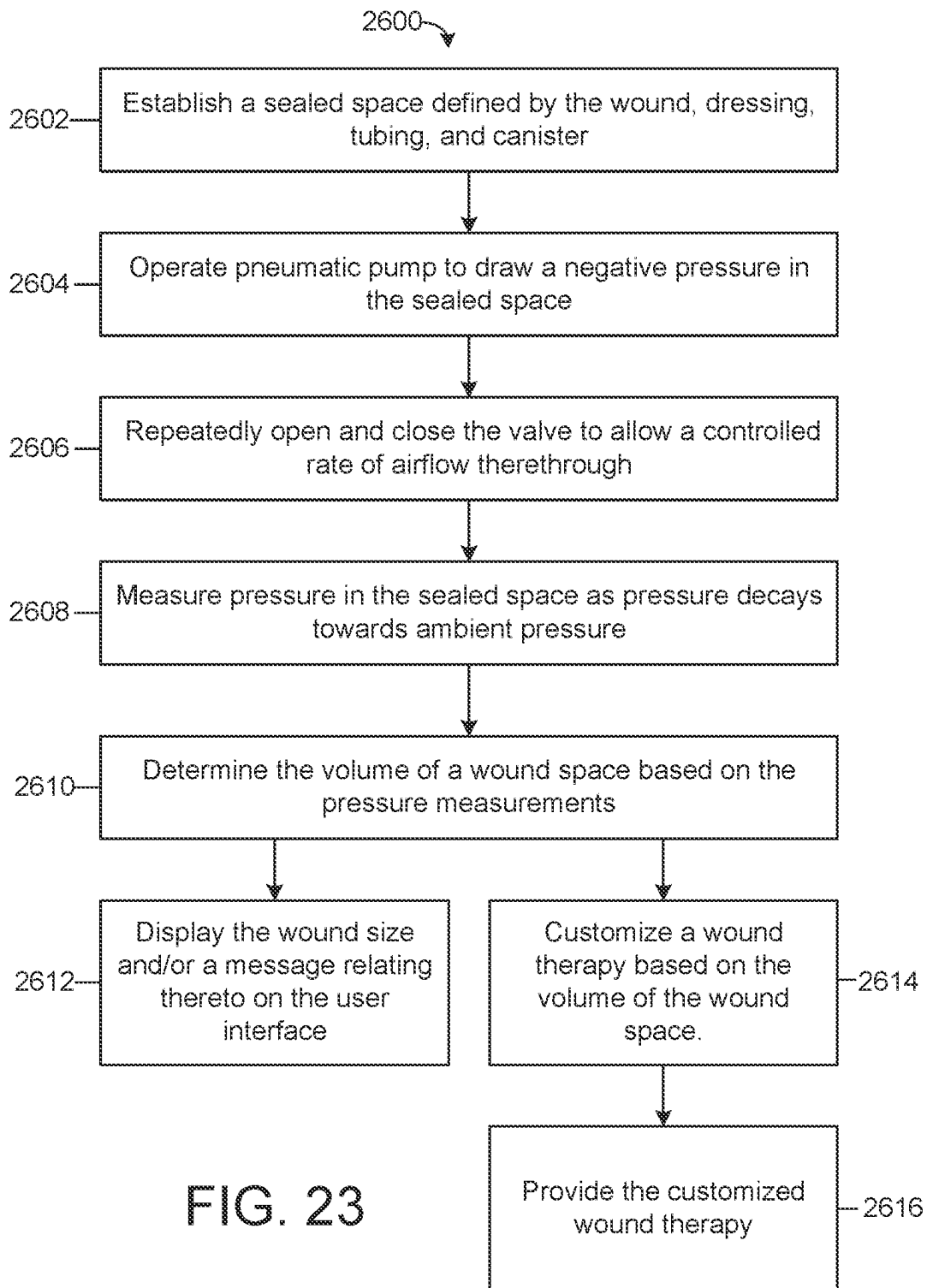


FIG. 23

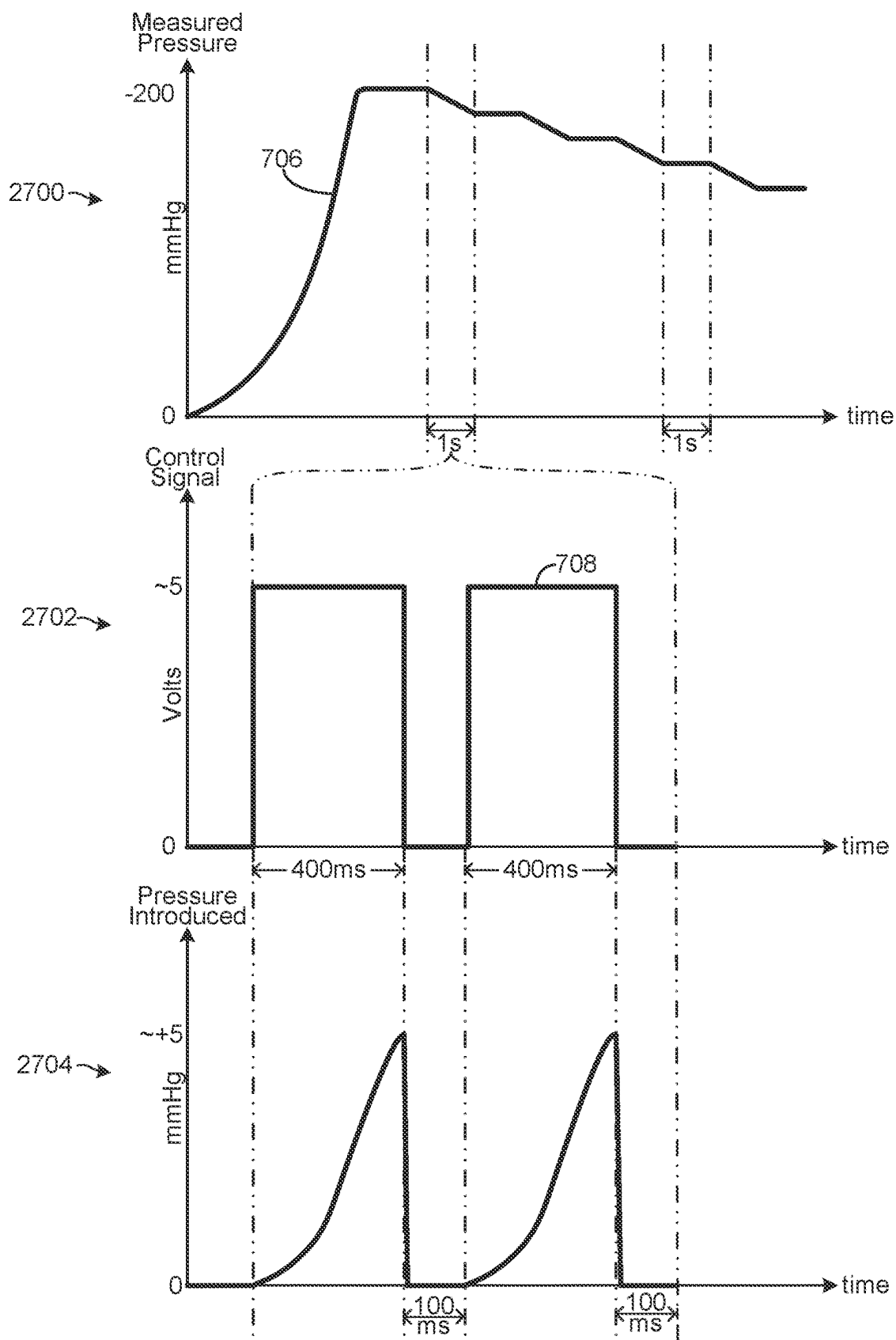


FIG. 24

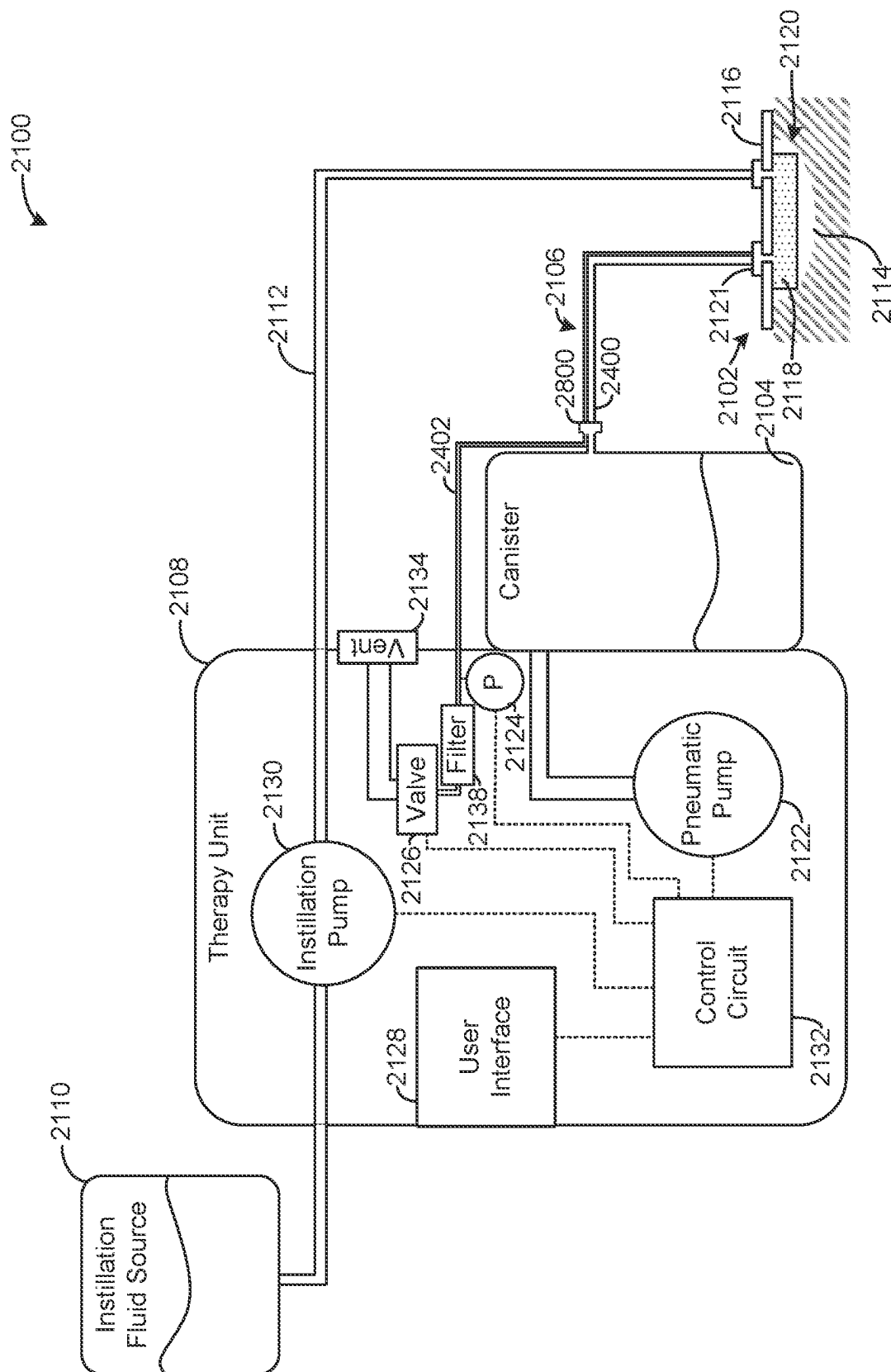


FIG. 25

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**WOUND THERAPY SYSTEM WITH
INTERNAL ALTERNATING ORIFICE****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a divisional of U.S. patent application Ser. No. 16/363,928, filed on Mar. 25, 2019, which claims the benefit of priority to U.S. Provisional Application No. 62/802,034, filed on Feb. 6, 2019, all of which are incorporated herein by reference in their entirety.

BACKGROUND

The present disclosure relates generally to a wound therapy system, and more particularly to a wound therapy system that provides negative pressure wound therapy (NPWT). NPWT refers to the creation of negative pressure (relative to atmospheric pressure) at a wound to promote healing of the wound. In a wound therapy system configured to provide NPWT, a dressing is typically sealed over a wound bed and placed in fluid communication with a pump operable to draw a negative pressure at the wound bed (i.e., in a wound space between the wound bed and the dressing). Because the dressing is sealed over the wound bed, often for a period of multiple days, it may be difficult to ascertain and monitor the progress of wound healing. One way to determine an amount of wound healing is based on a change in the amount of volume between the wound bed and the dressing (i.e., as the wound heals into the volume to occupy/consume part of the volume). Accordingly, systems and methods for volume determination in a wound therapy system may be advantageous.

In some cases, NPWT may be provided in coordination with instillation therapy and described as negative pressure and instillation wound therapy (NPIWT). Instillation therapy refers to the provision of instillation fluid (e.g., saline, antibiotic fluid) to the wound. One challenge in instillation therapy may be determining how much fluid to provide to the wound. It may be preferable to determine an amount of fluid to provide based on a size of the wound and/or a volume of available space adjacent the wound (i.e., between the dressing and the wound). Accordingly, systems and methods for volume determination in a wound therapy system may facilitate instillation therapy.

SUMMARY

One implementation of the present disclosure is a wound therapy system. The wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, tubing coupled to the dressing and fluidly communicable with the wound space, and a canister fluidly communicable with the tubing. The canister, the tubing, and the dressing define a sealed space that includes the wound space. A therapy unit is coupled to the canister. The therapy unit includes a pneumatic pump fluidly communicable with the sealed space, a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and controllable between an open position and a closed position, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve to repeatedly alternate between the open position and the closed position to allow a controlled rate of airflow through the valve, receive

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measurements of the pressure in the sealed space from the sensor, and determine a volume of the wound space based on the measurements of the pressure.

In some embodiments, the controlled rate of airflow is less than a restriction rate of a filter positioned between the valve and the canister.

In some embodiments, valve includes a solenoid valve. The control circuit is configured to control the valve to repeatedly alternate between the open position and the closed position by providing a voltage pattern to the solenoid valve. The voltage pattern includes a step function repeatedly stepping between approximately zero voltage and a non-zero voltage. The voltage pattern may remain at the non-zero voltage for no more than a maximum continuous duration of approximately 500 milliseconds.

In some embodiments, the voltage pattern includes a repeating pattern of approximately 400 milliseconds at a non-zero voltage, approximately 100 milliseconds at approximately zero voltage, approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage. The voltage pattern includes a first set of two periods of the repeating pattern, approximately one second at approximately zero voltage, and a second set of two periods of the repeating pattern. The voltage pattern may cause the solenoid valve to alternate between the open position and the closed position with a period of approximately 500 milliseconds.

In some embodiments, the control circuit is further configured to customize a customized wound therapy based on the volume of the wound space and control the therapy unit to provide the customized wound therapy. The customized wound therapy may include instillation therapy.

In some embodiments, the control circuit is configured to customize the instillation therapy by determining an amount of instillation fluid to supply to the wound space based on the volume of the wound space. The wound therapy system may include instillation tubing coupled to the dressing and fluidly communicable with the wound space, a source of the instillation fluid fluidly communicable with the instillation tubing, and an instillation pump controllable by the control circuit to provide the amount of the instillation fluid from the source to the wound space.

Another implementation of the present disclosure is a method of treating a wound. The method includes establishing a sealed space defined by a dressing, tubing, and a canister of a wound therapy system. The sealed space includes a wound space defined by the dressing and the wound. The method includes removing, with a pneumatic pump, air from the sealed space to establish a negative pressure in the sealed space and causing a solenoid valve to alternate between an open position and a closed position. The solenoid valve allows an airflow from a surrounding environment to the sealed space in the open position and prevents the airflow from the surrounding environment to the sealed space in the closed position. The method also includes measuring the pressure in the sealed space to generate pressure measurements, determining, based on the pressure measurements, a volume of the wound space, customizing a customized wound therapy based on the volume of the wound space, and providing the customized wound therapy to the wound.

In some embodiments, customizing a customized wound therapy includes determining an amount of an instillation fluid to be supplied to the wound space based on the volume of the wound space. Providing the customized wound

therapy to the wound includes controlling an instillation pump to supply the amount of the instillation fluid to the wound space.

In some embodiments, causing the solenoid valve to alternate between the open position and the closed position provides a controlled rate of airflow from the surrounding environment to the sealed space. The controlled rate of airflow is less than a restriction rate of a filter positioned between the canister and the solenoid valve.

In some embodiments, causing the solenoid valve to alternate between the open position and the closed position includes providing a voltage pattern to the solenoid valve. The voltage pattern may include a step function repeatedly stepping between approximately zero voltage and a non-zero voltage. The voltage pattern may include a repeating pattern of approximately 400 milliseconds at a non-zero voltage, approximately 100 milliseconds at approximately zero voltage, approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage.

In some embodiments, the voltage pattern includes a first set of two periods of the repeating pattern, approximately one second at approximately zero voltage, and a second set of two periods of the repeating pattern. The non-zero voltage causes the solenoid valve to be in the open position. A positive pressure of approximately 5 mmHg is provided to the sealed space during each 400 milliseconds at the non-zero voltage.

Another implementation of the present disclosure is a wound therapy system. The wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, first tubing coupled to the dressing and fluidly communicable with the wound space, a canister fluidly communicable with the first tubing, wherein the canister, the first tubing, and the dressing define a sealed space that includes the wound space, a pneumatic pump fluidly communicable with the sealed space, a sensor configured to measure a pressure in the sealed space, and a solenoid valve controllable between an open position and a closed position. The solenoid valve is configured to allow air to flow from a surrounding environment to the sealed space in the open position and prevent air from flowing from the surrounding environment to the sealed space in the closed position. The wound therapy system also includes instillation tubing coupled to the dressing and fluidly communicable with the wound space and a source of instillation fluid, an instillation pump coupled to the instillation tubing and controllable to supply an amount of the instillation fluid to the wound space, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space and provide a voltage pattern to the solenoid valve. The voltage pattern causes the solenoid valve to repeatedly alternate between the open position and the closed position. The control circuit is also configured to receive measurements of the pressure from the sensor, determine a volume of the wound space based on the measurements of the pressure, determine the amount of the instillation fluid based on the volume of the wound space, and control the instillation pump to supply the amount of the instillation fluid to the wound space.

In some embodiments, causing the solenoid valve to alternate between the open position and the closed position allows a controlled rate of airflow through the solenoid valve from the surrounding environment to the sealed space.

In some embodiments, the solenoid valve is positioned to allow the air to enter one or more outer lumens of the first

tubing. In some embodiments, a filter is positioned between the solenoid valve and the one or more outer lumens. Causing the solenoid valve to alternate between the open position and the closed position allows a controlled rate of airflow through the solenoid valve from the surrounding environment to the channel, and the controlled rate is less than a restriction rate of the filter.

In some embodiments, the instillation pump, the pneumatic pump, and the control circuit are housed within a therapy unit. In some embodiments, the solenoid valve is positioned within the therapy unit. In some embodiments, the solenoid valve is positioned outside the therapy unit and coupled to the first tubing.

Another implementation of the present disclosure is a therapy unit. The therapy unit includes a pneumatic pump fluidly communicable with a sealed space, a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and controllable between an open position and a closed position, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve to repeatedly alternate between the open position and the closed position to allow a controlled rate of airflow through the valve, receive measurements of the pressure in the sealed space from the sensor, and determine a volume of the sealed space based on the measurements of the pressure and the controlled rate, and provide a customized wound therapy based on the volume of the sealed space.

In some embodiments, the control circuit is configured to allow the controlled rate of airflow through the valve by controlling the valve to the open position for no longer than a maximum continuous duration of approximately 500 milliseconds.

Another implementation of the present disclosure is a wound therapy system. The wound therapy system includes a pneumatic pump fluidly communicable with a canister and tubing comprising a first lumen and a second lumen. The first lumen is configured to facilitate the flow of fluid from a dressing to the canister and the second lumen configured to facilitate measurement of a pressure at the dressing. The wound therapy system also includes a sensor configured to measure a pressure in the second lumen, a valve positioned between a second lumen and a surrounding environment and controllable between an open position and a closed position, a filter positioned between the valve and the second lumen, and a cap removeably coupleable to the tubing. The cap provides fluid communication between the first lumen and the second lumen when the cap is coupled to the tubing. The wound therapy system also includes a control circuit configured to, while the cap is coupled to the tubing, operate the pump to remove air from the canister, control the valve to the open position, receive measurements of the pressure in the second lumen from the sensor, and determine, based on the measurements of the pressure in the second lumen, a flow rate through the filter.

In some embodiments, the control circuit is configured to, while the cap is removed from the tubing and the dressing is coupled to the tubing, determine a volume of a wound space based on the flow rate through the filter and additional measurements of the pressure from the sensor. In some embodiments, the control circuit is configured to provide a customized wound therapy based on the volume of the wound space.

BRIEF DESCRIPTION OF THE FIGURES

Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood

by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 is a partial block diagram of a negative pressure wound therapy system including a therapy device coupled to a wound dressing via tubing, according to an exemplary embodiment.

FIG. 2 is a block diagram illustrating the negative pressure wound therapy system of FIG. 1 in greater detail, according to an exemplary embodiment.

FIG. 3 is a block diagram illustrating the negative pressure circuit, the removed fluid canister circuit and the wound site circuit of the negative pressure wound therapy system of FIG. 1 in greater detail, according to an exemplary embodiment.

FIG. 4 is a block diagram illustrating a negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 5 is a flowchart of a method of using a negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 6A is a flowchart of method of instilling an initial quantity of fluid to a wound site using the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 6B illustrates a negative pressure wound therapy system applied to a desired wound site to be treated, prior to the instillation of an initial volume of fluid to the wound site according to an exemplary embodiment.

FIG. 6C illustrates the negative pressure wound therapy system of FIG. 6B following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 6D illustrates the negative pressure wound therapy system of FIG. 6C during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 6C, according to an exemplary embodiment.

FIG. 6E illustrates the negative pressure wound therapy system of FIG. 6B following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 6F illustrates the negative pressure wound therapy system of FIG. 6E during venting of the negative pressure wound therapy system following the application of the second negative pressure as shown in FIG. 6E, according to an exemplary embodiment.

FIG. 6G illustrates the instillation of fluid to the wound site using the wound therapy system of FIG. 6B, according to an exemplary embodiment.

FIG. 7 illustrates a negative pressure wound therapy system applied to a wound site following an initial instillation of fluid to the wound site, according to an exemplary embodiment.

FIG. 8A is a flowchart of method of instilling an additional quantity of fluid to a wound site using the negative pressure wound therapy system of FIG. 7, according to an exemplary embodiment.

FIG. 8B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 8C illustrates the negative pressure wound therapy system of FIG. 8B during venting of the negative pressure

wound therapy system following the application of the first negative pressure as shown in FIG. 8B, according to an exemplary embodiment.

FIG. 8D illustrates the negative pressure wound therapy system of FIG. 7 following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 8E illustrates the negative pressure wound therapy system of FIG. 8D during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 8D, according to an exemplary embodiment.

FIG. 9A is a flowchart of method of instilling an additional quantity of fluid to a wound site to the negative pressure wound therapy system of FIG. 7, according to an exemplary embodiment.

FIG. 9B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 9C illustrates the negative pressure wound therapy system of FIG. 9B during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 9B, according to an exemplary embodiment.

FIG. 9D illustrates the negative pressure wound therapy system of FIG. 7 following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 9E illustrates the negative pressure wound therapy system of FIG. 9D during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 9D, according to an exemplary embodiment.

FIG. 10A is a flowchart of a method of determining whether sufficient dead space is present in a negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 10B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 10C illustrates the negative pressure wound therapy system of FIG. 10B during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 10B, according to an exemplary embodiment.

FIG. 11 is a flowchart of a process for monitoring the healing progression of the wound site over time, according to an exemplary embodiment.

FIG. 12 is a flowchart of a method of instilling an initial quantity of fluid to a wound site using the negative pressure wound therapy system, according to an exemplary embodiment.

FIG. 13 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

FIG. 14 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

FIG. 15 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

FIG. 16A is a block diagram of a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

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FIG. 16B illustrates the negative pressure wound therapy system comprising a tubeset module of FIG. 16A, according to an exemplary embodiment.

FIG. 17 is a flowchart of a fully automated method of operating a tubeset module, according to an exemplary embodiment.

FIG. 18 is a block diagram of a negative pressure and instillation wound therapy (NPIWT) system, according to an exemplary embodiment.

FIG. 19 is a cross-sectional illustration of a solenoid valve of the NPIWT system of FIG. 18 in a closed position, according to an exemplary embodiment.

FIG. 20 is a cross-sectional illustration of the solenoid valve of FIG. 19 in an open position, according to an exemplary embodiment.

FIG. 21 is a cross-sectional illustration of tubing of the NPIWT system of FIG. 18, according to an exemplary embodiment.

FIG. 22 is a flowchart of a process for managing blockages of the tubing of FIG. 21 using the solenoid valve of FIGS. 19-20, according to an exemplary embodiment.

FIG. 23 is a flowchart of a process for volume determination by the NPIWT system of FIG. 18, according to an exemplary embodiment.

FIG. 24 is a collection of graphs illustrating various aspects of the process of FIG. 23, according to an exemplary embodiment.

FIG. 25 is a block diagram of the NPIWT system of FIG. 18 with a removable cap, according to an exemplary embodiment.

DETAILED DESCRIPTION OF THE FIGURES

Overview

Referring generally to FIGS. 1-17, a wound therapy system is shown according to various exemplary embodiments. The wound therapy system may include a therapy device and a wound dressing. The therapy device may include an instillation fluid canister, a removed fluid canister, a valve, a pneumatic pump, an instillation pump, a tubeset module and a controller. The wound dressing can be applied to a patient's skin surrounding a wound. The therapy device can be configured to deliver instillation fluid to the wound and provide negative pressure wound therapy (NPWT) by maintaining the wound at negative pressure. Components of the wound therapy device, the wound dressing, and the wound site form a negative pressure circuit.

The controller can be configured to operate the pneumatic pump, the instillation pump, the tubeset module and/or other controllable components of the therapy device. In some embodiments, the controller estimates the volume of the wound based on a comparison of observed dynamic pressure responses to negative pressure being applied to the entirety of the negative pressure circuit and negative pressure being applied to a selected portion of the negative pressure circuit. Based on the comparison of the observed dynamic responses, the controller may be configured to determine a quantity of instillation fluid to be delivered to the wound site.

The tubeset module comprises one or more elements that are actuatable, controllable or which may otherwise be engaged by the controller, with the selective communication of the controller with the tubeset module being configured to allow the controller to, among other functions, effectuate and monitor various dynamic pressure responses in all of and/or in parts of the negative pressure circuit as needed to

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estimate the volume of the wound, determine a quantity of instillation fluid to be delivered to the wound site and/or perform any other number of functions that may be related to the use of the NPWT system 100.

According to some embodiments, the volume relative to the wound site determined by the controller may relate to the dead space at the wound site (i.e. the available space within a drape layer applied about the wound site into which instillation fluid may be delivered). In some such embodiments, the controller may be configured to determine a quantity of instillation fluid to be delivered to the wound site based on a predetermined percentage of the calculated dead space volume at the wound site (e.g., 20%, 50%, 80%, etc.). The controller can then operate the tubeset module and instillation pump to deliver the determined volume of instillation fluid to the wound. By basing the quantity of instillation fluid to be delivered to the wound site on a calculated volume of the dead space at the wound site, the negative pressure system may be configured to provide for more efficient and more precise delivery of instillation fluid, which may reduce the risk of leakage resulting from over-delivery of instillation fluid and the risk of ineffective wound site treatment resulting from under-delivery of instillation fluid.

In some embodiments, the controller may additionally, or alternatively, measure and monitor volumes relative to the wound site at a plurality of times during wound treatment, with the controller determining healing progression of the wound site based on changes in the measured volume relative to the wound site over the course of NPWT treatment. By monitoring the healing progression of the wound site, the controller may be configured to alert a user if the healing of the wound site is not progressing as intended or expected. These and other features of the wound therapy system are described in detail below.

Wound Therapy System

Referring now to FIG. 1, a negative pressure wound therapy (NPWT) system 100 is shown according to an exemplary embodiment. NPWT system 100 is shown to include a therapy device 102 fluidly connected to a wound dressing 112 via tubing 108 and 110. As will be described in more detail below, according to various embodiments a tubeset module 300 may be operably connected to the tubing 108 and/or 110.

According to various embodiments, a wound dressing 112 may be placed on or within the wound site 114 and adhered or sealed to a patient's skin 116 surrounding a wound site 114 using drape layer 117. Several examples of wound dressings 112 which can be used in combination with NPWT system 100 are described in detail in U.S. Pat. No. 7,651,484 granted Jan. 26, 2010, U.S. Pat. No. 8,394,081 granted Mar. 12, 2013, and U.S. patent application No. 14/087,418 filed Nov. 22, 2013. The entire disclosure of each of these patents and patent applications is incorporated by reference herein.

As illustrated by the block diagram of FIG. 2, in general the therapy device 102 includes a pneumatic pump 120, an instillation pump 122, a filter 128, and a controller 118. Pneumatic pump 120 can be fluidly coupled to removed fluid canister 106 (e.g., via conduit 136) and can be configured to draw a vacuum within removed fluid canister 106 by pumping air out of removed fluid canister 106. In some embodiments, pneumatic pump 120 is configured to operate in both a forward direction and a reverse direction. For example, pneumatic pump 120 can operate in the forward direction to pump air out of removed fluid canister 106 and

decrease the pressure within removed fluid canister 106. Pneumatic pump 120 can operate in the reverse direction to pump air into removed fluid canister 106 and increase the pressure within removed fluid canister 106. Pneumatic pump 120 can be controlled by controller 118, described in greater detail below.

Therapy device 102 can be configured to provide negative pressure wound therapy by reducing the pressure at wound site 114. Therapy device 102 can draw a vacuum at wound site 114 (relative to atmospheric pressure) by removing wound exudate, air, and other fluids from wound site 114. Wound exudate may include fluid that filters from a patient's circulatory system into lesions or areas of inflammation. For example, wound exudate may include water and dissolved solutes such as blood, plasma proteins, white blood cells, platelets, and red blood cells. Other fluids 121 removed from wound site 114 may include instillation fluid 105 previously delivered to wound site 114. Instillation fluid 105 can include, for example, a cleansing fluid, a prescribed fluid, a medicated fluid, an antibiotic fluid, or any other type of fluid which can be delivered to wound site 114 during wound treatment. Instillation fluid 105 may be held in an instillation fluid canister 104 and controllably dispensed to wound site 114 via tubing 108. In some embodiments, instillation fluid canister 104 is detachable from therapy device 102 to allow removed fluid canister 106 to be refilled and replaced as needed.

Instillation pump 122 can be fluidly coupled to instillation fluid canister 104 via upstream instillation tubing 108a and fluidly coupled to wound dressing 112 via downstream instillation tubing 108b. Instillation pump 122 can be operated to deliver instillation fluid 105 to wound dressing 112 and wound site 114 by pumping instillation fluid 105 through upstream instillation tubing 108a and downstream instillation tubing 108b. Instillation pump 122 can be controlled by controller 118, described in greater detail below. According to some embodiments, an instillation tubing valve 109 valve configured to allow for flow only in the direction from the instillation fluid canister 104 to the wound site 114 (e.g. via a one-way valve or a valve configured to be selectively switched by a user and/or by the controller 118 to a closed position prior to the application of negative pressure to the wound site 114) may generally be provided at a location along a portion of the downstream instillation tubing 108b. As will be described in more detail below, according to various embodiments, the instillation tubing valve 109 may be provided as part of the tubeset module 300.

Filter 128 can be positioned between removed fluid canister 106 and pneumatic pump 120 (e.g., along conduit 136) such that the air pumped out of removed fluid canister 106 passes through filter 128. Filter 128 can be configured to prevent liquid or solid particles from entering conduit 136 and reaching pneumatic pump 120. Filter 128 may include, for example, a bacterial filter that is hydrophobic and/or lipophilic such that aqueous and/or oily liquids will bead on the surface of filter 128. Pneumatic pump 120 can be configured to provide sufficient airflow through filter 128 that the pressure drop across filter 128 is not substantial (e.g., such that the pressure drop will not substantially interfere with the application of negative pressure to wound site 114 from therapy device 102).

Removed fluid canister 106 may be a component of the therapy device 102 configured to collect wound exudate and other fluids 121 removed from wound site 114. In some embodiments, removed fluid canister 106 is detachable from therapy device 102 to allow removed fluid canister 106 to be

emptied and replaced as needed. A lower portion of removed fluid canister 106 may be filled with wound exudate and other fluids 107 removed from wound site 114, whereas an upper portion of removed fluid canister 106 may be filled with air. Therapy device 102 can be configured to draw a vacuum within removed fluid canister 106 by pumping air out of removed fluid canister 106. The reduced pressure within removed fluid canister 106 can be translated to wound dressing 112 and wound site 114 via tubing 110.

As shown in FIG. 1, disposed along tubing 110 at a location between the removed fluid canister 106 and the wound site 114 is a tubing valve 111 configured to selectively permit and prevent fluid flow between the removed fluid canister 106 and the wound site 114. The tubing valve 111 may be defined by any number of different structures (e.g. spring-biased; duck-bill; clamp; check-valve, etc.) configured to allow for the selective control of fluids through the tubing 110, and may include valves that are configured to be selectively opened and/or closed by a user, in response to a sensed stimulus (e.g. a predetermined threshold pressure), or by the controller 118. As will be described in more detail below, according to various embodiments, the tubing valve 111 may be provided as part of the tubeset module 300.

Referring to the block diagram of FIG. 3, when the tubing valve 111 is in an open, flow configuration, removed fluid canister 106, tubing 110 (i.e. both upstream tubing portion 110a and downstream tubing portion 110b), conduit 136 extending between pneumatic pump 120 and removed fluid canister 106, the portion of downstream instillation tubing 108b extending between the drape layer 117 and instillation tubing valve 109, and wound site 114 are fluidly connected to define a negative pressure circuit 200. Referring further to FIG. 3, when the tubing valve 111 is in a closed, no-flow configuration, the removed fluid canister 106, conduit 136 and an upstream tubing portion 110a of the tubing 110 extending between the removed fluid canister 106 and the tubing valve 111 define a removed fluid canister circuit 202 that is fluidly isolated from a wound site circuit 204 defined by the wound site 114, a downstream tubing portion 110b of tubing 110 extending between the tubing valve 111, a portion of downstream instillation tubing 108b extending between the drape layer 117 and instillation tubing valve 109, and the wound site 114. As will be discussed in more detail below, the volumes of the tubing 110, conduit 136, and portion of downstream instillation tubing 108b extending between the drape layer 117 and instillation tubing valve 109 define known volumes which can be easily subtracted from or otherwise factored into calculations of volume(s) relative to the wound site 114.

Referring again to FIG. 1, according to some embodiments, also provided along and operably fluidly connected to tubing 110 at a location upstream of tubing valve 111 and downstream of removed fluid canister 106 is a calibrated leak system 113 defined by a vent 113a formed through an outer wall of the tubing 110, the vent 113a being selectively closeable by a vent valve 113b. Also forming a part of calibrated leak system 113 may be a flow detector 113c configured to measure airflow through the vent 113a. As will be described in more detail below, calibrated leak system 113 is configured to selectively control and measure airflow between tubing 110 and the ambient environment surrounding therapy device 102. According to various embodiments, calibrated leak system 113 can be selectively opened to allow airflow into tubing 110 at a known, predetermined rate. As will be described in more detail below, according to various embodiments, calibrated leak system 113 may be provided as part of the tubeset module 300.

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As will be described in more detail below, when both the vent valve **113b** and the tubing valve **111** are closed, operation of the pneumatic pump **120** may be configured to draw a vacuum in only the removed fluid canister circuit **202** portion of the negative pressure circuit **200** (such as, e.g., illustrated in FIG. **6E**). When the vent valve **113b** is closed and the tubing valve **111** is open, operation of the pneumatic pump **120** may be configured to draw a vacuum in the entirety of the negative pressure circuit **200** (such as, e.g., illustrated in FIG. **6C**). When the vent valve **113b** is open and the tubing valve **111** is closed, airflow from the environment around therapy device **102** may enter through the vent **113a** of the calibrated leak system **113** and fill the vacuum within the removed fluid canister circuit **202** (such as, e.g., illustrated in FIG. **6F**). As illustrated, e.g., by FIG. **6D**, when both the vent valve **113b** and the tubing valve **111** are open, airflow from the environment around therapy device **102** may enter through the vent **113a** of the calibrated leak system **113** and fill the vacuum within the entirety of the negative pressure circuit **200**. As will be understood, according to various embodiments, the opening and/or closing of the vent valve **113b** and/or tubing valve **111** may be effectuated manually or automatically, e.g., using tubeset module **300**.

Although the calibrated leak system **113** has been disclosed as being positioned in-line with a portion of the tubing **110** extending between the wound site **114** and the removed fluid canister **106**, according to some embodiments, such as, e.g., illustrated in FIG. **4**, the calibrated leak system **113** may be instead formed in-line with conduit **136**. The operation of the calibrated leak system **113** of the embodiment of FIG. **4** is similar to the operation of the calibrated leak system **113** illustrated in FIG. **1**, with the calibrated leak system **113** of FIG. **4** being configured to provide a path through which air from the ambient environment may flow into and fill portions or the entirety of the negative pressure circuit **200** following the creation of a vacuum within a portion or entirety of the negative pressure circuit **200**. As will be understood, according to various embodiments, any of the methods or systems illustrated or disclosed herein which incorporate a calibrated leak system **113** embodiment as illustrated in FIG. **1** may be modified with a calibrated leak system **113** embodiment as illustrated in FIG. **4**.

As illustrated by the block diagram of FIG. **2**, according to various embodiments, the controller **118** may be configured to operate various components of therapy device **102**. In particular, as will be described in more detail below, according to various embodiments, the controller **118** may be configured to control the various components of the NPWT system **100** to execute one or more volume determination procedures via which, e.g., a quantity of instillation fluid **105** to be delivered to the wound site **114** may be determined, the healing progression of the wound site **114** may be tracked, etc. According to various embodiments, the controller **118** may be configured such that these procedures may be performed with minimal user intervention and/or input.

According to various embodiments, therapy device **102** may include a variety of sensors. For example, in some embodiments, therapy device **102** may include pressure sensor **115a** and/or **115b** located in-line in the upstream tubing portion **110a** and/or downstream tubing portion **110b**, which are configured to measure pressure at the removed fluid canister **106** and/or wound site **114**. Pressure measurements recorded by pressure sensor(s) **115a** and/or **115b** can be communicated to controller **118**. According to various

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embodiments, controller **118** may use the pressure measurements from pressure sensor(s) **115a** and/or **115b** as inputs to various pressure testing operations and control operations performed by controller **118**. As will be described in more detail below, according to various embodiments, the pressure sensor(s) **115a** and/or **115b** may be provided as part of the tubeset module **300**.

In some embodiments, therapy device **102** includes a user interface **126**. User interface **126** may include one or more buttons, dials, sliders, keys, or other input devices configured to receive input from a user. User interface **126** may also include one or more display devices (e.g., LEDs, LCD displays, etc.), speakers, tactile feedback devices, or other output devices configured to provide information to a user. User interface **126** can also display alerts generated by controller **118**. For example, controller **118** can generate a “no canister” alert if removed fluid canister **106** is not detected.

In some embodiments, therapy device **102** includes a data communications interface **124** (e.g., a USB port, a wireless transceiver, etc.) configured to receive and transmit data.

Communications interface **124** may include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications external systems or devices. In various embodiments, the communications may be direct (e.g., local wired or wireless communications) or via a communications network (e.g., a WAN, the Internet, a cellular network, etc.). For example, communications interface **124** can include a USB port or an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, communications interface **124** can include a Wi-Fi transceiver for communicating via a wireless communications network or cellular or mobile phone communications transceivers.

Methods of Use

Referring to FIG. **5**, a flowchart of a method **500** of using NPWT system **100** according to an exemplary embodiment is shown. As will be discussed in more detail with reference to FIGS. **6A-6G**, initial set up of the NPWT system **100** and a delivery of an initial amount of instillation fluid **105** to a wound site **114** being treated by the NPWT system **100** occurs at step **502**.

As shown at step **504**, according to various embodiments, it may be desirable to deliver additional instillation fluid **105** to the wound site **114** following the instillation of an initial amount of instillation fluid **105** to the wound site **114**. As will be understood, the determination at step **504** of when and if additional instillation fluid **105** is to be delivered to the wound site **114** may be based on any number of various factors, including e.g. elapsed time from a prior instillation; type of wound site **114**; desired course of wound site **114** treatment; sensed conditions related to the wound site **114**, etc., and may be decided automatically by the controller **118**, or may be based on user input.

If it is determined at step **504** that additional fluid is to be delivered, at step **506** the dead space **119** at the wound site **114** is determined according to any of the methods as will be described below. According to various embodiments (described in more detail below), at step **506**, the controller **118** may be configured to determine the dead space **103** at the wound site **114** prior to such delivery of additional instillation fluid **105**, irrespective of: whether the quantity of instillation fluid **105** previously instilled to the wound site **114** is known; the presence of non-absorbed instillation fluid

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105 and/or wound exudate in the space defined between the wound site 114 and the drape layer 117; whether the volume of any contents 107 in the removed fluid canister 106, the volume of the removed fluid canister 106 itself, and/or the volume of any contents 107 previously emptied from the removed fluid canister 106 are known; whether the removed fluid canister 106 has been replaced with a different-sized removed fluid canister 106 during the course of the NPWT treatment; changes to the shape/size/volume of the wound site 114; etc.

At step 508, the quantity of additional instillation fluid 105 to be delivered to the wound site 114 is calculated. According to various embodiments, the quantity of additional instillation fluid 105 delivered to the wound site 114 may be based on the volume of the dead space determined at step 506. For example, in some embodiments, the controller 118 may calculate the volume of instillation fluid 105 to be delivered to wound site 114 by multiplying the volume of dead space determined at step 506 by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid 105 delivered to the wound site 114 does not exceed the available space within the drape layer 117 (i.e. dead space), thereby minimizing the risk of inadvertent leakage from the wound dressing 112/drape layer 117. In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8.

In addition to being used to calculate instillation fluid 105 volumes, in some embodiments, the NPWT system 100 may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site 114 over time. Accordingly, in some embodiments, method 500 may optionally include the step 510 of estimating wound site 114 volume, and using the estimated volume to track healing progress of the wound site 114, discussed in more detail with reference to FIG. 11 below.

In some embodiments, it may be desired to remove instillation fluid 105 previously instilled to a wound site 114 from the wound site 114 at some time following the delivery of the instillation fluid 105 to the wound site 114. Accordingly, it may be advantageous to confirm, prior to instilling instillation fluid 105 to the wound site 114, that the dead space in the removed fluid canister 106 will be sufficient to receive the removed instillation fluid 105 and/or any additional fluid 121 (e.g. wound exudate) from the wound site 114 prior to delivering the additional instillation fluid 105 to the wound site 114. As such, method 500 may optionally include step 512 at which the volume of additional instillation fluid 105 calculated at step 508 is compared to the dead space of the removed fluid canister 106 (measured, e.g., during the determination of dead space at the wound site 114 at step 506), with an alarm being presented to the user at step 514 if the instillation fluid 105 to be delivered exceeds the dead space of the removed fluid canister 106. If the instillation fluid 105 to be delivered does not exceed the dead space of the removed fluid canister 106 (or if step 512 is not included as part of method 500), the calculated instillation fluid 105 is delivered to the wound site 114, with some or all of steps 504, 506, 508, 510, 512, 514, 516 being repeated any number of additional times over the course of the NPWT treatment.

Referring to FIG. 6A a flowchart detailing the steps of a method 600 for an initial set up of NPWT system 100 and for delivery of an initial amount of instillation fluid 105 to a wound site 114 entailed in step 502 of the method 500 of FIG. 5 is shown according to one embodiment. At step 602, a NPWT system 100 (such as, e.g., illustrated in FIG. 1) is

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provided, with the drape layer 117 and wound dressing 112 being positioned at the desired wound site 114 to be treated, as shown, e.g., in FIG. 6B.

Once the set-up of the NPWT system 100 at step 502 is complete, the determination of the dead space 119 available at the wound site 114 into which instillation fluid 105 may be delivered may begin at step 604 with the controller 118 operating the pneumatic pump 120 to establish a first desired negative pressure within the entirety of the negative pressure circuit 200, such as, e.g., illustrated in FIG. 6C.

In embodiments in which the tubing valve 111 comprises a normally-closed pressure sensitive valve that is openable in response to an applied, predetermined threshold negative pressure, the first desired negative pressure generated by the controller 118 at step 604 may be equal to or greater than the predetermined threshold pressure required to open the tubing valve 111, so as to ensure that the vacuum applied by the pneumatic pump 120 is applied across the entirety of the negative pressure circuit 200. In some embodiments, the threshold pressure required to open the tubing valve 111 may be a pressure of approximately negative 125 mmHg, with the controller 118 being configured to apply at step 604 a first negative pressure that is equal to or greater than negative 125 mmHg.

Alternatively, in embodiments in which the opening/closing of the tubing valve 111 is controlled manually or in direct response to a signal from the controller 118 (using, e.g., a tubeset module 300 as described below), the negative pressure delivered at step 604 may generally include any desired range of negative pressures, with step 604 including verification by the user and/or controller that the tubing valve 111 is in an open, flow orientation prior to the negative pressure being applied by the pneumatic pump 120. As illustrated, e.g., in FIG. 6C, according to various embodiments, the instillation tubing valve 109 and the vent valve 113b may be configured to be set to closed configurations during the application of negative pressure to the negative pressure circuit 200.

As illustrated by FIG. 6D, at step 606, following the attainment of the desired first negative pressure within the negative pressure circuit 200 (as, e.g., measured and reported to the controller 118 by pressure sensor 115a and/or pressure sensor 115b), the operation of the pneumatic pump 120 is stopped, and the vent valve 113b is opened to allow air from the ambient environment surrounding the therapy device 102 to flow through the vent 113a and into the negative pressure circuit 200. According to various embodiments, the opening of the vent valve 113b at step 606 may be effectuated manually by a user or in response to instructions from the controller 118 being transmitted to the tubeset module 300. In yet other embodiments, the calibrated leak system 113 may be formed without a vent valve 113b (i.e. the vent 113a defines a constant leak within the tubing 110), such that air from the ambient environment surround the therapy device 102 will flow into the negative pressure circuit 200 without requiring any user and/or controller 118 intervention.

As air from the ambient environment flows in to the negative pressure circuit 200, parameters related to the flow of air through the vent 113a into the negative pressure circuit 200 are monitored (e.g. via flow detector 113c, pressure sensor 115a, pressure sensor 115b, etc.), with the measured parameters subsequently being used by the controller 118 at step 612 to determine the volume of the negative pressure circuit 200. According to various embodiments, the parameters related to the flow of air into the negative pressure circuit 200 may include, e.g., the rate of flow of air into the

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negative pressure circuit 200 (as measured, e.g., by flow detector 113c), the duration of time required for pressure within the negative pressure circuit 200 to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent 113a and/or following operation of the pump 120 being ceased, the changing pressure (as, e.g., measured by pressure sensor 115a and/or pressure sensor 115b) within the negative pressure circuit 200 as the pressure increases from the negative pressure applied at step 604 to the predetermined pressure, etc.

Once the pressure within the negative pressure circuit 200 has increased to a desired pressure and the measurement of the desired parameters has been completed by the controller 118, the controller 118 may be configured operate pneumatic pump 120 to establish a second desired negative pressure within the removed fluid canister circuit 202 portion of the negative pressure circuit 200 at step 608, such as, e.g., illustrated in FIG. 6E. In embodiments in which the tubing valve 111 comprises a normally-closed pressure sensitive valve that is openable in response to an applied, predetermined threshold negative pressure, the second desired negative pressure generated by the controller 118 at step 608 may be less than the predetermined threshold pressure required to open the tubing valve 111, so as to ensure that the vacuum applied by the pneumatic pump 120 at step 608 is applied across only the removed fluid canister circuit 202 portion of the negative pressure circuit 200. For example, in some embodiments, the threshold negative pressure required to open the tubing valve 111 may be approximately negative 125 mmHg, with the controller 118 being configured to apply a negative pressure at step 608 that is less than negative 125 mmHg, such as, e.g., a pressure of approximately negative 50 mmHg.

Alternatively, in embodiments in which the opening/closing of the tubing valve 111 is controlled manually or in direct response to a signal from the controller 118, the negative pressure delivered at step 608 may generally include any desired range of negative pressures, with step 608 including verification by the user and/or controller that the tubing valve 111 is in a closed, no-flow orientation prior to the negative pressure being applied by the pneumatic pump 120. As will be understood, in such embodiments, the second negative pressure applied by the controller 118 at step 608 to the removed fluid canister circuit 202 may include a pressure that is equal to or different from the negative pressure that is applied by the controller 118 at step 604 to the negative pressure circuit 200. As illustrated, e.g., in FIG. 6E, according to various embodiments, the instillation tubing valve 109 and the vent valve 113b may be configured to be set to closed configurations (either manually or automatically, e.g., using tubeset module 300) during the application of negative pressure to the removed fluid canister circuit 202 at step 608.

As illustrated by FIG. 6F, at step 610, following the attainment of the desired second negative pressure within the removed fluid canister circuit 202 (as, e.g., measured and reported to the controller 118 by pressure sensor 115a and/or pressure sensor 115b), the operation of the pneumatic pump 120 is stopped, and air from the ambient environment surrounding the therapy device 102 is allowed to flow through the vent 113a and into the removed fluid canister circuit 202. As air from the ambient environment flows into the removed fluid canister circuit 202, parameters related to the flow of air through the vent 113a and into the removed fluid canister circuit 202 are monitored, with the measured parameters subsequently being used by the controller 118 to calculate the volume of the removed fluid canister circuit

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202 at step 612. According to various embodiments, the parameters related to the flow of air into removed fluid canister circuit 202 may include, e.g., the rate of flow of air into the removed fluid canister circuit 202 (as measured, e.g., by flow detector 113c), the duration of time required for pressure within the removed fluid canister circuit 202 to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent 113a and/or ceasing operation or the pump 120 at step 610, the pressure (as, e.g., measured by pressure sensor 115a and/or pressure sensor 115b) within the removed fluid canister circuit 202 as the pressure increases from the negative pressure applied at step 608 to the predetermined pressure; etc.

At step 612, the controller 118 may be configured to determine the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 based on the parameters measured at steps 606 and 610. According to some embodiments, the controller 118 may base these volume calculations on stored relationships between various measured parameter values and corresponding volumes. These relationships between measured parameter measurements and corresponding volumes that are stored by the controller 118 may include various functions, models, lookup table, etc., and may be based on pre-existing information input and stored by the controller 118, or on information obtained and processed by the controller 118 during an optional, initial training procedure conducted by the controller 118 prior to the use of the NPWT system 100 to treat wound site 114 (e.g. prior to the initiation of method 500; as part of the initial setup and initial instillation of instillation fluid of step 502; etc.). One non-limiting examples of embodiments of training procedures by which such relationships may be generated by the controller 118 are outlined in related, U.S. Provisional Application 62/650,132, filed Apr. 17, 2018 and titled WOUND THERAPY SYSTEM WITH WOUND VOLUME ESTIMATION, the entire disclosure of which is incorporated by reference herein.

Using the determined volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200, the controller 118 may determine the volume of the dead space 119 at the wound site 114 (i.e. the portion of the interior space defined between the wound site 114 and the lower surface of the drape layer 117 that is not occupied by the wound dressing 112 and/or any instillation fluid 105/other fluid) by subtracting the volume of the removed fluid canister circuit 202 from the volume of the negative pressure circuit 200. According to various embodiments, the determination of the volume of the dead space 119 at the wound site 114 at step 614 may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 to account for/factor in the known volumes of the downstream tubing portion 110b and the portion of the downstream instillation tubing 108b extending between the drape layer 117 and the instillation tubing valve 109 into the determination of the volume of the dead space 119 at the wound site 114.

At step 614, an initial quantity of instillation fluid 105 that is to be delivered to the wound site 114 is calculated. According to various embodiments, the calculated initial quantity of instillation fluid 105 that is delivered to the wound site 114 may be based on the volume of the dead space 119 calculated by the controller 118 at step 612. For example, in some embodiments, the controller 118 may calculate the initial volume of instillation fluid 105 to be delivered to the wound site 114 by multiplying the volume

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of dead space **119** calculated at step **612** by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid **105** delivered to the wound site **114** does not exceed the available space within the drape layer **117** (thereby minimizing inadvertent leakage from the wound dressing **112**/drape layer **117**. In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8. However, it is contemplated that the fluid instillation factor can have any value in various alternative embodiments.

As noted previously with reference to step **510**, in addition to being used to calculate a quantity of instillation fluid **105** to be delivered during any stage of treatment using NPWT system **100** and under any number of different conditions (e.g. allowing for the calculation of additional instillation fluid **105** to be delivered at step **516** even if the removed fluid canister **106** has been emptied, or entirely replaced with a different sized removed fluid canister **106** during the course of treatment), in some embodiments the NPWT system **100** may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site **114** over time. Accordingly, in some embodiments, at step **616**, an initial baseline wound site **114** volume estimate may optionally be determined (via, e.g., a method as described with regards to FIG. **11** below) and stored by the controller **118**, which may be used as a reference point against which future wound site **114** volume estimates may be compared to track healing progression of the wound site **114**.

For reasons similar to those described with reference to step **512** of the method **500** of FIG. **5**, according to some embodiments, at step **618** the amount of initial instillation fluid **105** that is to be delivered calculated at step **614** may be compared to a determined dead space **103** of the removed fluid canister **106** to determine whether the dead space within the removed fluid canister **106** will be sufficient to collect any fluids **121** from the wound site **114** (including non-absorbed instillation fluid **105**) following the delivery of instillation fluid **105** at step **516**. As will be understood, in embodiments in which the NPWT system **100** has not been operated prior to the use of the NPWT system **100** at step **602**, the volume of the removed fluid canister **106** should be empty, such that the dead space **103** of the removed fluid container **106** should be equal to the volume of the removed fluid canister **106**. If the volume of the removed fluid canister **106** is not known and/or if removed fluid **107** is present in the removed fluid canister **106** at step **602**, the dead space **103** of the removed fluid container may be calculated by subtracting the known volumes of conduit **136** and the upstream tubing portion **110a** from the volume of the removed fluid canister circuit **202** determined at step **614**. Similar to step **514**, at step **620** an alarm may be presented to a user if the initial volume of instillation fluid **105** to be delivered calculated at step **614** exceeds the dead space **103** of the removed fluid canister **106**. Otherwise, if the volume of the initial instillation fluid **105** to be delivered does not exceed the dead space **103** of the removed fluid canister **106**, the calculated instillation fluid **105** is delivered to the wound site **114** at step **622**, as shown, e.g., in FIG. **6F**.

Referring to FIG. **7**, a NPWT system **100** according to one embodiment is shown at a point in time subsequent to a decision to instill additional instillation fluid **105** to the wound site **114** at step **504** of the method **500** of FIG. **5**, but prior to the determination of wound dead space at the wound site at step **506**. As shown in FIG. **7**, at the time immediately preceding the determination of dead space at the wound site

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114 at step **506**, a quantity of fluid **121** (e.g. non-absorbed instillation fluid **105** from a prior instillation, wound exudate, etc.) may be present in the space between the drape layer **117** and the wound site **114**, with the remaining space between the drape layer **117** and the wound site **114** defining an initial dead space **119a**. As also shown in FIG. **7**, according to some embodiments, an initial quantity of removed fluid **107** may be present in the removed fluid canister **106** at the time immediately preceding the start of step **506**, with the remaining volume of the removed fluid canister **106** being defined by an initial dead space **103a**. As will be understood, according to some embodiments, no fluid may be present at either the wound site **114** and/or in the removed fluid canister **106** at the time immediately preceding step **506**, in which embodiments the quantities of each of the fluid **121** in the wound space and the removed fluid **107** in the removed fluid canister **106** would be equal to zero.

As noted above, a quantity of fluid **121** may be present at the wound site **114** immediately prior to the initiation of step **506**. According to some embodiments, it may not be desired and/or required to remove fluid **121** from the wound site (e.g. non-absorbed instillation fluid **105** from prior instillations, wound exudate, etc.) prior to the delivery of additional instillation fluid **105** to the wound site **114** at step **516** of the method **500** of FIG. **5**. Accordingly, in some embodiments of method **500**, the additional instillation fluid **105** instilled to the wound site at step **516** may be delivered to the wound site **114** irrespective of any fluid **121** that may be present at the wound site **114**.

Referring to FIGS. **8A-8E**, one embodiment of a method **800** of determining an amount of dead space at a wound site **114** which may be used at step **506** of the method **500** of FIG. **5** in embodiments in which fluid **121** from the wound site **114** is not removed from the wound site **114** prior to instilling additional instillation fluid **105** is illustrated. In particular, according to the method **800** of FIGS. **8A-8E**, as no fluid **121** is displaced from the wound site **114** during the method **800** (i.e. step **506**), the final dead space into which the additional instillation fluid **121** will be instilled will be the same initial dead space **119a** at the wound site that is present immediately prior to the initiation of step **506** (i.e. the dead space **119a** shown in FIG. **7**).

As shown by the flowchart in FIG. **8A**, the method **800** of determining dead space is substantially the same as the method **600** of calculating the dead space **119** upon initial instillation of instillation fluid **105** to the wound site **114** at step **502** (which is discussed in more detail with reference to FIGS. **6A-6G**). In particular, similar to steps **604** and **606**, the method **800** of FIG. **8A** also includes steps **802** and **804** (shown, e.g., in FIGS. **8B** and **8C**, respectively) during which negative pressure is applied to and removed from the negative pressure circuit **200**. Similar to steps **608** and **610** of the method **600** of FIG. **6A**, the method **800** of FIG. **8** also includes steps **806** and **808** (shown, e.g., in FIGS. **8D** and **8E**, respectively) during which negative pressure is applied to and removed from the removed fluid canister circuit **202**. Also similar to the method **600** of FIG. **6A**, in the method **800** of FIGS. **8A-8E**, the application and subsequent removal of negative pressure to the negative pressure circuit **200** of steps **802** and **804** may be performed either prior to or after the application and subsequent removal of negative pressure to the removed fluid canister circuit **202** of steps **806** and **808**.

As noted above, the method **800** of FIGS. **8A-8E** may be performed in substantially the same manner as the method **600** described with references to FIG. **6A** above. However,

whereas, as described above with reference to the method of FIGS. 6A-6E, according to various embodiments, any range of negative pressures may generally be applied to the negative pressure circuit 200 at step 604 of method 600, the negative pressure applied to the negative pressure circuit 200 at step 802 of the method 800 must be limited to negative pressures that will not result in the fluid 121 at the wound site 114 being displaced into the removed fluid canister 106.

Following the completion of step 808, the controller 118 may be configured to calculate the volume of the dead space 119a at the wound site 114 (which corresponds to the maximum volume of additional instillation fluid 105 that may be delivered to wound site 114) at step 508 of method 500 of FIG. 5. More specifically at step 508, after calculating the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 based on the parameters measured at steps 804 and 808 (in a manner similar to that described with reference to step 612 of the method 600 of FIGS. 6A-6G), the dead space 119a at the wound site 114 may be calculated based on subtracting the measured volume of the removed fluid canister circuit 202 from the measured volume of the negative pressure circuit 200, with the volume of the removed fluid canister circuit 202 of the method 800 of FIGS. 8A-8E being defined by the dead space 103a of the removed fluid canister 106, conduit 136 and upstream tubing portion 110a; and the volume of the negative pressure circuit 200 being defined by the volume of the removed fluid canister circuit 202 (i.e. dead space 103a of the removed fluid canister 106, conduit 136 and upstream tubing portion 110a), the downstream tubing portion 110b, dead space 119a of the wound site 114 and the portion of downstream instillation tubing 108b extending between the drape layer 117 and instillation tubing valve 109.

According to various embodiments, in embodiments of method 500 in which the determination of the volume of the dead space 119a at the wound site 114 at step 508 is based on measured parameters related to the removed fluid canister circuit 202 and negative pressure circuit 200 obtained using the method 800 of FIGS. 8A-8E, step 508 may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 to account for/ factor in the known volumes of the downstream tubing portion 110b and the portion of the downstream instillation tubing 108b extending between the drape layer 117 and the instillation tubing valve 109 into the determination of the volume of the dead space 119a at the wound site 114.

Although, as described above, in some embodiments of method 500, additional instillation fluid 105 may be delivered at step 516 without first removing any remaining fluid 121 at the wound site 114, according to other embodiments, it may be desirable to remove fluid 121 from the wound site 114 prior to the delivery of additional instillation fluid 105.

Referring to FIGS. 9A-9E, one embodiment of a method 900 of determining an amount of dead space at a wound site 114 which may be used at step 506 of the method 500 of FIG. 5 in embodiments in which it is desired to remove fluid 121 from the wound site 114 prior to instilling additional instillation fluid 105 is illustrated. In particular, according to the method 900 of FIGS. 9A-9E, any fluid 121 initially at the wound site 114 immediately prior to step 506 (e.g. as shown in FIG. 7) is displaced from the wound site 114 during the method 900 (i.e. step 506), such the final dead space 119b into which the additional instillation fluid 121 will be instilled will be greater than the initial dead space 119a at the wound site that is present immediately prior to the initiation

of step 506 by an amount generally corresponding to a volume of the fluid 121 displaced from the wound site 114 to the removed fluid canister 106 during the method 900.

As shown by the flowchart in FIG. 9A, the method 900 of determining dead space is substantially the same as the method 600 of calculating the dead space 119 upon initial instillation of instillation fluid 105 to the wound site 114 at step 502 (discussed in more detail with reference to FIGS. 6A-6G). In particular, similar to steps 604 and 606, the method 900 of FIG. 9A also includes steps 902 and 904 (shown, e.g., in FIGS. 9B and 9C, respectively) during which negative pressure is applied to and removed from the negative pressure circuit 200. Similar to steps 608 and 610 of the method 600 of FIG. 6A, the method 900 of FIG. 9A also includes steps 906 and 908 (shown, e.g., in FIGS. 9D and 9E, respectively) during which negative pressure is applied to and removed from the removed fluid canister circuit 202.

However, unlike the method 600 of FIG. 6A in which the application and subsequent removal of negative pressure to the negative pressure circuit 200 at steps 604 and 608 may be performed either prior to or after the application and subsequent removal of negative pressure to the removed fluid canister circuit 202 of steps 610 and 612, in the method 900 of FIG. 9A, the application and subsequent removal of negative pressure to the negative pressure circuit 200 at steps 902 and 904 is performed prior to the application and subsequent removal of negative pressure to the removed fluid canister circuit 202 of steps 906 and 908. Additionally, whereas, as described above with reference to the method of FIGS. 6A-6E, according to various embodiments, any range of negative pressures may generally be applied to the negative pressure circuit 200 at step 604 of method 600, the negative pressure applied to the negative pressure circuit 200 at step 902 of the method 900 of FIG. 9A must be sufficient to cause the displacement of fluid 121 from the wound site 114 into the removed fluid canister 106.

Following the completion of step 908, the controller 118 may be configured to calculate the volume of the final dead space 119b at the wound site 114 (which corresponds to the maximum volume of additional instillation fluid 105 that may be delivered to wound site 114) at step 508 of method 500 of FIG. 5. More specifically at step 508, after calculating the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 based on the parameters measured at steps 904 and 908 (in a manner similar to that described with reference to step 612 of the method 600 of FIGS. 6A-6G), the final dead space 119b at the wound site 114 may be calculated based on subtracting the measured volume of the removed fluid canister circuit 202 from the measured volume of the negative pressure circuit 200, with the volume of the removed fluid canister circuit 202 of the method 800 of FIGS. 9A-9E being defined by the final dead space 103b of the removed fluid canister 106 (with the final dead space 103b of the removed fluid canister 106 being generally equal to the difference between an initial dead space 103a within the removed fluid canister 106 and the volume of fluid 121 displaced into the removed fluid canister 106 from the wound site 114 at step 802, as shown, e.g., in FIG. 9B), conduit 136 and upstream tubing portion 110a; and the volume of the negative pressure circuit 200 being defined by the volume of the removed fluid canister circuit 202 (i.e. final dead space 103b of the removed fluid canister 106, conduit 136 and upstream tubing portion 110a), the downstream tubing portion 110b, final dead space 119b of the wound site 114 and the portion of downstream instillation tubing 108b extending between the drape layer 117 and instillation tubing valve 109.

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According to various embodiments, in embodiments of method 500 in which the determination of the volume of the dead space 119 at the wound site 114 at step 508 is based on measured parameters related to the removed fluid canister circuit 202 and negative pressure circuit 200 obtained using the method 900 of FIGS. 9A-9E, step 508 may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit 202 and the negative pressure circuit 200 to account for/ factor in the known volumes of the downstream tubing portion 110b and the portion of the downstream instillation tubing 108b extending between the drape layer 117 and the instillation tubing valve 109 into the determination of the volume of the dead space 119a at the wound site 114.

In some embodiments of method 500 of FIG. 5 in which fluid 121 from the wound site 114 is removed prior to the instillation of additional instillation fluid 105 at step 516, it may be desirable to ensure that the initial dead space 103a in the removed fluid canister 106 immediately prior to beginning the step of determining dead space at the wound site at step 506 is sufficient to hold fluid 121 that will be displaced from the wound site 114 into the removed fluid canister during step 506, so as to avoid the risk of removed fluid canister 106 overflow.

Accordingly, in some embodiments of method 500 in which fluid 121 from the wound site 114 is removed prior to the instillation of any additional instillation fluid 105 at step 516, the method of step 506 of determining dead space at the wound site 114 (e.g., such as described with reference to the method 900 of FIGS. 9A-9E) may include determining whether there is sufficient dead space at the removed fluid canister 106 to hold the fluid 121 from the wound site 114 that may be displaced into the removed fluid canister 106 as part of the method of determining dead space at the wound site 114.

Illustrated in FIGS. 10A-10C is one embodiment of such a method that may be used to minimize the risk of overflow of the removed fluid canister 106 during step 506 in which dead space at the wound site 114 is being determined (e.g., via method 900 as described in FIGS. 9A-9E). At steps 1002 and 1004 (shown in FIGS. 10B and 10C, respectively) negative pressure is applied to and removed from the removed fluid canister circuit 202 to determine the initial dead space 103a in the removed fluid canister 106 prior to beginning step 506 (e.g. as shown in FIG. 7). In general, the steps 1002 and 1004 of the method 1000 of FIGS. 10A-10E may be performed in a manner substantially similar to the manner in which steps 608 and 610 of the method 600 of FIGS. 6A-6G are performed. At step 1006, the volume of the removed fluid canister circuit 202 is calculated based on the parameter measured at step 1004 (in a manner similar to that described with reference to step 612 of the method 600 of FIGS. 6A-6G). Once the volume of the removed fluid canister circuit 202 has been calculated, the known volumes of the conduit 136 and the upstream tubing portion 110a may be subtracted from the calculated removed fluid canister circuit 202 to determining the volume of the initial dead space 103a in the removed fluid canister 106 (i.e. the maximum volume of fluid 121 displaced from the wound site 114 that the removed fluid canister 106 may hold).

Once the volume of the initial dead space 103a has been calculated at step 1006, at step 1008, the controller 118 may be configured to estimate the volume of the fluid 121 at the wound site 114 at the time immediately preceding the determination of dead space at the wound site 114 at step 506. The volume of the fluid 121 at the wound site 114 may be based on any number of different factors and variables

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such as, e.g., stored values of quantities of instillation fluid 105 previously delivered to the wound site 114, stored values of fluid 121 previously removed from the wound site, elapsed time (e.g. from a prior instillation, a prior removal of fluid 121, etc.), etc., with the controller 118 at step 1008 further being configured to compare this estimated volume of fluid 121 to the initial dead space 103a calculated at step 1006, alerting the user to empty the removed fluid canister 106 at step 1010 if the controller 118 determines that the estimated fluid 121 volume exceeds the calculated initial dead space 103a. If the calculated initial dead space 103a is sufficient to hold the estimated volume fluid 121 from the wound site 114, at step 1012 the controller 118 may be configured to begin the step 506 of determining dead space at the wound site 114, e.g., according to method 900 as described with reference to FIGS. 9A-9E.

As noted above, according to some embodiments of method 500, it may be advantageous to monitor changes in the volume of the wound site 114 to track the progress of the healing of the wound site 114 at an optional step 510.

In general, the volume of the wound site 114 is defined by the entirety of the interior extending between the wound site 114 and the drape layer 117 attached to the skin 116 about the wound site 114. At various points during treatment using the NPWT system 100, located within and defining the volume of the wound site may be any one of, and any combination of: the wound dressing 112, fluid 121, and/or dead space 119. As will be understood, unless the wound dressing 112 is replaced during treatment, the volume of the wound site 114 volume occupied by the wound dressing 112 will generally remain unchanged over the course of treatment, whereas the portion of the wound site 114 volume occupied by the fluid 121 and/or dead space 119 may change with time.

Referring to FIG. 11, a block diagram illustrating one embodiment of a method 1100 of tracking wound site 114 healing progression which may be used at step 510 of the method 500 of FIG. 5 is illustrated. At step 1102, an initial volume of the wound site 114 is estimated and recorded by the controller 118 at a point in time prior to an initial instillation of instillation fluid 105 to the wound site 114, and may serve as a baseline against which subsequent wound site 114 volume estimates are compared to track healing progress. According to various embodiments, estimation of the initial volume of the wound site 114 at step 1102 may be performed according to (or as) step 616 of method 600 described with reference to FIGS. 6A-6G.

At step 1104, the estimated volume of the wound site 114 is determined and recorded at one or more additional times during treatment (e.g., once per day) following the estimation of the initial wound site 114 volume at step 1102, with the times at which such one or more wound site 114 volumes are estimated and the values of the determined wound site 114 volume being stored as data points within the memory of therapy device 102 and/or presented to a user as an output of therapy device 102 (e.g., via communications interface 124 or user interface 126). In some embodiments, the estimated wound volume can be plotted as a function of time.

The additional wound site 114 volume estimates determined at one or more additional times over the course of treatment at step 1104 may be estimated according to any number of different processes. For example, according to some embodiments, the wound site 114 volume estimates recorded at step 1104 may be based on the final dead space volume at the wound site 114 calculated, e.g., at step 508 of

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method 500 and/or using method 900 as described with reference to FIG. 5 and/or 9A-9E, respectively.

As shown at step 510 of FIG. 5 and step 616 of FIG. 6A, according to some embodiments, the wound site 114 volume estimates at steps 1102 and/or steps 1104 may be performed in conjunction with method of delivering of instillation fluid 105 to the wound site 114. However, as will be understood, according to other embodiments the determination of and recording of some, all, or none of the wound site 114 volume estimates at steps 1102 and/or steps 1104 may be performed independent of any delivery of instillation of instillation fluid 105 to the wound site 114.

As additional wound site 114 volume estimates are obtained at steps 1104, at step 1106, changes in the estimated wound site 114 volumes over time may be used to determine healing progression of the wound site 114. For example, step 1106 may include comparing wound site 114 volume estimates obtained at step 1104 to one or more previous estimates of the wound site 114 volume (obtained at either step 1104 or step 1102) to identify a change in the wound site 114 volume. In some embodiments, step 1106 may additionally include determining a rate at which the wound site 114 is healing based on the changes in the estimated wound site 114 volume over time. In some embodiments, step 1106 may include extrapolating or predicting a time at which wound site 114 will be fully healed based on the series of wound site 114 volume estimates stored by the controller 118. For example, step 1106 may include predicting a time at which the estimated wound site 114 volume will reach zero (or another threshold value) based on the initial wound site 114 volume estimate obtained at step 1002 and the series of additional wound site 114 volume estimates obtained at step 1004.

According to some embodiments, instead of, or in addition to, a calibrated leak system 113 being provided which is located upstream of the tubing valve 111, the NPWT system 100 may include a calibrated leak system 113 located downstream of the tubing valve 111. In general, such embodiments in which a calibrated leak system 113 is located downstream of the tubing valve 111 may operate in a manner substantially similar to the various methods described with reference to FIGS. 1-11. However, in contrast to the step of monitoring the pressure decay within the removed fluid canister circuit 202, determining a volume of the removed fluid canister circuit 202 based on the monitored pressure decay, and subsequently using the determined volume of the removed fluid canister circuit 202 to calculate wound site 114 volume, in the method 1200 of FIG. 12, pressure decay is instead monitored within the wound site circuit 204, with the determined volume of the wound site circuit 204 subsequently being used to calculate the dead space 103 in the removed fluid canister 106.

For example, referring to FIG. 12, a flowchart detailing the steps of a method 1200 for an initial set up of NPWT system 100 and delivery of an initial amount of instillation fluid 105 to a wound site 114 is shown according to one embodiment in which a calibrated leak system 113 of the NPWT system 100 is positioned downstream of the tubing valve 111. In general, steps 1202, 1204 and 1206 of the embodiment of the method 1200 of FIG. 12 may be performed in a manner substantially similar to that as described with reference to steps 602, 604 and 606 of the method 600 of FIG. 6A.

At step 1208, the controller 118 is configured to initiate operation of the pump 120 to apply a second negative pressure (which may be equal to or different from the negative pressure applied by the controller 118 at step 1204)

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to the negative pressure circuit 200. According to various embodiments, the instillation tubing valve 109 and the vent valve 113b may be configured to be set to closed configurations during the application of negative pressure to the negative pressure circuit 200 at step 1208. In embodiments in which a controller 118 controlled tubeset module 300 is used, the controller 118 may be configured to instruct the tubeset module 300 to effectuate the closing of the instillation tubing valve 109 and/or the vent valve 113b.

At step 1210, following the attainment of the desired second negative pressure within the negative pressure circuit 200 (as, e.g., measured and reported to the controller 118 by pressure sensor 115a and/or pressure sensor 115b), the tubing valve 111 is closed so as to define a wound site circuit 204, the operation of the pneumatic pump 120 is stopped, and air from the ambient environment surrounding the therapy device 102 is allowed to flow through the vent 113a of the and into the wound site circuit 204. As air from the ambient environment flows into the wound site circuit 204, parameters related to the flow of air through the vent 113a and into the wound site circuit 204 are monitored, with the measured parameters subsequently being used by the controller 118 to calculate the volume of the wound site circuit 204 at step 1212. According to various embodiments, the parameters related to the flow of air into wound site circuit 204 may include, e.g., the rate of flow of air into the wound site circuit 204 (as measured, e.g., by flow detector 113c), the duration of time required for pressure within the wound site circuit 204 to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent 113a and/or ceasing operation of the pump 120 at step 1210, the pressure (as, e.g., measured by pressure sensor 115b) within the wound site circuit 204 as the pressure increases from the negative pressure applied at step 1208 to the predetermined pressure; etc.

At step 1212, the controller 118 may be configured to determine the volume of the wound site circuit 204 based on the parameters measured during step 1208. According to some embodiments, the controller 118 may base this wound site circuit 204 volume calculation on stored relationships between various measured parameter values and corresponding volumes. These relationships between measured parameter measurements and corresponding volumes that are stored by the controller 118 may include various functions, models, lookup table, etc., and may be based on pre-existing information input and stored by the controller 118, or on information obtained and processed by the controller 118 during an optional, initial training procedure conducted by the controller 118 prior to the use of the NPWT system 100 to treat wound site 114 (e.g. prior to the initiation of method 500; as part of the initial setup and initial instillation of instillation fluid of step 502; etc.). One non-limiting examples of embodiments of training procedures by which such relationships may be generated by the controller 118 are outlined in related, U.S. Provisional Application 62/650,132, filed Apr. 17, 2018 and titled WOUND THERAPY SYSTEM WITH WOUND VOLUME ESTIMATION, the entire disclosure of which is incorporated by reference herein.

Using the determined volume of the wound site circuit 204, the controller 118 may determine the volume of the dead space 119 at the wound site 114 (i.e. the portion of the interior space defined between the wound site 114 and the lower surface of the drape layer 117 that is not occupied by the wound dressing 112 and/or any instillation fluid 105/ other fluid) by subtracting or otherwise adjusting the calculated volume of the wound site circuit 204 to account

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for/factor in the known volumes of the downstream tubing portion **110b** and the portion of the downstream instillation tubing **108b** extending between the drape layer **117** and the instillation tubing valve **109** into the determination of the volume of the dead space **119** at the wound site **114**.

At step **1214**, an initial quantity of instillation fluid **105** that is to be delivered to the wound site **114** is calculated. According to various embodiments, the calculated initial quantity of instillation fluid **105** that is delivered to the wound site **114** may be based on the volume of the dead space **119** calculated by the controller **118** at step **1212**. For example, in some embodiments, the controller **118** may calculate the initial volume of instillation fluid **105** to be delivered to the wound site **114** by multiplying the volume of dead space **119** calculated at step **1212** by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid **105** delivered to the wound site **114** does not exceed the available space within the drape layer **117** (thereby minimizing inadvertent leakage from the wound dressing **112**/drape layer **117**. In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8. However, it is contemplated that the fluid instillation factor can have any value in various alternative embodiments.

As noted previously with reference to step **510**, in addition to being used to calculate a quantity of instillation fluid **105** to be delivered during any stage of treatment using NPWT system **100** and under any number of different conditions (e.g. allowing for the calculation of additional instillation fluid **105** to be delivered at step **516** even if the removed fluid canister **106** has been emptied, or entirely replaced with a different sized removed fluid canister **106** during the course of treatment), in some embodiments the NPWT system **100** may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site **114** over time. Accordingly, in some embodiments, at step **1216**, an initial baseline wound site **114** volume estimate may optionally be determined (via, e.g., a method as described with regards to FIG. **11**) and stored by the controller **118**, which may be used as a reference point against which future wound site **114** volume estimates may be compared to track healing progression of the wound site **114**.

At step **1218** the dead space **103** of the removed fluid canister **106** may be calculated to determine whether the dead space within the removed fluid canister **106** will be sufficient to collect any fluids **121** from the wound site **114** (including non-absorbed instillation fluid **105**) following the delivery of instillation fluid **105** at step **516**. As will be understood, in embodiments in which the NPWT system **100** has not been operated prior to the use of the NPWT system **100** at step **1202**, the volume of the removed fluid canister **106** should be empty, such that the dead space **103** of the removed fluid container **106** should be equal to the volume of the removed fluid canister **106**.

The dead space **103** of the removed fluid container may be calculated by subtracting the known volumes of conduit **136** and the upstream tubing portion **110a** from a volume of the removed fluid canister circuit **202** determined by subtracting the volume of the wound site circuit **204** calculated at step **1212** from a determined volume of the negative pressure circuit **200**. As will be understood, the volume of the negative pressure circuit **200** may be determined in a manner similar to the method via which the volume of the wound site circuit **204** is determined at step **1212**.

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Similar to step **514**, at step **1220** an alarm may be presented to a user if the initial volume of instillation fluid **105** to be delivered calculated at step **1214** exceeds the dead space **103** of the removed fluid canister **106**. Otherwise, if the volume of the initial instillation fluid **105** to be delivered does not exceed the dead space **103** of the removed fluid canister **106**, the calculated instillation fluid **105** is delivered to the wound site **114** at step **1222**.

As will be understood, in some NPWT system **100** embodiments in which a calibrated leak system **113** is provided both upstream and downstream of the tubing valve **111**, the NPWT system **100** may be operated to estimate wound site **114** volume and/or estimate dead space **103** at the removed fluid canister **106** according to a method that is the same as or similar to the method **600** of FIG. **6A** or a method that is the same as or similar to the method **1200** of FIG. **12**.

In other embodiments of NPWT system **100** in which both an upstream and downstream calibrated leak system **113** are provided, the NPWT system **100** may be operated to estimate wound site **114** volume and/or estimate dead space **103** at the removed fluid canister **106** according to a method that is the same as or similar to the method **600** of FIG. **6A** and a method that is the same as or similar to the method **1200** of FIG. **12**. For example, according to some embodiments, a modified method of operating a NPWT system **100** having both upstream and downstream calibrated leak systems **113** may include the steps of: monitoring pressure decay within the negative pressure circuit **200** (such as, e.g., described with reference to step **606** of the method **600** of FIG. **6** and/or step **1206** of the method **1200** of FIG. **12**); monitoring pressure decay within the removed fluid canister circuit **202** (such as, e.g., described with reference to step **610** of the method **600** of FIG. **6**); and monitoring pressure decay within the wound site circuit **204** (such as, e.g., described with reference to step **1210** of the method **1200** of FIG. **12**).

In such embodiments, the determination of wound site **114** volume based on direct measurement (e.g. using the method **1200** of FIG. **12**) may be compared to wound site **114** volume calculated based on indirect measurement (e.g. using the method **600** of FIG. **6A**) and the determination of dead space **103** at the removed fluid canister **106** based on direct measurement (e.g. using the method **600** of FIG. **6A**) may be compared to dead space **103** volume calculated based on indirect measurement (e.g. using the method **1200** of FIG. **12**). The controller **118** in such embodiments may be configured to generate an alarm or alert in response to a discrepancy between the direct and indirect measurements of wound site **114** volume and/or dead space **103** at the removed fluid canister **106**. By providing such redundancy to the wound site **114** and/or removed fluid canister **106** dead space **103** calculations, such embodiments may be configured to allow the NPWT system **100** to provide more accurate and reliable results.

As will be understood, according to various embodiments, the controller **118** may be programmed to allow the NPWT system **100** to determine volume relative to the wound site **114** using any or all of the methods described herein. Accordingly, while in some embodiments the controller **118** may optionally be preprogrammed to automatically determine a volume of instillation fluid **105** to be delivered according to a particular method (e.g. the method **900** embodiment illustrated in FIGS. **9A-9E**), the controller **118** may optionally also allow a user to select any of the other modes of calculating a volume relative to the wound site **114** based on whether the user desires to, e.g.: remove fluid **121** from the wound site **114** prior to instillation of additional instillation fluid **105**; verify sufficient dead space **103a** in the

removed fluid canister **106** prior to determining the dead space at the wound site **114**; verify sufficient dead space **103b** in the removed fluid canister **106** prior to the instillation of a calculated quantity of additional instillation fluid **105** to be delivered to the wound site **114**; monitor changes in the wound site **114** volume to track healing progression; etc.

Tuberset Module

Although in some arrangements, some or all of the calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109**, or other NPWT system **100** components may be configured to be manually operated/actuated/utilized by a user, as noted above, according to various embodiments, some or all of these components may alternatively be configured to be operated/actuated/utilized by the controller **118**, without requiring any user assistance to do so. In such a manner, implementation of the system for/method of determining a volume of instillation fluid to be delivered to a wound site, estimating a volume of a wound, monitoring healing progression of a wound, and/or other use of the NPWT system **100** may be fully automated using the controller **118**, allowing for easier use of the NPWT system **100**.

By providing the NPWT system **100** with an automated manner via which controller **118** may control or otherwise interact with one or more of the calibrated leak system **113**, tubing valve **111**, instillation tubing valve **109**, and/or other component(s) of the NPWT system **100**, the tuberset module **300** may increase the accuracy of the NPWT system **100**. For example, in light of the ability of the controller **118** to utilize the tuberset module **300** to independently actuate (i.e. without user intervention) the operation of the calibrated leak system **113**, the tubing valve **111** and/or the instillation tubing valve **109** elements, the controller **118** may be configured to increase the rate at which data related to instillation fluid volume estimation, wound site volume estimation, wound site **114** healing progression monitoring, and/or other functions of the NPWT system **100** is gathered. By increasing the data points used to provide such information, the reliability of the information provided by the controller **118** may thereby be increased. Similarly, the obviation or minimization of user involvement provided by the tuberset module **300** may facilitate (and thereby increase the likelihood of) the usage of a dual calibrated leak system **113** arrangement as described with reference to FIG. **12** above, thus also increasing the reliability of the NPWT system **100**.

In general, the tuberset module **300** comprises a housing element **304** containing a power source **301**, a communications interface **302**, and one or more actuatable elements **303** configured to be controlled by the controller **118**. In some embodiments, the tuberset module **300** may optionally also comprise one or more additional non-actuatable elements **305**, such as, e.g., pressure sensor **115a** and/or pressure sensor **115b**. According to embodiments in which calibrated leak system **113** is not defined by a vent valve **113b** and only comprises a non-actuatable vent **113a**, the non-actuatable element(s) **305** may comprise such a calibrated leak system **113** formed without a vent valve **113b**.

As will be understood, according to some embodiments, some or all of the actuatable elements **303** may be configured so as to be self-actuating. In some such embodiments, the actuatable element **303** may comprise an internal actuator that is in operably connected (via a wired, wireless, or any other type of connection) to the power source **301** and/or

communications interface **302** of the tuberset module **300**, via which instructions received from the controller **118** and/or power are relayed to the actuator of the actuatable element **303**. In other such embodiments, such self-actuation actuatable element **303** may individually comprise one or both of a power source and/or communications interface (in addition to the power source **301** and/or communications interface **302** of the tuberset module). In such embodiments, the instructions from the controller **118** may be received directly by the communication interface of the actuatable element **303** from the controller, or may be received indirectly by the communication interface of the actuatable element **303** from the communication interface **302** of the tuberset module **300**. In other embodiments, some or all of the actuatable elements **303** may be configured to be actuated by any number of different types of, or combinations of known actuators that are contained by the housing element **303**, with the actuators of the housing element **303** being configured to effectuate actuation of the one or more actuatable elements **303** in response to instructions received from the controller **118**.

The power source **301** may comprise any number of, and combination of, sources of energy that are configured to supply sufficient energy to the communications interface **302**, actuatable element(s) **303** and/or non-actuatable elements **305** contained by the housing element **304** as required for operation of the NPWT system **100**. In some embodiments in which some or all of the tuberset module **300** is integrated into the therapy device **102**, the power provided by the power source **301** of the housing element **304** may comprise a power source of the therapy device **102**.

The communications interface **302** may comprise any number of, and combination of, wired and/or wireless connections via which the tuberset module **300** may receive communications (such as, e.g., actuation signals) from the controller **118**. According to some embodiments, the communications interface **302** may optionally also be configured to send information to and/or receive information from the controller **118**, other tuberset module **300** housing elements **304** (such as, e.g., information regarding the status of the one or more actuatable elements **303** and/or non-actuatable elements **305** of the tuberset module **300**), and/or other sources. In some embodiments in which some or all of the tuberset module **300** is integrated into the therapy device **102**, the communications interface **302** of the housing element **304** may be defined by a portion of a communications interface of the therapy device **102**.

According to some arrangements, tuberset module **300** may be defined by a single housing element **304**, with each of actuatable elements **303** (e.g. upstream and/or downstream calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109**, etc.) and/or non-actuatable elements **305** that are to be controlled/utilized by the controller **118** forming a part of the single, integral housing element **304**. In other embodiments, the tuberset module **300** may be defined by a plurality of separate and distinct housing elements **304**, with each housing element **304** formed with one or more of the various actuatable elements **303** and/or non-actuatable element **305** that are to be controlled/utilized by the controller **118**.

According to various arrangements, the one or more housing elements **304** defining the tuberset module **300** may be provided as a separate, discrete, individual component of the NPWT system **100**, which may subsequently be attached to or otherwise incorporated into one or more of the other components of a new or existing NPWT system **100**. In other arrangements, some or all of the one or more housing

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elements 304 defining the tubeset module 300 may be provided as an integrated part of one or more of the other components of the NPWT system 100.

For example, in some arrangements, some or the entirety of the tubeset module 300 may be integrated into the wound dressing 112, with the portion of the tubeset module 300 provided with the wound dressing 112 being configured to be removed from the NPWT system 100 with the removal of the wound dressing 112. Upon removal of the integrated wound dressing 112/tubeset module 300, the entire wound dressing 112/tubeset module 300 may be disposed of. Alternatively, the tubeset module 300 may be removed from the wound dressing 112 prior to disposal of the wound dressing 112 and optionally reused with another wound dressing 112 and/or other NPWT system 100 component.

In other arrangements, some or the entirety of the tubeset module 300 may be integrated into the removed fluid canister 106, with the portion of the tubeset module 300 provided with the removed fluid canister 106 being removed from the NPWT system 100 with the removal of the removed fluid canister 106 from the NPWT system 100. In some such embodiments, the tubeset module 300 may be monolithically formed with the removed fluid canister 106, while in other embodiments; the tubeset module 300 may be non-integrally formed with the removed fluid canister 106.

According to another arrangement, the tubeset module 300 may be configured to be integrated in-line with one or both of the tubing 108 and/or 110. In such embodiments, attachment adapters 400 may be provided on one or both of the tubeset module 300 and/or tubing 108 and/or 110 to facilitate a fluid tight attachment of the tubeset module 300 to the tubing 108 and/or 110. According to some embodiments, the attachment adapters 400 may be provided on the tubeset module 300, with the attachment adapters 400 being configured to be able to form a fluid tight attachment directly with one or both of the tubing 108 and/or 110, allowing NPWT systems formed without a tubeset module 300 and/or calibrated leak system 113, tubing valve 111 and/or the instillation tubing valve 109 to be retrofitted with a tubeset module 300 so as to provide a NPWT system 100 as disclosed herein.

In some arrangements, some or the entirety of the tubeset module 300 may be integrated into the housing of the therapy device 102. In such embodiments, the efficiency of using the NPWT system 100 may be increased, as by incorporating a tubeset module 300 including some or all of the calibrated leak system 113, tubing valve 111 and/or the instillation tubing valve 109 into the housing of the therapy device 102, the time to setup the NPWT system 100 may be reduced as compared to the time that would otherwise be required to setup up a NPWT system 100 in which some or all of the calibrated leak system 113, tubing valve 111 and/or instillation tubing valve 109 were provided as separate and discrete components of the NPWT system 100. Additionally, by incorporating the tubeset module 300 into the housing of the therapy device 102, a NPWT system 100 as described herein may be provided irrespective of the particular tubing, removed fluid canister, wound dressing, or other component (s) that are provided to define a NPWT system 100 for treatment of a wound site 114.

Referring to FIGS. 13-16, various embodiments of a tubeset module 300 configured to allow for partially or fully automated control of the NPWT system 100 using the controller 118 are shown. As will be understood, although reference has been made to the controller 118 being provided as part of the therapy device 102, it is to be understood that, according to various arrangements, the controller 118

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may be provided separate and remote from the therapy device 102 and/or NPWT system 100 (e.g., by a remote medical provider). In such embodiments, the remotely provided controller 118 may be configured to communicate directly with the tubeset module 300 and/or indirectly with the tubeset module 300 via a communications interface provided by the therapy device 102.

As illustrated by the NPWT system 100 embodiment of FIG. 13, in some arrangements, the tubeset module 300 is provided as a single, integrated housing element 304 containing actuatable elements 303 comprising a calibrated leak system 113, a tubing valve 111 and an optional instillation tubing valve 109. According to some embodiments, one or both of the tubing valve 111 and the optionally provided instillation tubing valve 109 may comprise the same or distinct clamps. As shown in FIG. 13, also contained within the housing element 304 is a power source 301 configured to actuate the actuatable element 303 in response to instructions being received from the controller 118 via the communications interface 302.

Although in the embodiment illustrated in FIG. 13 a single, integral tubeset module 300 is shown as being in-line with both tubing 108 and tubing 110, according to other arrangements (now shown) it is to be understood that a first housing element 304 comprising the calibrated leak system 113 and a tubing valve 111 may be provided in-line with tubing 110, while an optional, second housing element 304 comprising instillation tubing valve 109 may be provided in-line with tubing 110.

As illustrated by the NPWT system 100 of FIG. 13, in some embodiments, the tubeset module 300 may be formed integrally with upstream tubing portion 110a and/or upstream instillation tubing 108a. According to some such embodiments, the upstream tubing portion 110a and/or upstream instillation tubing 108a with which the tubeset module 300 is integrally formed may in turn be formed integrally with the therapy device 102. In such embodiments, the tubeset module 300 is configured to be removably attached to the downstream tubing portion 110b and/or downstream instillation tubing 108b formed integral with the wound dressing 112, such that following use of the NPWT system 100 with a first wound dressing 112, the therapy device 102 with integrated upstream tubing portion 110a and/or upstream instillation tubing 108a and tubeset module 300 may be reused with a new, second wound dressing 112. In other embodiments, such as, e.g., illustrated by the NPWT system 100 of FIG. 14, some or all of the tubeset module 300 may alternatively be formed integrally with the wound dressing 112, with the tubeset module 300 being configured to be removed from the NPWT system 100 with the removal of the wound dressing 112.

Referring to the NPWT system 100 of FIG. 15, according to some embodiments, the tubeset module 300 may comprise a first housing element 304 containing a calibrated leak system 113 and an optional instillation tubing valve 109, pressure sensor 115a and/or pressure sensor 119 positioned in-line with the tubing 108 and/or 110. A second housing element 304 comprising a tubing valve 111 may be spaced from the first housing element 304. As shown in FIG. 14, according to some arrangements, the second housing element 304 may be integrated into the removed fluid canister 106. In other embodiments, the second housing element 304 may alternatively be incorporated into the therapy device 102, or may be provided at a second location in-line with the tubing 110.

Referring to FIG. 16A, a block diagram of a NPWT system 100 according one embodiment is shown. As illus-

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trated by the NPWT system 100 of FIG. 16A, according to some embodiments, fluid communication between some or all of the negative pressure circuit 200 and the ambient environment may be provided by a purge valve system 450 provided along the instillation tubing 108 as an alternative to, or in addition to, calibrated leak 113. According to various embodiments, the purge valve system 450 may comprise a structure similar to that of calibrated leak system 113 (including calibrated leak system 113 embodiments comprising any combination of vent 113a, vent valve 113b and/or flow detector 113c components).

As also illustrated by FIG. 16A, in such NPWT system 100 embodiments, the tubing valve 111 and/or instillation tubing valve 109 may be replaced by a valve assembly 460 that is fluidly attached to the tubing 110 at a junction between the upstream tubing portion 110a and downstream tubing portion 110b and that is attached to the instillation tubing assembly at a junction between the upstream tubing 108a and downstream tubing 108b, and which is actuable to a variety of positions. In a first position, the valve assembly 460 may permit fluid flow from the pneumatic pump 120 to the wound site 114 via tubing 110 and from the instillation pump 104 to the wound site 114 via instillation tubing 108. In a second position, the valve assembly 460 may permit fluid flow from the pneumatic pump 120 to the wound site 114 via tubing 110 while blocking fluid flow from the instillation pump 104 to the wound site 114 via instillation tubing 108. In a third position, the valve assembly 460 may block fluid flow from the pneumatic pump 120 to the wound site 114 via tubing 110 while permitting flow from the instillation pump 104 to the wound site 114 via instillation tubing 108. In a fourth position, the valve assembly 460 may fluidly connect the upstream tubing portion 110a with the upstream instillation tubing 108a, resulting in the isolation of the downstream tubing portion 110b, the downstream tubing 108, and the wound dressing 112 from the remainder of the therapy device 102.

When in the first configuration, the valve assembly 460 defines a negative pressure circuit 200 is defined by the tubing 136, the fluid canister 106, the tubing 110, the wound site 114 and the portion of the instillation tubing extending between the wound site 114 and the purge valve 450. When in the fourth configuration, the valve assembly 460 defines a removed fluid canister circuit 202 is defined by the tubing 136, the fluid canister 106, the upstream tubing portion 110a, and the portion of the upstream tubing 108a extending between the valve assembly 460 and the purge valve 450 and a wound site circuit 204 defined by the downstream tubing portion 110b, the wound site 114, and the downstream tubing 108b.

As will be understood, the valve assembly 460 and the purge valve 450 of the NPWT system 100 of FIG. 16A may be operated in a manner similar to the operation of the tubing valve 111, calibrated leak, and/or instillation tubing valve 109 as described with reference to any of the methods described herein for determining wound site volume, estimating a volume of fluid to be instilled, monitoring wound healing progression, or performing any other functions using the NPWT system 100.

Referring to FIG. 16B, a tubeset module 300 configured for used with a NPWT system 100 incorporating a purge valve 450 (such as, e.g. shown in FIG. 16A) is shown according to one embodiment. As illustrated by FIG. 16B embodiments in which the purge valve 450 is provided as a discrete component of the therapy device 112 capable of being automatically actuated by the controller 118, the tubeset module 300 may comprise only a single actuable

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element 303, defined by valve assembly 460. As will be understood, in other embodiments, (such as, e.g. where the purge valve 450 provided as part of the therapy device is not automatically actuable by the controller 118), the purge valve 450 may be provided as a part of a tubeset module 300 that is partially or entirely integrated into the therapy device 112.

Although in the NPWT system 100 embodiment illustrated in FIG. 16A the purge valve 450 is illustrated as being provided as part of the therapy device 112, according to other embodiments, the purge valve 450 may be alternatively, or additionally, provided as a part of the upstream tubing 108a. In such embodiments, the purge valve 450 may accordingly be provided as an actuable element 303 of the tubeset module 300.

As will be understood, the controller 118 may be configured to effectuate any number of different operations using the NPWT system 100 based on the selective, fully automated actuation of/interaction with some or all of the actuable elements 303 and/or non-actuable elements 305 of a tubeset module 300 according to any number of different methods and protocols. According to various embodiments, the order and/or combination of instructions transmitted by controller 118 to the tubeset module 300 and/or the information received by the controller 118 from the tubeset module 300 may be configured to automatically operate the tubeset module 300 in a manner that allows the controller 118 to automatically effectuate one or more of the methods 500, 600, 800, 900, 1000, 1100, 1200, etc. described herein.

Represented in FIG. 17 is one method 1700 via which the controller 118 may utilize a tubeset module 300 containing actuable elements 303 comprising a tubing valve 111, instillation tubing valve 109, and calibrated leak system 113 and non-actuable element(s) 305 comprising one or both of pressure sensor 115a and/or pressure sensor 115b to automatically control the NPWT system 100 to determine dead space 119 at a wound site 114 according to a method such as, e.g., described with reference to the method 500 of FIG. 5 and the method 600 of FIG. 6A.

At step 1701, in response to the controller 118 being initiated to determine dead space at a wound site 114 (such as, e.g., at step 506 of the method 500 of FIG. 5), the controller 118 may initiate communication with the tubeset module 300 to confirm that the instillation tubing valve 109 and vent valve 113b of the calibrated leak system 113 are closed, and that the tubing valve 111 is opened. If the instillation tubing valve 109 and/or vent valve 113b are open, the controller 118 may instruct the tubeset module 300 to effectuate actuation of the instillation tubing valve 109 and/or vent valve 113b into a closed configuration. Similarly, if the tubing valve 111 is detected by the controller 118 as being closed, the controller 118 may transmit instructions to the tubing valve 111 via the communications interface 302 to actuate opening of the tubing valve 111.

Once the controller 118 has received, via the communications interface 302, confirmation that the instillation tubing valve 109 and vent valve 113b are closed and the tubing valve 111 is open, the controller 118 may be configured to initiate operation of the pneumatic pump 120 to apply negative pressure to the negative pressure circuit 200 (such as, e.g., described with reference to step 604 of the method 600 of FIG. 6A). During operation of the pneumatic pump 120, the controller 118 at step 1703 may be configured to receive from the pressure sensor 115a and/or pressure sensor 115b pressure readings corresponding to the pressure within the negative pressure circuit 200. As will be understood, the

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pressure readings received by the controller 118 at step 1703 may be received continuously, at predetermined intervals, and/or in response to specific requests for pressure readings transmitted by the controller 118 to the tubeset module 300 via communications interface 302.

In response to receiving pressure readings from the tubeset module 300 indicative of the pressure within the negative pressure circuit 200 having reached a threshold pressure, the controller 118 at step 1705 may be configured to stop operation of the pneumatic pump 120 and transmit to the tubeset module 300 an actuation signal configured to cause the opening of the vent valve 113b.

At step 1707, the controller 118 may be configured to receive from the pressure sensor 115a and/or pressure sensor 115b pressure readings corresponding to pressure decay within the negative pressure circuit 200, such as, e.g., described with reference to step 606 of FIG. 6A. The pressure readings received by the controller 118 at step 1707 may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller 118 to the tubeset module 300 via communications interface 302.

Once the controller 118 has received pressure readings from the tubeset module 300 indicative of the pressure within the negative pressure circuit 200 having reached a threshold pressure (such as, e.g., ambient pressure), the controller 118 at step 1709 may be configured to effectuate, using the tubeset module 300, the actuation of the closing of the tubing valve 111 and the vent valve 113b in advance of the application of negative pressure to the resultant removed fluid canister circuit 202 (such as, e.g., during step 608 of the method 600 of FIG. 6A).

At step 1711, the controller 118 once again may be configured to receive pressure readings from the tubeset module 300. The pressure readings received by the controller 118 at step 1711 may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller 118 to the tubeset module 300 via communications interface 302. In response to receiving pressure readings from the tubeset module 300 indicative of the pressure within the removed fluid canister circuit 202 having reached a threshold pressure, the controller 118 at step 1713 may be configured to stop operation of the pneumatic pump 120 and transmit to the tubeset module 300 an actuation signal configured to cause the opening of the vent valve 113b.

At step 1715, the controller 118 may be configured to receive from the pressure sensor 115a pressure readings corresponding to pressure decay within the removed fluid canister circuit 202, such as, e.g., described with reference to step 610 of FIG. 6A. The pressure readings received by the controller 118 at step 1715 may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller 118 to the tubeset module 300 via communications interface 302.

According to some embodiments, following step 1715, at step 1717, the controller 118 may be configured to actuate, using the tubeset module 300, the opening of the instillation tubing valve 109, in advance of the instillation of instillation fluid to the wound site 114 (such as, e.g., described with reference to step 516 of the method 500 of FIG. 5 and/or step 622 of the method 600 of FIG. 6A).

Wound Therapy System with Internal Alternating Orifice

Referring now to FIG. 18, an NPIWT system 2100 is shown, according to an exemplary embodiment. The

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NPIWT system 2100 includes a dressing 2102 fluidly communicable with a canister 2104 via first tubing 2106 and a therapy unit 2108 coupled to the canister 2104. As shown in FIG. 18, the NPIWT system 2100 also includes an instillation fluid source 2110 fluidly communicable with the dressing 2102 via the therapy unit 2108 and second tubing 2112. The NPIWT system 2100 and components thereof may correspond to, may be implemented with, may be combined, and/or may otherwise provide various features of the systems and methods described above with reference to FIGS. 1-17. It should be understood that the present disclosure contemplates various combinations of the embodiments shown in the drawings.

The dressing 2102 is shown as applied to a wound bed 2114. The dressing 2102 includes a drape 2116 sealed over the wound bed 2114 and a foam layer 2118 positioned between the drape 2116 and the wound bed 2114. In various embodiments, the dressing 2102 may include various layers and features. The drape 2116 may be made of a substantially air-impermeable material (e.g., a polyurethane-based material) and may include an adhesive border that allows the drape to be sealed to a patient's skin around the wound bed 2114. The foam layer 2118 may include a manifold layer that allows airflow therethrough and facilitates the distribution of negative pressure across the wound bed 2114. A wound space 2120 that includes the open volume (i.e., through which air may flow) in the foam layer 2118 and otherwise situated between the drape 2116 and the wound bed 2114 is thereby established.

The first tubing 2106 extends from the dressing 2102 to the canister 2104. A cross-section of the first tubing 2106 is shown in FIG. 21, according to an exemplary embodiment. As described in detail with reference to FIG. 21, the first tubing 2106 includes an inner lumen 2400 and one or more outer lumens 2402. The inner lumen 2400 provides for the flow of fluid from the wound space 2120 into the canister 2104. The one or more outer lumens 2402 are fluidly communicable with a pressure sensor 2124 to facilitate measurement of the pressure at the wound space 2120. The one or more outer lumens 2402 are also fluidly communicable with a valve 2126 as described below. It should be understood that, while described as inner and outer in the examples herein, any geometrical arrangement of multiple lumens may be used in various embodiments. A connection pad (e.g., low pressure interface) 2121 is coupled to the drape 2116 and facilitates connection of the first tubing 2106 to the dressing 2102.

The canister 2104 is configured to collect wound exudate (e.g., fluid, other debris) removed from the wound space 2120 via the first tubing 2106. The canister 2104 is fluidly communicable with the wound space 2120 via the first tubing 2106. The canister 2104, the first tubing 2106, and the dressing 2102 thereby define a sealed space that includes the wound space 2120.

The therapy unit 2108 is coupled to the canister 2104 and includes a pneumatic pump 2122 fluidly communicable with the sealed space, a sensor 2124 configured and positioned to measure pressure in the sealed space, a valve 2126 positioned between the sealed space and an environment, a user interface 2128, and an instillation pump 2130 coupled to the second tubing 2112. The therapy unit 2108 also includes a control circuit 2132 communicably and operably coupled (e.g., capable of exchanging electronic signals with) the pneumatic pump 2122, the sensor 2124, the valve 2126, the user interface 2128, and the instillation pump 2130.

The pneumatic pump 2122 is controllable by the control circuit 2132 and operable to pump (e.g., draw, remove) air

from the canister **2104**, the first tubing **2106**, and the wound space **2120** (i.e., from the sealed space). The pneumatic pump **2122** may thereby create a negative pressure in the sealed space relative to atmospheric pressure, for example between 25 mmHg and 175 mmHg. The pneumatic pump **2122** may create a pressure differential that causes fluid and debris to be drawn out of the wound space **2120**, through the first tubing **2106**, and into the canister **2104**.

The sensor **2124** is positioned and configured to measure the pressure in the sealed space. As shown in FIG. **18**, the pressure sensor **2124** is positioned to measure pressure via one or more outer lumens **2402**. In other embodiments a sensor **2124** may be included to measure pressure elsewhere in the sealed space (e.g., in the canister **104**). The sensor **2124** provides pressure measurements to the control circuit **2132** (e.g., digital values, analog signals). The control circuit **2132** may be configured to receive the pressure measurements from the sensor **2124** and use the pressure measurements in a control loop to generate control signals for the pneumatic pump **2122** that cause the pneumatic pump **2122** to maintain a desired pressure in the sealed space or provide a desired pattern of pressure in the sealed space.

The user interface **2128** may include a display screen, a touch screen, a speaker, a button, a switch, or any other element capable of providing information to a user or receiving input from a user. In some embodiments, the control circuit **2132** is configured to generate a graphical user interface and cause the graphical user interface to be displayed on the user interface **2128**. The graphical user interface may include various information about the NPIWT provided by the NPIWT system **2100**, for example relating to the pressure in the sealed space, an amount of instillation fluid to be provided, a schedule of negative pressure and instillation cycles, and/or a size of the wound space **2120**. The user interface **2128** may allow a user to input commands and settings relating to the operation of the therapy unit **2108**. The control circuit **2132** may receive such inputs from the user interface **2128** and control the therapy unit **2108** in accordance with the inputs.

The instillation pump **2130** is configured to cause instillation fluid to be transported from the instillation fluid source **2110** to the wound space **2120** via second tubing **2112**. The instillation pump **2130** may be controllable by the control circuit **2132** to provide a desired amount of the instillation fluid to the wound space **2120**, provide instillation fluid to the wound space **2120** at a desired rate, prevent instillation fluid from flowing to the wound space **2120**, or otherwise control the flow of instillation fluid to the wound space **2120**. The instillation pump may include a peristaltic pump or some other type of pump.

The valve **2126** is controllable between an open position and a closed position. As shown in FIG. **18**, the valve **2126** is located at an interior of the therapy unit **2108** in pneumatic communication with a surrounding environment (e.g., ambient air) via a vent **2134** positioned along an exterior of the therapy unit **2108**. The valve **2126** is also shown as communicable with the one or more outer lumens **2402** of the first tubing **2106**. A filter **2138** is located between the canister **2104** and the pump **2122**. When the valve **2126** is in the open position, air may flow between the surrounding environment and the sealed space via the filter **2138**. When the valve **2126** is in the closed position, air is prevented from flowing therethrough. As shown in FIGS. **19-20** and described in detail with reference thereto, the valve **2126** may be a solenoid valve. In various other embodiments, other types of valves may be included. As described in detail below, the valve **2126** may be controllable to allow a sudden

surge ("blast") of air therethrough in a manner intended to clear a blockage in the one or more outer lumens **2402** of the first tubing **2106**. The valve **2126** may also be controllable to allow a controlled rate of airflow therethrough to facilitate determination of a volume of the wound space **2120**.

The filter **2138** is configured to prevent contaminants from moving from the surrounding environment to the wound space **2120** via the valve **2126** and the one or more outer lumens **2402**. The filter **2138** thereby protects the wound **2114** from infection or other complications. The filter **2138** restricts the rate of flow of air from the surrounding environment into the sealed space through the filter **2138** (e.g., by creating a pressure drop across the filter **2138** due to the filter media, contaminants trapped in the filter media, etc.) to a maximum of a restriction rate of the filter **2138**. The restriction rate may be difficult to ascertain, may vary over time, or may be different in different instances of the filter **2138** (i.e., differing across multiple therapy units **108**).

In the embodiments shown, the restriction rate of the filter is less than a typical rate of airflow through the valve **2126** when the valve **2126** is held in the open position for an extended amount of time (e.g., 500 milliseconds or greater). Accordingly, the difficulty in determining the restriction rate of the filter **2138** leads to a difficulty in determining a rate of airflow into the sealed space when the valve **2126** is held in the open position for an extended amount of time.

The control circuit **2132** is configured to control the operation of the therapy device **2108**. For example, as described in detail below, the control circuit **2132** is configured to control the pneumatic pump **2112** to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve **2126** to provide a controlled leak to the sealed space, receive pressure measurements from the sensor **2122**, determine a volume of the wound space **2120** based on the pressure measurements, and customized a wound therapy based on the volume of the wound space **2120**. In some embodiments, the control circuit **2132** is also configured to detect a potential blockage of a lumen of the first tubing **2106**, control the valve **2126** to the open position to allow a blast of air therethrough, keep the valve **2126** open while the blast of air clears the blockage, and control the valve **2126** to return to the closed position. These and other features of the control circuit **2132** are described in detail below.

Referring now to FIGS. **19-20**, cross-sectional views of the valve **2126** are shown, according to exemplary embodiments. In the embodiments shown, the valve **2126** is a solenoid valve. FIG. **19** shows the valve **2126** in the closed position and FIG. **20** shows the valve in the open position. It should be understood that FIGS. **19-20** show one of many possible embodiments of the valve **2126**.

The valve **2126** includes an inlet **2200** pneumatically communicable with the surrounding environment via the vent **2134**, an outlet **2202** pneumatically communicable with the sealed space via channel **2136**, a solenoid **2206**, a plunger **2204** extending axially through the solenoid **2206** and substantially centered in the solenoid, a stopper **2205** coupled to the plunger **2204**, and a spring **2208** coupled to the plunger **2204**. The solenoid **2206** has a positive lead **2210** and a negative lead **2212** shown as operably coupled (e.g., conductively coupled) to the control circuit **2132**.

The solenoid **2206** includes a coil of wire through which the plunger **2204** extends. When a current flows through the solenoid (e.g., when a voltage differential is applied across the solenoid **2206**, a magnetic field is created in the solenoid **2206**. The magnetic field is substantially aligned with a central axis of the solenoid. The plunger **2204** is made of a

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magnetic material, such that the magnetic field causes movement of the plunger **2204** when voltage is applied across the solenoid **2206**.

As shown in FIG. **19**, a voltage of approximately zero volts is applied across the solenoid **2206**. That is, the control circuit **2132** prevents a voltage difference between the positive lead **2210** and the negative lead **2212**. Accordingly, approximately zero current is created in the solenoid **2206**, and approximately zero magnetic field is created by the solenoid **2206**. The spring **2208** exerts a force on the plunger **2204** that holds the stopper **2205** adjacent the inlet **2200**. The stopper **2205** prevents air from entering the valve **2206** via the inlet **2200**. Airflow from the vent **2134** to the channel **2136** is thereby prevented (i.e., the valve **126** is in the closed position).

As shown in FIG. **20**, a non-zero voltage is applied across the solenoid (e.g., approximately 5 volts). That is, the control circuit **2132** provides a control signal to the valve **2126** by creating a voltage differential between the positive lead **2210** and the negative lead **2212** of the solenoid **2206**. It should be understood that, in various embodiments, various values of the non-zero voltage may be required to operate the valve **2126**. When the control circuit **2132** provides a non-zero voltage to the valve **2126** (i.e., across the solenoid **2206**), a magnetic field is created that causes the plunger **2204** to compress the spring **2208** and move the stopper **2205** away from the inlet **2200**. Air may then flow from the vent **2134** through the valve **2126** to the sealed space (i.e., the valve **2126** is in the open position).

When the non-zero voltage is removed (i.e., when the voltage differential between the positive lead **2210** and the negative lead **2212** is brought to approximately zero), the magnetic field goes to zero and the spring **2208** forces the plunger **2204** and stopper **2205** back to the closed position shown in FIG. **19**. Thus, the valve **2126** may be controlled to repeatedly alternate between the closed position shown in FIG. **19** and the open position shown in FIG. **20** by alternating between an approximately zero voltage and an approximately non-zero voltage. Example voltage patterns for controlling the valve **2126** to provide a controlled rate of airflow therethrough are described in detail below.

Referring now to FIG. **21**, a cross-sectional view of the first tubing **2106** is shown, according to an exemplary embodiment. In the embodiment shown, the first tubing **2106** includes an inner lumen **2400** and four outer lumens **2402**. That is, the first tubing **2106** is shown to include five separate lumens (e.g., channels, bores, pathways) through which air, fluid, and/or other debris may flow. Preferably, fluid and debris flows primarily through the inner lumen **2400**, while air flows through the outer lumens **2402**. At or near the canister **2104**, the path of the outer lumens **2402** is separated from the path of the inner lumen **2400** as shown in FIG. **18**. The inner lumen **2400** is connected to the inner volume of the canister to allow fluid and debris from the wound space **2120** to be collected in the canister. The outer lumens **2402** are connected to the sensor **2124** to facilitate the measurement and monitoring of pressure at the wound space.

The connection pad **2121** may include grooves and other physical features configured to direct fluid and debris towards the inner lumen **2400** and away from the outer lumens **2402**. However, fluid and debris may occasionally reach one or more of the outer lumens **2402** and cause a blockage of the one or more of the outer lumens **2402**. A blockage of the inner lumen **2400** may also occur. As described below with reference to FIG. **22**, the valve **2126** may be controlled to allow a blast of air to be released

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through the outer lumens **2402** to clear the fluid or other blockage from the outer lumens **2402**, i.e., by pushing air and fluid back towards the dressing **2102** and out of the outer lumens **2402**.

Referring now to FIG. **22**, a flowchart of a process **2500** for clearing blockages in the first tubing **2106** is shown, according to an exemplary embodiment. At step **2502**, a potential blockage of one or more outer lumens **2402** is determined. As one example, the control circuit **2132** may detect a blockage based on pressure measurements from the sensor **2124**. As another example, the control circuit **2132** assumes a potential blockage exists after a predetermined time period, i.e., such that steps **2504-2508** are triggered at a predetermined frequency.

At step **2504**, the valve **2126** is opened to allow a blast of air therethrough. For example, the control circuit **2132** may provide a non-zero voltage to the solenoid **2206** of the valve **2126**. The control circuit **2132** may cause the valve **2126** to be held in the open position for an extended time period, i.e., longer than the periods shown in FIG. **22** and discussed with reference thereto below. For example, in one embodiment the non-zero voltage is provided for approximately 500 milliseconds to hold the valve **126** open for approximately 500 milliseconds. When the valve **2126** is held open, a blast of air may flow therethrough at a high airflow rate due to the pressure differential between the surrounding environment (ambient air) and the sealed space. This blast of air may flow into a blocked outer lumen **2402** and push any blockage out of the first tubing **2106** towards the dressing **2106**. Blockages in the outer lumens **2402** may thereby be periodically cleared to allow free airflow through the outer lumens **2402**, for example to ensure that the measurements of the pressure sensor **2124** accurately represent the pressure at the wound space **2120**.

At step **2506**, the solenoid valve **2508** is closed. For example, the control circuit **2132** causes approximately zero voltage to be provided across the solenoid **2206**. Airflow from the environment to the sealed space is prevented. The pneumatic pump **2122** may be operated to reestablish a desired negative pressure at the wound space **2120**.

Referring now to FIG. **23**, a flowchart of a process **2600** for wound volume determination and wound therapy customization is shown, according to an exemplary embodiment. The process **2600** may be carried out by the NPIWT system **100** of FIG. **18**.

At step **2602**, a sealed space defined by the wound **2114**, the dressing **2102**, the first tubing **2106**, and the canister **2104** is established. The sealed space includes the wound space **2120**. In other words, the dressing **2102** is applied to the wound **2114** with the drape **2116** sealed over the wound **2114** and the foam layer **2118** (or other layers included in the dressing **2102** in various embodiments) to define the wound space **2120**. The first tubing **2106** is coupled to the drape **2116** in fluid communication with the wound space **2120** via the connection pad **2121**. The first tubing **2106** is also coupled to the canister **2104** in fluid communication with the canister **2104**.

At step **2604**, the pneumatic pump **2122** is operated to draw a negative pressure in the sealed space. That is, the control circuit **2132** provides a control signal to the pneumatic pump **2122** that causes the pneumatic pump to remove air from the sealed space. The control circuit **2132** may receive pressure measurements from pressure sensor **2124** and cause the pneumatic pump **2122** to cease operation when a desired negative pressure is achieved (e.g., -125 mmHg) and/or otherwise control the pneumatic pump **2122**

based on the pressure measurements to provide a desired negative pressure or pattern of desired negative pressures.

At step **2606**, the valve **2126** is repeatedly opened and closed (e.g., "cycled") to allow a controlled rate of airflow therethrough. The control circuit **2132** may provide a control signal to the valve **2126** that causes the valve **2126** to repeatedly open and close. For example, in an embodiment where the valve **2126** is a solenoid valve, for example as shown in FIGS. **19-20**, at step **2606** the control circuit **2132** provides a voltage pattern to the valve **2126**. That is, the control circuit **2132** may repeatedly alternate a voltage differential across between the positive lead **2210** and the negative lead **2212** between approximately zero volts and a non-zero voltage (e.g., approximately five volts). For example, the voltage pattern may include a step function that repeatedly steps between approximately zero voltage and the non-zero voltage.

In one example, as shown in FIG. **24** and described in detail with reference thereto, the voltage pattern may include a repeating pattern of approximately 400 milliseconds at the non-zero voltage, approximately 100 milliseconds at approximately zero voltage, approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage. The voltage pattern may thereby cause the valve **2126** to alternate between the open position and the closed position with a period of approximately 500 milliseconds. In some embodiments, the voltage pattern may include a first set of two repetitions of the repeating pattern, followed by approximately one second at approximately zero voltage, followed by a second set of two repetitions of the repeating pattern. In preferred embodiments, the non-zero voltage is provided, in each repetition, for no more than a maximum continuous duration of approximately 500 milliseconds.

By controlling the valve **2126** to repeatedly alternate between the open position and the closed position, a controlled rate of airflow is allowed therethrough across the repetitions. That is, a lower rate of airflow is allowed through the valve **2126** as compared to holding the valve **2126** open for an extended or indefinite amount of time (e.g., as described for process **500**), for example 500 milliseconds or longer. The controlled rate may be customized by altering the voltage pattern. Additionally, the controlled rate may be known based on the voltage pattern. For example, the controlled rate may be predetermined by bench testing for each of one or more voltage patterns. In preferred embodiments, the controlled rate is less than a restriction rate of the filter **2138**.

At step **608**, the pressure in the sealed space is measured as the negative pressure in the sealed space decays towards ambient pressure (i.e., approaches approximately atmospheric pressure). The controlled airflow through the valve **2126** allows air to enter the sealed space and causes the pressure in the sealed space to decay towards ambient pressure. The sensor **2124** may measure the pressure in the sealed space and provide the pressure measurements to the control circuit **2132**. The control circuit **2132** may record (store, save) the pressure measurements. In some embodiments, the control circuit **2132** may collect the pressure measurements to form a pressure decay curve.

At step **610**, the volume of the wound space **2120** is determined based on the pressure measurements. For example, based on the known controlled rate of airflow through the valve **2126** and the measured pressure decay curve, the volume of the sealed space may be determined. The volume of the wound space may then be determined by removing a volume of the canister and tube from the total

volume of the sealed space. In some cases, one or more additional valves, sensors, etc. are included to facilitate generation and collection of data for use in wound size determination. Various methods for calculating wound size are possible in various embodiments, for example as described with reference to FIGS. **1-17**.

At step **2612**, the wound size (e.g., the volume of the wound space **2120**) and/or a message relating thereto is displayed on the user interface **2128**. For example, the control circuit **2130** may cause a graphical user interface that includes the wound size to be displayed on a screen of the user interface **2128**. As another example, the control circuit **2130** may determine one or more warnings, progress reports, or other wound-related message based on the wound size and control the user interface **2128** to display the warning, report, or other message. For example, the user interface **2128** may display a graphical representation of change in the volume of the wound space over time.

At step **2614**, a wound therapy is customized based on the volume of the wound space. In some embodiments, the control circuit **2130** automatically customizes a wound therapy based on the determined volume of the wound space **2120**. In other embodiments, a user is facilitated in customizing a wound therapy based on the volume of the wound space **2120** based on information displayed on the user interface **2128**.

In the example shown, the control circuit **2130** automatically customizes instillation by automatically determining an amount of instillation fluid to be supplied to the wound space **2120** based on the determined volume of the wound space **2120**. For example, the control circuit **2130** may multiply the determined volume of the wound space **2120** by a scaling factor to determine the amount of instillation fluid to be supplied to the wound space **2120**. As another example, the control circuit **2130** may determine the amount of instillation to be supplied as equal to the volume of the wound space **2120**. Various calculations are possible for various applications, wound types, instillation fluid types, patient and/or caregiver preferences, etc.

At step **2616**, the customized wound therapy is provided. For example, the control circuit **2130** may control the instillation pump **2130** to provide the determined amount of instillation fluid from the instillation fluid source **2110** to the wound space **2120**. Instillation therapy may thereby be tailored to meet the needs of the healing wound in real time. Various other customized therapies are possible in various embodiments.

Referring now to FIG. **24**, a collection of graphs illustrating the operation of the NPIWT system **2100** is shown, according to an exemplary embodiment. FIG. **24** shows a measured pressure graph **2700**, a control signal graph **2702**, and an introduced pressure graph **2704**. The pressure graph **2700** illustrates a change in pressure in the sealed space over time as measured by the pressure sensor **2124**. As illustrated by the pressure graph **2700**, a measured pressure line **2706** approaches a desired negative pressure (shown as -200 mmHg) as the pneumatic pump **2122** is operated to draw air out of the sealed space. The measured pressure then decays as the valve **2126** is controlled to allow a controlled rate of airflow therethrough.

The control signal graph **2702** illustrates a voltage pattern applied to the valve **2126** (i.e., across the solenoid **2206**). As shown, a control signal **2708** alternates between approximately zero voltage and a non-zero voltage, shown as approximately five volts. As shown, the control signal includes approximately 400 milliseconds at the non-zero voltage, approximately 100 milliseconds at approximately

zero voltage, another approximately 400 milliseconds at the non-zero voltage, and another approximately 100 milliseconds at approximately zero voltage. After these two repetitions (i.e., after two periods of 500 milliseconds), the control signal may include one second at approximately zero voltage as illustrated in FIG. 24. It should be understood that various other frequencies and periods for a voltage pattern may be used in various embodiments. As one possible additional example, in an alternative embodiment the voltage pattern alternates between approximately 200 milliseconds at a non-zero voltage and 50 milliseconds at approximately zero voltage for three or more repetitions (e.g., four repetitions), followed by approximately one second at approximately zero voltage before repeating the voltage pattern. In various embodiments, the non-zero voltage is repeatedly provided for a duration between a minimum continuous duration of approximately 50 milliseconds and a maximum continuous duration of approximately 500 milliseconds, with alternating periods of approximately zero voltage.

The introduced pressure graph 2704 illustrates the amount of pressure let into the sealed space over time. In the example shown, approximately 5 mmHg is introduced into the sealed space for each 400 millisecond segment of non-zero voltage in the control signal 708. The introduced pressure graph 2704 illustrates that the pressure decay in the sealed space may be managed by the alternating pattern of the valve 2126 (i.e., of the control signal 2708). For example, the introduced pressure graph 2704 indicates that a lag time may exist between the beginning of a non-zero voltage period and a point in time corresponding to peak rate of pressure reduction or peak rate of airflow through the valve 2126.

Referring now to FIG. 8, the NPIWT system 2100 of FIG. 18 is shown in an alternative embodiment. In the embodiment shown, the volume of the sealed space and/or the wound space can be determined by first determining the restriction rate of the filter 2138 (i.e., the rate of airflow through the filter 2138) as part of a calibration process before therapy is started. As shown in FIG. 18, a removable cap (structure, cover, joint) 2800 is placed proximate a point wherein the inner lumen 2400 and the outer lumens 2402 come together to form the first tubing 2106 (e.g., at a port of the canister 2104). For example, the first tubing 2106 may be disconnected at this point and replaced by the removable cap 2800 as shown in FIG. 8. The removable cap 2800 connects the inner lumen 2400 and the outer lumens 2402 and causes air to flow directly therebetween (i.e., without passing through the dressing 2102) during a process for determining the restriction rate of the filter 2138. The removable cap 2800 may then be removed and the first tubing 2106 connected in the configuration described above.

To determine the restriction rate of the filter 2138 while the removable cap 2800 is applied as in FIG. 8, the valve 2126 is closed and the pneumatic pump 2122 is run to remove air from the canister 2104. The valve 2126 may then be opened for an indefinite amount of time to allow air to flow back into the canister 2104 while the pressure sensor 2124 measures the change in pressure over time. Based on a known volume of the canister 2104 and the change in pressure over time while the valve 2126 is open, the control circuit 2132 may calculate the rate of airflow through the filter. The cap 2800 facilitates this process by ensuring that the unknown volume of the wound space does not influence the rate of change of the pressure during such a process.

Configuration of Exemplary Embodiments

As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a

broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure. Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, calculation steps, processing steps, comparison steps, and decision steps.

The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The

order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

The “circuit” may also include one or more processors communicably coupled to one or more memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc. In some embodiments, the one or more processors may be external to the apparatus, for example the one or more processors may be a remote processor (e.g., a cloud based processor). Alternatively or additionally, the one or more processors may be internal and/or local to the apparatus. In this regard, a given circuit or components thereof may be disposed locally (e.g., as part of a local server, a local computing system, etc.) or remotely (e.g., as part of a remote server such as a cloud based server). To that end, a “circuit” as described herein may include components that are distributed across one or more locations. The present disclosure contemplates methods, systems and program products on any machine-readable media for

accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

What is claimed is:

1. A control circuit for a negative pressure therapy system, the control circuit communicatively coupled to a pump that is fluidly communicable with a sealed space, a sensor fluidly coupled to the sealed space, and a valve fluidly coupled between a sealed space and a surrounding environment, the control circuit configured to:

actuate the pump to draw fluid from the sealed space, repeatedly actuate the valve between an open position and a closed position, allowing a controlled rate of airflow through the valve;

receive measurements of a pressure in the sealed space generated by the sensor during a pressure decay in response to the controlled rate of airflow through the valve; and

in response to the measurements of the pressure and the controlled rate of airflow, determine a volume of the sealed space.

2. The control circuit of claim 1, wherein the controlled rate of airflow is less than a restriction rate of a filter fluidly coupled between the valve and a canister.

3. The control circuit of claim 1, wherein the valve comprises a solenoid valve and the control circuit is further configured to actuate the valve by providing a voltage pattern to the solenoid valve.

4. The control circuit of claim 3, wherein the voltage pattern comprises a step function between a zero voltage and a non-zero voltage.

5. The control circuit of claim 4, wherein the voltage pattern remains at the non-zero voltage for no more than a maximum continuous duration of 500 milliseconds.

6. The control circuit of claim 5, wherein voltage pattern comprises a repeating pattern of 400 milliseconds at a non-zero voltage, 100 milliseconds at zero voltage, 400 milliseconds at the non-zero voltage, and 100 milliseconds at zero voltage.

7. The control circuit of claim 6, wherein the voltage pattern comprises a first set of two periods of the repeating pattern, one second at zero voltage, and a second set of two periods of the repeating pattern.

8. The control circuit of claim 3, wherein the voltage pattern causes the solenoid valve to alternate between the open position and the closed position with a period of 500 milliseconds.

9. The control circuit of claim 1, wherein the control circuit is further configured to determine a customized wound therapy based on the volume of the sealed space and to provide the customized wound therapy. 5

10. The control circuit of claim 9, wherein the customized wound therapy comprises instillation therapy. 10

11. The control circuit of claim 10, wherein the control circuit is configured to customize the instillation therapy by determining an amount of instillation fluid to supply to the sealed space based on the volume of the sealed space.

12. The control circuit of claim 11, wherein the control circuit is further configured to actuate an instillation pump to provide the amount of the instillation fluid to the sealed space. 15

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